

Somewhere Over the Horizon: causal contact under the antipodal identification of horizons, and why it needs four dimensions

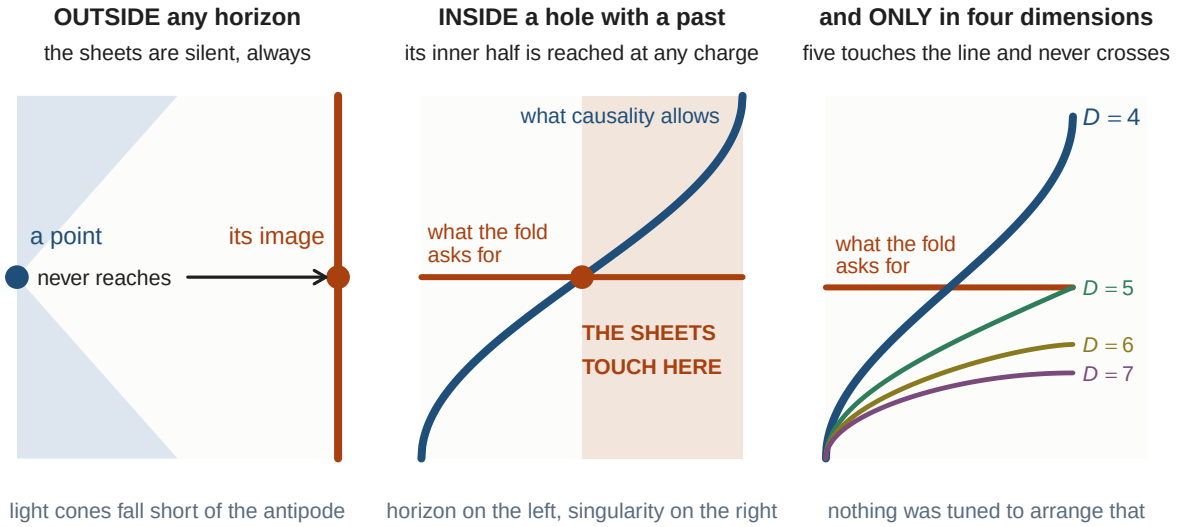
B. H. Wiseman

Linnet Labs, Sydney, Australia (independent researcher; no external funding)

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Companion to Separate Ways and the Upside Down: the classical world and the field equations from a CPT fold, and the neutrino line that would break it.

Where the two sheets can touch, and where they cannot



A CPT-folded universe is sealed at every horizon by a theorem, and opens in one place only:
inside a black hole that has a past, in the inner half of an uncharged one.

Graphical abstract. Left: outside any horizon a point and its fold image are spacelike separated, because reaching the antipode takes a conformal time of π and the spacetime lasts π , so both light cones fall short and the two sheets are silent. Centre: inside a hole possessing a past the separation turns timelike, and the angle a causal curve can sweep, $4 \arcsin \sqrt{x}$ across the interior, first meets the half-turn the antipodal map asks for at the midpoint, so contact fills the inner half at any charge. Right: the same sweep in D dimensions clears that with a factor of two to spare in four, equals it in five where no curve attains it, and falls short above. No parameter enters either side of the comparison.

Abstract

The antipodal, or CPT, identification of a spacetime pairs every point with an image. At a cosmological horizon it underlies the CPT-symmetric universe of Boyle, Finn and Turok, and at a black hole the elliptic interpretation of Gibbons and of Sanchez and Whiting. We ask where a point can be causally connected to its own image. Outside every horizon it cannot: the two are spacelike separated and their commutator vanishes. Inside a black hole with a past it can, but reaching the image takes half a turn of the transverse sphere as well as a fall, and causality caps the turn at $\pi/(D-3)$ per leg whatever the mass. Two legs give $2\pi/(D-3)$ against the π required, so in four dimensions contact fills the inner half of the interior at any charge, and in five or more large dimensions it never occurs. In that region the identified spacetime has closed causal curves. At our own horizon the same map gives a reciprocal law for correlators that holds

in every state, with all even multipoles vanishing, and it is the square root of the thermal transformation. Three ways of treating the image region as a second copy of our own space fail, on positivity, the first law and the binary pulsar. How the two regions join at a black-hole horizon is not fixed. The one join that absorbs no momentum is adopted, and it predicts zero ringdown damping-time deviation, testable at twice present exposure. Within the Kerr-Newman-de Sitter family only maximal holes are mapped to themselves, out to a spin $a/M = 1.10092$.

1. Introduction

Every observer sits inside a horizon. Ours is cosmological: the expansion carries distant things away faster than their light can return, and what lies beyond that surface is out of reach permanently. Nothing is trapping anything, so the cause differs from a black hole's, while the causal structure is the same one and so is the mathematics that describes it. The horizon is not far away in the usual sense. It surrounds you.

CPT swaps matter for antimatter, reflects space and runs time backwards. Identifying each point of a spacetime with its image under it is an old idea with several forms. Boyle, Finn and Turok take CPT to hold of the universe and not only of the physics inside it, so the Big Bang has a far side, a mirror sheet [20,21]. At a black hole the same identification is the elliptic interpretation of Gibbons [29] and of Sanchez and Whiting [30], taken up again by 't Hooft [40] and, for the charged hole, by Strauss, Whiting and Franzen [41], and Harlow and Numasawa gauge it [6]. The cosmology paper this accompanies [39] fixes the map, a wedge reflection composed with the antipodal map of the transverse sphere, and calls the result the fold. There Einstein's field equations come out of that symmetry with the metric a fixed background at every step, so that derivation did not need a quantum theory of gravity, and the dark matter's mass is capped at 491.6 PeV.

This paper asks one question of the fold: where, if anywhere, can a point be causally connected to its own image? The answer and what follows from it are listed here, and each item says whether it is proved, calculated, adopted or open.

1. **Outside every horizon, nowhere** (proved; below, and §5.1): a point and its image are spacelike separated, so the cross-sheet commutator vanishes identically. The two sheets share correlations, as any two spacelike separated regions do, and cannot signal.
2. **Inside a black hole with a past, the inner half** (proved for Schwarzschild, a lower bound at charge; §5.1, A.15): reaching the image takes half a turn of the transverse sphere as well as a fall, and causality caps the turn at $2\pi - 4\arcsin\sqrt{r/2M}$, which reaches π at $r = M$. At charge the same calculation gives $r \leq (r_+ + r_-)/2$, and whether contact reaches further out is open. The fold is a free involution, so this region is where the quotient carries closed causal curves.
3. **Four large dimensions and no others** (proved; §5.2): a Schwarzschild interior in D dimensions supplies $2\pi/(D-3)$ of turning against the π required. Four clears it, five equals it and never attains it, six and above fall short. Compact extra dimensions are constrained, not forbidden.
4. **A reciprocal law at our own horizon** (exact; §3, A.10): write R for the ratio of the correlation between a point and a reflected point to the correlation between that point and an unreflected point. Then R times its own reflection is exactly one, for every state, every metric on the horizon and every free involution, so $\log R$ is odd and every even multipole vanishes. The fold's map is the square root of the thermal transformation, which fixes the two-sheet squeeze at $\tanh r = e^{-\beta\omega/2}$.
5. **The mirror sheet is not a copy of our own space** (three tests; §3.1 to §3.3): three readings of it as a single copy fail, on positivity, on the first law, and against the orbital decay of the Hulse-Taylor binary pulsar.
6. **What happens at contact** (calculated in model geometries; §5.3, A.18, A.19): the image term diverges as $\delta^{-3/2}$ at the contact radius and turns imaginary inside it, which is a decay rate and so pair creation. Its stress defocuses the curves that would make contact, which supports chronology protection here without proving it.

7. **The seam** (adopted; §4, A.11): the fold does not fix how the sheets join at a black-hole horizon, and of the family of joins exactly one absorbs no momentum, the transparent seam, which is adopted on that stated premise. It predicts an ordinary Kerr ringdown with zero damping-time deviation, which about twice the present exposure can test. A.14 also gives the response of a ringdown to a horizon reflectivity, and that holds for any reflecting-horizon model.
8. **Which holes have a past** (counted within one family; §6): a hole formed by collapse has no second exterior, and none of this describes it. Within the Kerr-Newman-de Sitter family the holes the fold maps to themselves are the maximal ones alone, out to a spin $a/M = 1.10092$. Such a hole would not grow faster for having two sides. None is known to exist.

Items 1 to 5 are forced by geometry and take nothing from the cosmology paper beyond the map. Items 6 to 8 are conditional, and each states its condition. Section 7 records what the fold requires of the matter sector, conditional on the cosmology paper's production model, and Section 8 shows why counting the radiation does not decide whether the mirror sheet is a copy of our degrees of freedom or a place carrying its own.

One structural consequence comes before any result, because several of the results are instances of it. The cosmology paper's §2.1 splits every field into the average of its two sheet readings and their difference,

$$\Phi_c = \frac{\Phi + \Theta\Phi}{2}, \quad \Phi_q = \Phi - \Theta\Phi,$$

which are the closed-time-path variables and obey $[\Phi_c, \Phi_c] = [\Phi_q, \Phi_q] = 0$ and $[\Phi_c, \Phi_q] = i\Delta$. Since $\Theta^2 = 1$, the average is exactly the fold-even part of the field and the difference exactly the fold-odd part. The fold's parity is therefore the classical-quantum split itself. What survives the quotient is the classical content, and what the quotient discards is where the commutator lives.

Two of this paper's results are instances of that, and reading them together is the point of saying it here. Section 1's silence is the statement that the cross-sheet commutator vanishes outside a horizon, so the quantum variable has nothing to act on there and the classical world is classical for a reason. And the two modes that destabilise the maximal hole of Section 6 are fold-odd, which is to say they are quantum-sector modes: they do not appear in the classical geometry at all, which is why the fold removes them without anything being tuned. A.8 then ties the two together. The fold's map is a half-period thermal shift, so at a horizon in equilibrium the cross-sheet correlator is the direct one shifted by $i\beta/2$, which fixes the two-sheet squeeze at $\tanh r = e^{-\beta\omega/2}$ with nothing left to choose. So the quantum half of the split has an amplitude, and that amplitude is a temperature.

The obvious objection to a mirror sheet is that nobody has ever seen one. This paper's first result answers that objection with a calculation. The cosmology paper's §2.1 builds its algebra on the premise that a point and its fold image are causally disjoint. On the geometry we actually inhabit that premise is free: elliptic de Sitter space puts every point spacelike to its own antipode, so the commutator between a field at a point and the same field at the folded image vanishes identically. The two sheets are correlated everywhere and can signal nowhere. The silence is a theorem.

If the seal is exact, the useful question is whether it breaks anywhere. There is one kind of place. Carry the same construction to a black hole, where the fold's map is the wedge reflection $J : (U, V) \mapsto (-U, -V)$, and

$$dU = 2U, \quad dV = 2V \quad \implies \quad dU dV = 4UV,$$

so the radial part of the separation between a point and its image turns timelike exactly when $UV > 0$. That region is the interior of the hole, and nothing else. Whether the full four-dimensional separation follows is a second question, taken up below. The cross-sheet commutator is identically zero outside a horizon and nonzero inside, and it changes over at the horizon because UV changes sign there, with nothing tuned to put it there. Operators that fail to commute can signal, so if the two halves of a folded universe can touch anywhere at all, the only place is inside a black hole, where no outside observer can ever look.

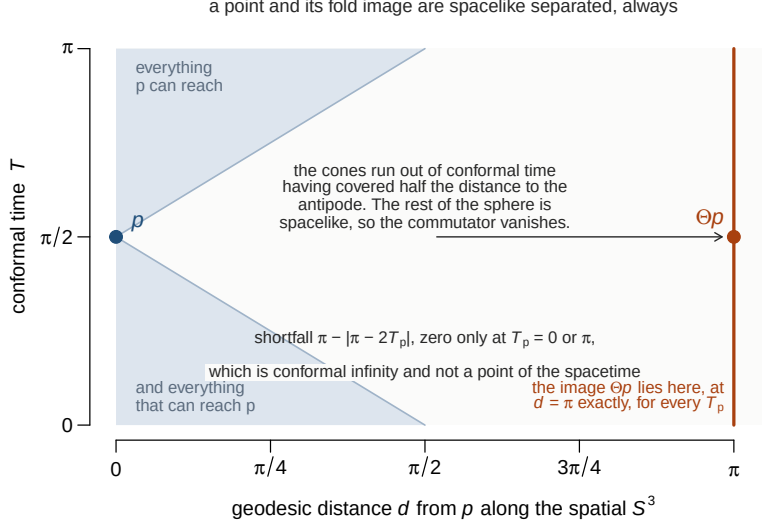


Figure 1: Why. de Sitter is conformal to a piece of the Einstein static cylinder, with round three-spheres for spatial slices and a conformal time running over an interval of length π . The antipodal map of the embedding is the antipodal map of that S^3 together with $T \mapsto \pi - T$, and two points antipodal on a sphere are at geodesic distance π whatever their position, so the image of p sits on the right-hand edge at every conformal time. The horizontal axis is therefore distance along S^3 rather than the polar angle, because the conformal square hides the angular part of the separation and that part is the whole argument. Radial null rays are then lines at forty-five degrees, and both of p 's cones run out of conformal time having covered less than half of the way round: the future one at $T = \pi$ and the past one at $T = 0$. The shortfall is $\pi - |\pi - 2T_p|$, which is positive for every point of the spacetime and reaches zero only at conformal infinity. Equivalently and with no picture, $Z = X \cdot Y / \ell^2$ is one at coincidence and exceeds one for timelike separation, and $Z(X, -X) = -1$. A map turning through only half as much of the sphere fails this.

2. When the time directions lose coherence

Before any of that, the two sheets have to be two. Hartle and Hawking's wavefunction is real-valued, and a real wavefunction contains both an expanding and a contracting WKB branch [1]. Inhomogeneous modes can decohere them, as in the work of Halliwell, Hawking and Kiefer [2-5]. Here the two branches are compared by an explicit overlap instead. Two settings appear below and they are not the same calculation, the closed-de Sitter throat, and the radiation-dominated bang the cosmology paper adopts. Each carries its own notion of time, so a threshold quoted in one does not transfer to the other, and the sections below say which is in use.

The closed-de Sitter calculation uses standard no-boundary branch states and does not require the matter model of Boyle, Finn and Turok, whose cosmology the cosmology paper adopts. Identifying those branches with the CPT-related copies is an additional step, and the next paragraphs separate what supports it from what does not. The cosmology paper's own numbers do not depend on the outcome, since that paper adopts its cosmology as a separate input and uses none of the patch algebra.

What supports it is that one map does both jobs. In the embedding, α sends $X \mapsto -X$, so it carries the right static patch $X_1 > |X_0|$ to the left one and simultaneously sends $X_0 \mapsto -X_0$, hence $t \mapsto -t$, exchanging the contracting and expanding phases of $a(t) = \cosh Ht/H$. Both exchanges are the same involution acting on two coordinates at once, so the assignment of the right patch and the expanding branch to one sheet, and the left patch and the contracting branch to the other, requires no second map and makes no choice twice.

One further thing follows, and one does not. An antiunitary relates the two patch algebras, conjugation composed with a geometric map. What relates the two branch states is also conjugation, §2.1 obtains them as complex conjugates because the Wheeler-DeWitt operator is real. Both are conjugation, and α performs it. Its time reversal sends $\eta \rightarrow -\eta$, and the environment's frequency function obeys $\Omega_n(-\eta) = \overline{\Omega_n(\eta)}$.

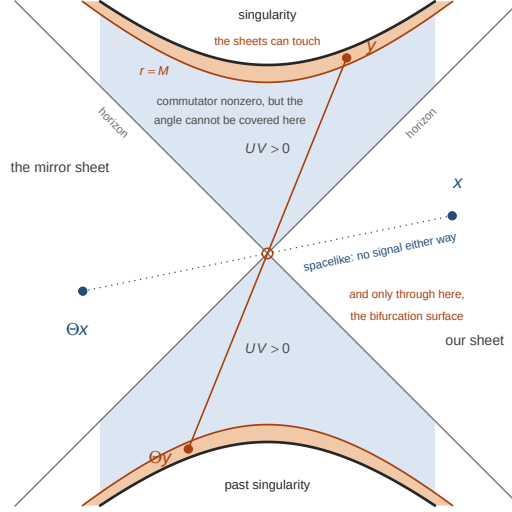


Figure 2: Kruskal diagram of an eternal hole. The fold’s wedge reflection is the point reflection through the origin, so it swaps the two exteriors and swaps the two interiors. Under it the null coordinates scale as $dU = 2U$ and $dV = 2V$, so the cross-sheet commutator goes as UV : it vanishes identically in the exteriors, where a point and its image Θx are spacelike separated, and is nonzero in the shaded interior regions, where y and Θy are not. Contact does not open at the horizon, because reaching the image’s causal region also means covering half a turn of the transverse sphere, and A.15 shows causality allows that only at $r \leq M$. The inner shaded band is that region, and both boundaries are exact. The curve that achieves it passes through the bifurcation surface at the centre, which is the only route the geometry allows.

That identity is algebraically empty on its own and we do not offer it as evidence, every function of the form $(\text{even} + i\text{odd})/\text{even}$ satisfies it. What is not empty is that Ω_n has that form. Its real part carries the η dependence only through an even denominator and its imaginary part is odd, and the reason is narrower than the reality of the Wheeler-DeWitt operator, which alone gives only the empty conjugate relation. A real equation together with a real datum at $\eta = 0$ produces the form: $\Omega_n(0) = (n^2 - 1)/n$ is real because of the no-boundary condition at the throat, and $a(\eta) = 1/(H \cos \eta)$ is even there. Given the form, the Gaussian $\psi_n \propto \exp(-\Omega_n \chi^2/2)$ on one branch is the antilinear image of the other’s.

That the no-boundary condition holds exactly is checkable, and it settles what kind of object these states are: $\Omega_n(0) = (n^2 - 1)/n$ is real at every n with the Wronskian normalised to one, which is the Euclidean condition, and the Euclidean state on closed de Sitter is Bunch-Davies. The two branch states are therefore the same Bunch-Davies state conditioned on the two branches, and no state on any other space, and its restriction to a static patch is KMS at $T = H/2\pi$ by Gibbons-Hawking, which A.6 already uses.

What is still not shown is the identification itself, and the obstruction is specific. The two pairs are different partitions of the same manifold. The patch pair is $X_1 > |X_0|$ and its image, the branch pair is $X_0 > 0$ and $X_0 < 0$, and the right static patch meets both branches. One involution exchanging the members of each does not make them one pair. Conditioning a state on a WKB branch is also not the same operation as restricting it to a region, and the calculation below settles the question in the negative.

The right static patch is the causal diamond of a worldline and reaches past and future infinity: solving $X_1^2 = H^{-2} + X_0^2 - (X_2^2 + X_3^2 + X_4^2)$, the patch condition $X_1 > |X_0|$ bounds only the transverse part, so the patch contains points at every global time. The branch label is $\text{sign}(\dot{a})$, hence $\text{sign}(X_0)$, and the patch straddles both signs in equal measure. An operator localised in the patch therefore has support on both branches, and no element of the patch algebra determines the branch label.

So the branch conditioning is not an operation on the patch algebra at all. It acts on the gravitational wavefunction, whose split is by global time; the restriction acts on the matter algebra, whose split is by

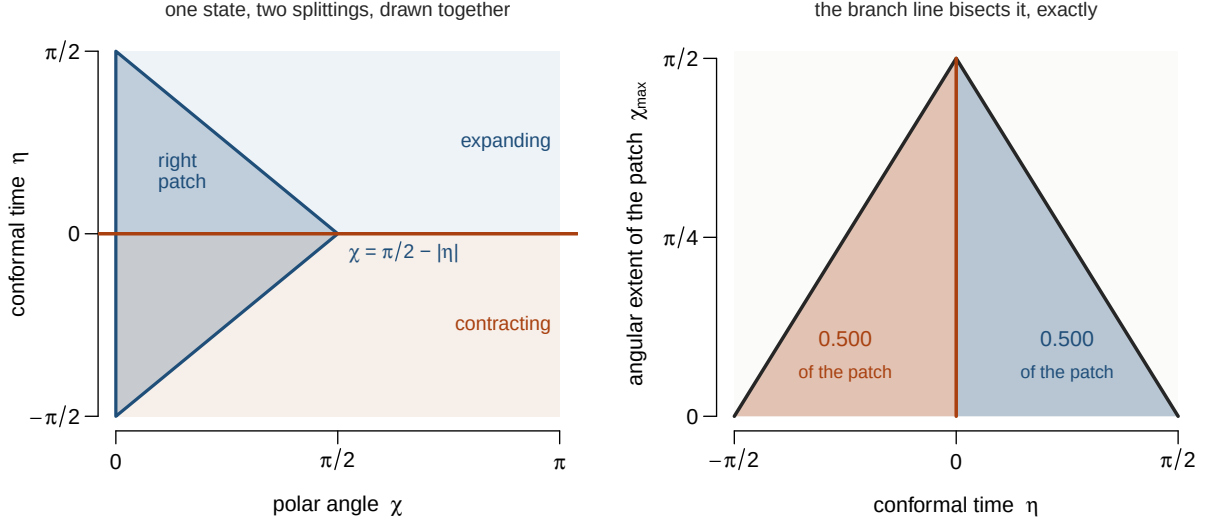


Figure 3: One state, split two ways, and the two ways cross. Left: closed de Sitter in conformal coordinates, global time η running up and the polar angle χ measured from the observer’s worldline. The right static patch is the diamond $\chi \leq \pi/2 - |\eta|$, which is a split by causal position, while the branch label is the sign of $\dot{a}$ and so of η , a split by global time. The branch line runs straight through the patch. Right: the patch’s angular extent against global time, with the two halves shaded. The conformal four-volume divides 0.500 to 0.500, and it is exact, because the patch condition and the metric are both even in η , so $\eta \rightarrow -\eta$ carries each half onto the other. All four cells are occupied, which is why an operator localised in the patch has support on both branches and no element of the patch algebra fixes the branch label. One involution exchanges the members of each pair, and that is not enough to make the two pairs one pair.

causal position. The two do not fail to commute so much as fail to compose, and the identification cannot be made this way.

What survives is exact and is what we claim. There is one state, the Bunch-Davies state, and it carries two splittings: by causal position into the two patches, and by global time into the two branches. α exchanges the members of each. The patch restriction is KMS at $T = H/2\pi$ and the branch conditioning returns complex conjugates. Those are true statements about a single state with two splittings that one involution relates, and the splittings cross. The horizon results of §§2 and 3 use the first splitting, and the cosmology paper’s results use neither, taking their inputs from the adopted cosmology instead, so nothing below rests on an identification we have just shown cannot be made. Nothing has to be asserted to match, because the contracting branch’s environment state is α ’s image of the expanding one. One structure appears in two sectors, and no map from a region to a solution is required.

The limit is the setting. This is the free nonconformal environment of §2.1 on closed de Sitter, and the check is that the antilinear lift acts as conjugation on that environment’s frequency function. It has not been carried to the interacting matter sector at a radiation bang, which is where §2.1, and the cosmology paper’s §3 numbers, live, and where the same argument would need the corresponding mode functions. The step is therefore exhibited on the free de Sitter environment and open at the bang.

2.1 A closed-de Sitter overlap

Closed FRW minisuperspace with $\Lambda = 3H^2$ has a classically allowed region beginning at the no-boundary throat $aH = 1$. For N_f minimally coupled massless scalar fields, the exact mode function gives

$$\Omega_n = \frac{n(n^2 - 1) - i \tan \eta \sec^2 \eta}{\tan^2 \eta + n^2}, \quad \text{Im } \Omega_n = -\frac{\dot{a} (aH)^2}{\dot{a}^2 + n^2}.$$

Both branches carry complex-conjugate Gaussian environment states. With $A \equiv aH$, their overlap is

$$\log |\langle E_- | E_+ \rangle| = -\frac{N_f}{4} \sum_{n \geq 2} n^2 \log \left[1 + \frac{A^4(A^2 - 1)}{n^2(n^2 - 1)^2} \right],$$

with large- A asymptote $-(\pi/12)N_f A^3$. Appendix A.7 derives the sum. Summed over all harmonics, for one minimally coupled scalar, the overlap reaches e^{-1} at $aH = 1.95376$, $Ht = 1.28984$.

That is the ungauged number, and the question is whether gauging changes it. The cosmology paper's Appendix C.2 adopts Harlow and Numasawa's argument that a spacetime inversion cannot be a global symmetry and must be gauged [6]. The fold has a linear involution available to gauge, and it is not Θ : Θ is antilinear and so is the modular conjugation, so their composition is linear and implements $P_\perp$. Gauging it restricts physical states to its invariant sector. Care is needed over which involution acts. $P_\perp$ is $\text{diag}(+1, +1, -1, -1, -1)$ in the embedding, so on a closed slice it sends $(X_1, X_2, X_3, X_4) \mapsto (X_1, -X_2, -X_3, -X_4)$: it fixes one spatial direction and reverses three, and is not the antipodal map of S^3 , which reverses all four. It therefore does not multiply a harmonic level by a sign, it splits it. Degree- k harmonics decompose under the $SO(3)$ acting on (X_2, X_3, X_4) into $l = 0, \dots, k$, and $P_\perp$ acts on the l multiplet as $(-1)^l$, so the invariant dimension at level $n = k + 1$ is $\sum_{l \text{ even}} (2l + 1)$ rather than n^2 or nothing. At $n = 2$ the four harmonics split as one invariant and three not. That is the correct one-particle decomposition and it is not the mode count the overlap uses.

Restricting physical states removes no oscillator from the product. The branch states are Gaussian in each mode coordinate, $\psi_\pm(q) \propto e^{-(a \pm ib)q^2/2}$, and a Gaussian in q^2 cannot tell $+q$ from $-q$, so the projector $(1 + U(P_\perp))/2$ acts on it as the identity whatever the parity of the harmonic it belongs to. Each mode keeps its overlap factor $(1 + b^2/a^2)^{-1/4} < 1$ instead of having it replaced by one. Two quanta in an odd mode are invariant as well.

So the threshold is the ungauged one,

$$aH = 1.95376, \quad Ht = 1.28984,$$

which is the figure quoted above recomputed independently from the displayed sum. The two agree to five figures.

Three countings of the harmonics are available and one of them is right. Deleting the odd-parity oscillators moves the threshold to $aH = 2.43912$ and $Ht = 1.53984$, nineteen per cent later in proper time, and that counting is not adopted: deleting them needs a restriction on the field itself, where the states would be the natural place to look, and that is a stronger hypothesis than the one the cosmology paper's C.2 adopts and one this paper does not derive. Keeping whole levels of odd n gives 2.59538 instead, and that is the parity of the full antipodal map rather than of $P_\perp$, which is not the map the fold quotients by. Nothing has to be added back. Gauging a discrete group introduces sectors classified by $H^1(M; \mathbb{Z}_2)$, and the cosmology paper's C.2 establishes that this vanishes on the cover, so there is one bundle, no twisted sector, and the invariant subspace is the whole physical space. The cosmology paper's C.2 records that vanishing as a negative, against a holonomy route to the stabilising rule; here it is what makes the gauged answer clean. For the restricted counting, the ratio of restricted to unrestricted exponent tends to one half at large A , and both ways of dropping modes approach it, 0.4999 and 0.4998 at $A = 40$, so that limit does not distinguish which half has been dropped and cannot be used to choose between them.

No explicit $\hbar$ or G appears in the exponent, and the background Hubble rate sets the time scale. The threshold declares a measure of decoherence. It does not make time appear discontinuously and it does not select a branch. Conformally coupled free fields have real widths in this calculation and do not contribute to the separation. Both the environment and the chosen overlap threshold move that separation.

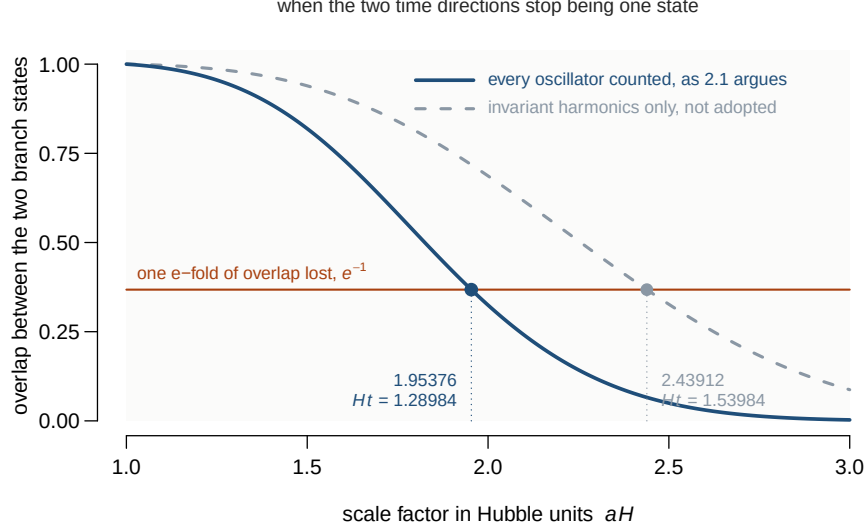


Figure 4: The overlap of the two branch states on closed de Sitter, against the scale factor in Hubble units. §2.1 gives $\log |\langle E_- | E_+ \rangle|$ as a sum over harmonics weighted by d_n , and the whole question is which d_n . Counting every oscillator gives n^2 and the solid curve; counting only the $P_\perp$ -invariant harmonics gives the sum of $2l + 1$ over even l , which at $n = 2$ is one against four, and the dashed curve. Both reach e^{-1} , one at $aH = 1.95376$ and the other at 2.43912 , a threshold nineteen per cent later in proper time. This subsection argues the solid one applies, because the branch states are Gaussian in each mode coordinate and a Gaussian in q^2 cannot tell $+q$ from $-q$, so the projector acts as the identity on every mode whatever the parity of its harmonic. The dashed reading is drawn for comparison and is not the one adopted.

The RP^3 harmonic restriction is a separate topology-dependent comparison, and it is easily run together with the comparison above. The cover does not acquire a restriction merely because its fields carry an antilinear CPT relation; an antilinear relation between two sheets restricts nothing on either. What restricts is the linear involution inside it, once gauged, and that has nothing to do with the topology of the quotient. One restriction comes of choosing a different spacetime, the other follows from a symmetry this paper already commits to.

3. Horizon law, and three tests of what the mirror sheet is

One statement about our own horizon is settled before any question about the mirror sheet arises, and it holds whichever answer that question gets. Let B be the bifurcation surface of the horizon, let $P_\perp$ be the free involution of B the fold supplies, and let W_B be the restriction of the two-point function to B . Write

$$R(x, y) = \frac{W_B(x, P_\perp y)}{W_B(x, y)}$$

for the correlation across a reflected separation over the correlation across the direct one. Replacing y by $P_\perp y$ exchanges the numerator with the denominator, since $P_\perp^2 = 1$, so

$$R(x, y) R(x, P_\perp y) = 1$$

identically. That substitution is the entire proof and nothing else enters it, so the law holds for every state, every correlator, every metric on B and every free involution. On a round B with an isotropic state $P_\perp$ carries an angle to its supplement, so $\log R$ is odd about ninety degrees, every even multipole of it vanishes identically, and the value at ninety degrees is one, which is the law's own fixed point.

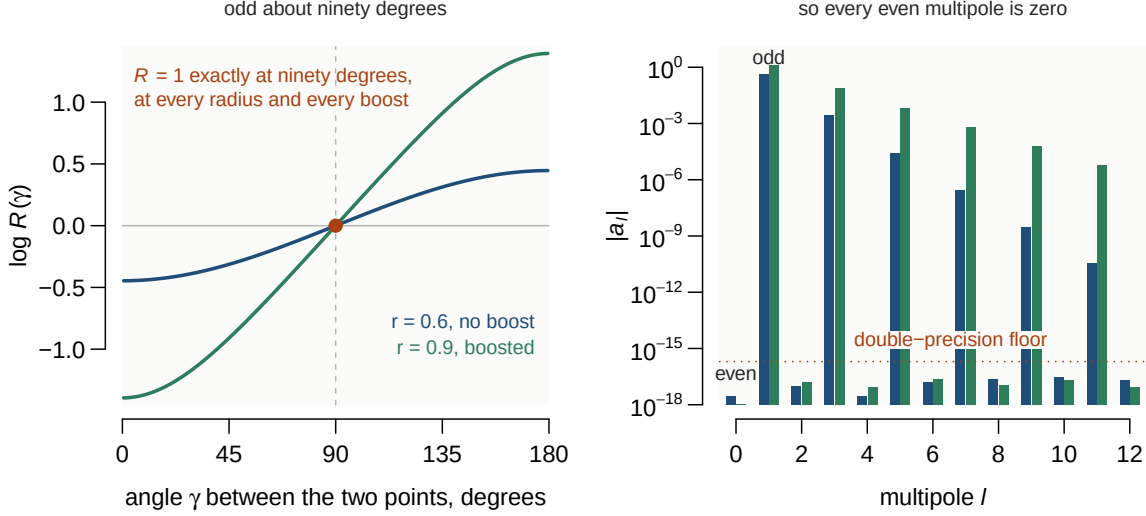


Figure 5: The law and what it kills, for the conformal scalar on de Sitter. A.10’s two kernels differ only in the sign of the transverse term, $Z_{\alpha,J} = -(1-r^2) \cosh \delta t \mp r^2 \cos \gamma$, so their ratio is $(c-b \cos \gamma)/(c+b \cos \gamma)$ with $c > b > 0$. Sending γ to its supplement swaps numerator and denominator, so $\log R$ is odd about ninety degrees, exactly, at every radius and every boost; the two curves are one unboosted case and one boosted one. Right: its Legendre coefficients. Since $P_\ell(-x) = (-1)^\ell P_\ell(x)$, every even multipole integrates an odd function against an even one and vanishes. The even bars are drawn at the panel floor because they are zero, the largest being 3×10^{-17} against odd coefficients of order one. A kernel without the antipodal symmetry, run as a control, puts its even coefficients at 0.52.

Being true of everything, the law forbids nothing by itself, and A.10 says so in those terms before saying what does carry the argument. Away from B the boost separation has not yet dropped out, and there, requiring the geometric image kernel to be a thermal half-period shift of the Bunch-Davies kernel fixes the period: the mismatch is 4×10^{-17} at $\beta = 2\pi/H$ and 0.115 ten per cent away on either side, a factor 2.6×10^{15} worse. No temperature was put in anywhere. What that does and does not establish is §3.4’s subject, A.10 has the derivation, and A.12 collects how far the law reaches beyond the surface. Its algebraic form on any algebra with an involution, the relative entropy it supplies as a measure of a state falling short of its own fold image, and what that measure says about §2.1’s branch weights.

Section 2 asked when the two time directions come apart. The rest of this section asks whether two of them exist at all. The cosmology paper’s §2.1 left three ways to use the fold’s antilinear map Θ : quotient spacetime by it, impose a reality condition on the quotient, or keep both sheets and pair the contour legs. They are not equivalent. Three tests below bear on the readings that make the mirror sheet a single copy of our own space, and the three come from unrelated parts of physics. The first is algebraic and asks which states the quotient admits; the second is thermodynamic and tests the first law; the third is dynamical, against a binary-pulsar measurement. Nothing connects the three arguments except the reading they select.

Each names the ansatz or response law it tests, and a failure constrains that choice, leaving every other quotient untouched. The scope matters and is kept throughout. What it does not diminish is that three independent lines of attack, from three different parts of physics, fall the same way.

3.1 A constant image-state ansatz

The first test is positivity. For the de Sitter-invariant conformally coupled scalar ansatz

$$W = \frac{1}{2}i\Delta + A G_{\text{sym}}(Z) + B G(-Z),$$

positivity requires $A \geq 1$ and $A^2 - B^2 \geq 1$. Hadamard normalisation at coincidence requires $A = 1$, and therefore $B = 0$. A nonzero constant image amplitude cannot satisfy both conditions. Positive image states outside the Hadamard class, including the alpha-vacuum family, are not excluded by positivity alone [7]. The result also leaves open smooth, state-dependent finite-frequency correlations outside this two-amplitude ansatz. Appendix A.8 gives the mode covariance and the two inequalities.

3.2 Entropy-temperature pairing

The second test is thermodynamic. The proposed elliptic pairing $(T_H, \mathcal{A}/8G)$, where $\mathcal{A}$ is the cover horizon area, fails the Smarr relation $M = 2TS$ and the first law $dM = T dS$. For example, in Schwarzschild units $G = M = 1$, $T_H = 1/(8\pi)$ and $\mathcal{A} = 16\pi$, the proposed pair gives $2TS = 1/2$ and $T_H d(\mathcal{A}/8G) = dM/2$. The two-copy pairing $(T_H, \mathcal{A}/4G)$ and 't Hooft's pairing $(2T_H, \mathcal{A}/8G)$ both give $2TS = M$ and the correct first law. The quotient two-point function used in the comparison has imaginary period $\beta_H/2 = 2\pi R$ at the central worldline, half the Hawking period, so its even sector realises 't Hooft's $2T_H$ rather than the two-copy T_H . Away from the centre, all harmonics contribute and no single temperature describes the response. What thermodynamics rejects is therefore the stated temperature-entropy assignment. The quotient geometry survives it.

3.3 A complete reciprocal time-symmetric response

Thermodynamics rejected the pairing without touching the geometry, so the third test is dynamical and looks at what the quotient does to radiation. The antipode reverses time orientation, and so exchanges the retarded and advanced kernels. Within the affine family $\alpha G_{\text{ret}} + (1 - \alpha)G_{\text{adv}}$, invariance selects

$$G_{\text{sym}} = \frac{G_{\text{ret}} + G_{\text{adv}}}{2}.$$

For the stationary problem considered here, reciprocal in the sense that the kernel is unchanged when its two arguments are swapped, this kernel is Hermitian. With a complete steady-state cycle, or vanishing all-history boundary terms, integration by parts gives zero total dissipative work. A.9 derives that vanishing, and checks the Wheeler-Feynman exchange of the retarded and advanced kernels under the antipode numerically alongside it. If this kernel is the binary's complete effective gravitational response, with no independent homogeneous drive or environmental contribution, the binary has no secular radiation damping. The observed orbital decay of PSR B1913+16 excludes that response, Weisberg and Huang report an observed-to-GR ratio of 0.9983 ± 0.0016 [8].

Quotient descent does not fix the complete response. An invariant homogeneous solution can change a lifted field without changing its source equation. Nor does antipodal parity imply local time parity: on $\mathbb{R} \times S^1$ with $A(t, \theta) = (-t, \theta + \pi)$, the homogeneous wave $h = \sin t \cos \theta$ is A-even, while $\partial_t h = \cos t \cos \theta$ is even in time at fixed position. A suitable local source can exchange work with this wave, with cancellation in the antipodal region. This scalar counterexample defeats the symmetry inference; it does not calculate a gravitational self-force.

A twisted Green kernel is odd under the antipode in each argument and even under its simultaneous action on both arguments. Its identity distribution is the twisted delta. There is no contradiction of the form $\delta = -\delta$.

One observational test survives, and within its premises it is exact. A complete reciprocal time-symmetric response has no secular dissipative work and fails the pulsar measurement. A more general quotient response requires its own boundary and source problem. Friedman and Higuchi's locally defined quotient theory makes the distinction between local physics and global identification explicit [9].

3.4 What the horizon law does not predict, and what would break it

Everything in Sections 2 and 3 is exact and none of it predicts a measurement, and this section states that plainly. Section 5’s contact result is forced too and is not covered by what follows; it predicts no measurement either, but for a different reason, which is that the region it describes is sealed behind a horizon.

Every cosmological observable is built from same-leg correlators, $W(x, y)$ or $W(x, x)$, galaxy clustering, the microwave background spectra, detector noise, the stochastic gravitational-wave background. The reciprocal law is a statement about the cross-leg kernel $W(x, P_\perp y)$, which lives in the doubled Hilbert space. That object is an influence functional only if the mirror sheet is a bath the observer is coupled to, and it is not, because we have access to one sheet. Two attempts to find an observable anyway both fail, and for compatible reasons: the galaxy-clustering identification compares the wrong object and is excluded outright if forced, and the discrepancy between the two kernels selects no axis, so it cannot produce a number-count dipole at any amplitude.

So the horizon law is valuable because it is exact, forced and independent of the state. Predicting is not what it does. It belongs in the family of Bisognano and Wichmann’s duality theorem [10] and of the thermofield-double structure of the eternal black hole [11]: results that fix what a theory must look like without telling an experimenter what to do. The testable number in this paper is Section 4’s ringdown null, which a running instrument is testing. It is not a consequence of the reciprocal law either.

The family resemblance is not an identity. Reading Θ as the Tomita-Takesaki modular conjugation would need a local net of algebras, a wedge, a Wightman vacuum and a modular flow implemented by boosts, and none of those is established here. What is claimed is a structural theorem about modular-like structure on a specific region, and no more.

The missing ingredients are specific. The three properties one would naturally check, that the involution squares to one, that it carries the algebra onto its commutant, and that it fixes the state, do not between them identify a modular conjugation. A finite standard pair shows it. Take the 2×2 matrices with the Hilbert-Schmidt inner product, M acting by left multiplication, $\rho = \text{diag}(0.8, 0.2)$, $\Omega = \rho^{1/2}$, and $\Theta(X) = PX^\dagger P$ with $P = \text{diag}(1, -1)$. All three properties hold to machine zero, and the relation that actually defines the conjugation, $S(A\Omega) = A^\dagger\Omega$ with $S = J\Delta^{1/2}$, fails by 1.7889. No choice of positive polar factor rescues it either, since ΘS has eigenvalues -2 and $-\frac{1}{2}$ and a positive operator has neither. The failure mode is a parity twist of J , which is exactly what $\Theta = J \circ P_\perp$ is by construction.

The same pair carries the repair. Θ is the modular conjugation of a *different* purification, $\Omega' = \rho^{1/2}P$, whose restriction to either factor is the same ρ . Against Ω' the twisted conjugation has zero defect and the untwisted one has 1.7889, the two have exchanged roles. The global vector has moved, overlapping the old one at 0.6, and a cross-sheet correlation has changed sign from $+0.4$ to -0.4 , while neither sheet’s own state has moved at all. A parity-twisted fold is therefore the modular conjugation of a parity-twisted purification, indistinguishable from the ordinary one on either sheet alone and differing only where the two sheets are correlated. That is this section’s silence stated in the other language.

Both statements are finite-dimensional. Neither is a continuum construction, neither supplies a Hadamard field state on the patch, and neither removes anything this subsection says is missing. What they do is replace “not established” with a specific thing to establish, and name the state it would have to be established for.

Two things would break the horizon law. The first is the wedge-reflection theorems. A.10 takes J to be fixed by them and $P_\perp$ to be whatever their composition leaves; if the fold’s setting does not meet their hypotheses, the transverse involution stops being forced and the whole “nothing is chosen” claim goes with it. The second is the Hadamard condition, which several results above assume and which the alpha-vacuum family does not satisfy, as 3.1 notes. Both are stated openly, and either one failing takes this half with it.

4. Minimal black-hole response

For black holes, the minimal implementation retains ordinary Kerr perturbation dynamics and the absorbing horizon boundary condition. We add no seam interaction. At the order treated, it therefore predicts no

additional horizon reflectivity and no fold-induced shift of the Kerr quasinormal spectrum.

That boundary condition is adopted. The obvious route to deriving it looks available and is not, and closing it takes a paragraph. Where the geometry is two-sided, which covers the cosmological horizon and an eternal hole, a constant reflectivity does put an image term between mirror-placed points at spacelike separation, and no admissible state is singular there. A.11 follows that route and closes it: neither way of counting the mode sum supports the conclusion, and in the one that gives a nonzero coefficient the transparent seam is the worst offender, where the argument needs it to be the only clean one. A hole formed by collapse has one exterior, so there are no mirror-placed points and the route does not even start. The absorbing condition is therefore the standard choice in both cases, as it is throughout the ringdown literature, and the prediction is the same. A.17 asks whether matter may transit the singular locus and make the Hawking temperature dispensable, and finds the obstruction is the fold's own.

The boundary condition is a physical commitment of this implementation. Covariance under the fold and conservation of energy do not select it. Energy crosses the seam at every reflectivity, and the allowed seam matrices include a continuous range of them. Momentum crosses freely at one only, and that is the narrowing A.11 uses, under a premise about the seam rather than a consequence of the fold. The remaining structure does not narrow that range, symmetry under the fold and closure of the constraint algebra admit the whole family. Nor does topology. On the de Sitter cover, which is simply connected with H^1 vanishing for every coefficient group, there are no periods to quantise a coefficient against (the cosmology paper's C.2). That argument is about the cover, and for a gauged spacetime involution the sectors live on the quotient, so the quotient has to be checked as well.

$P_\perp$ acts on the closed slice as $(X_1, X_2, X_3, X_4) \mapsto (X_1, -X_2, -X_3, -X_4)$, whose fixed set is $X_2 = X_3 = X_4 = 0$, two points, so it is not free there even though it is free on B . Writing $X_1 = \cos \theta$ and the rest as $\sin \theta \hat{n}$, the map is $\hat{n} \mapsto -\hat{n}$ at fixed θ , so the quotient is the suspension of $\mathbb{RP}^2$. The suspension isomorphism then gives $H^1 = 0$ there as well, which is what the argument needed. It also gives $H^2 = \mathbb{Z}_2$, which the degree-one argument never looked at; for the group $\mathbb{Z}_2$ itself $H^2(\mathbb{Z}_2; U(1)) = 0$, so no discrete torsion arises from the group, and whether the space's H^2 does anything here we leave open, asserting neither way; that argument belongs to $S^3 \times \mathbb{R}$ and we do not carry it to a Kerr exterior.

So no symmetry or topological argument on the content already present fixes the coefficient, and what would is either additional field content, of the kind a gauge holonomy or a compact scalar would supply, or a condition of a different type. Hadamard regularity is a condition of the second kind, and A.11 shows it does not fix the coefficient either. Transparency is adopted. A positive reason for that choice does exist and it is short. The scale-free seams form a one-parameter group under composition, and the transparent value is its identity: it composes with any other seam and leaves that seam where it was, and it is the only member that does. Adopting it therefore adds no boundary term to the action rather than adding a particular one, which is what having no seam means as an operation on the theory. A.11 has the composition law. Two requirements come out of this that any proposed seam law must meet, ours or anyone's.

The first is ultraviolet. A dimensionless coefficient in the corner term forces a frequency-independent reflectivity, and no physical interface reflects identically at arbitrarily short wavelength. Completing a seam action in the ultraviolet comes before fixing its coefficient for any seam that reflects. For the corner term the question does not arise, because nothing here reflects. Within the corner term r cannot depend on frequency, and the transparent member $\kappa = 1$, $t = 1$ is the one this paper adopts. A.11 records that the boundary condition is a commitment chosen from a range that fold covariance and energy conservation leave open, narrowed to the transparent point only by the premise that the seam absorbs no momentum. The requirement applies undiminished to the reflective proposals, which carry a scale the corner term does not and therefore still need their own matching law. A.11 also records what a seam carrying the horizon's own scale would commit to.

The second is structural, and it says what a seam would have to be, where the usual argument says what it cannot be. Iyer-Wald entropy survives at a bifurcation surface because the Noether charge is built from the Killing field and its derivative, the field vanishes there while the antisymmetrised derivative does not. A connection inherits that surviving piece through the second term of its Lie derivative; a matter field does not, since its Lie derivative is $\xi^a \partial_a$ of the field and dies with ξ . Computed in Kruskal, the one-form's Lie derivative

holds at $-\kappa_{\text{sg}}A_U(0)$ at every distance while the scalar's falls away as the surface is approached. So for the minimally coupled scalar term examined here, that Lie-derivative contribution vanishes at the bifurcation surface. It does not follow that any seam contributing to horizon entropy must be connection-valued: a nonminimal coupling $f(\phi)R/16\pi G$ gives $S = (4G)^{-1} \int_B f(\phi) dA$ through the curvature derivative in Wald's formula even when $\mathcal{L}_\xi \phi = 0$, as Iyer and Wald's dilaton example shows. The statement is restricted to the action studied. It remains the reason the Chern-Simons precedents, which are connection-built, look close and do not transfer. A dynamical matching law for a general rotating-horizon seam remains unconstructed here, and the same requirement falls on any proposal with nonzero reflection, which must supply that law before its waveform can be compared with the data.

Two constructions inside the CPT-symmetric programme bear on this commitment. Tzanavaris, Boyle and Turok's black mirror [12] identifies horizon points while keeping the exteriors distinct, a third reading: neither the one-copy quotient tested in 3.1 to 3.3 nor the two-sheet structure used here. Those tests target a quotient and do not bear on it. The difference that does is this. The black mirror makes the horizon a locus of identification; the fold identifies no points anywhere, and the fixed-point set of its map does not lie on the horizon. These are different boundary conditions on the same exterior dynamics, so they differ in the seam matching law, the horizon algebra and the entropy each supports. That is the law left unconstructed above. Constructing it decides between them; null results, which both readings survive, do not.

Amaro Seoane [13] derives a horizon reflectivity for that model equal to the generalised Boltzmann factor, fixed by T_H . Its reach is bounded by one thing. The constant-horizon condition gives $\omega = 0$ rather than a computed complex quasinormal spectrum, and the inspiral signature is spin-dependent, so no universal sign for a damping-time shift follows. The size of the reflectivity turns on a term that is easy to drop. The generalised Boltzmann factor carries the co-rotation term, $|R| = \exp[-|\omega - m\Omega_H|/2T_H]$, and dropping $m\Omega_H$ reverses the spin dependence. For the fundamental $l = m = 2$ mode, with the Berti-Cardoso-Will fit [14] $M\omega = 1.5251 - 1.1568(1 - a)^{0.1292}$ this paper already uses, the amplitude reflectivity rises with spin across the range that matters: 9.8×10^{-3} at $a = 0$, where Ω_H vanishes and the two expressions agree, then 0.148 at $a = 0.7$ and 0.444 at $a = 0.9$, where the horizon returns about a fifth of the incident energy, against the 10^{-6} of [13]. It is not monotone all the way to extremality on this fit, which gives 0.662 at $a = 0.95$ and 0.372 at $a = 0.99$, so the rise is stated for the interval calculated and not extrapolated past it. Dropping $m\Omega_H$ returns the 10^{-6} for the same mode.

No exclusion follows. Under the most generous damping model, in which all ringdown damping is horizon absorption and returning $|R|^2$ lengthens the damping time to $\tau/(1 - |R|^2)$, reproducing $\delta\hat{\tau}_{220} = 0.14$ needs $|R| = 0.35$. A reflectivity fixed by T_H supplies that at $a \gtrsim 0.9$ with no extra heat at all. The horizon it asks for is $0.77 T_H$ at $a = 0.9$ and $0.39 T_H$ at $a = 0.95$, cooler than Hawking. Dropping the co-rotation term would wrongly suggest that a T_H -fixed reflectivity cannot reach the reported excess. What can be said is narrower and cuts both ways: a reflectivity magnitude is not a quasinormal-mode calculation, so nothing here either excludes the black mirror or shows it accounts for the excess. The two readings differ somewhere reachable. At low spin the absorbing horizon adopted here returns nothing and a black mirror returns about 10^{-4} , a gap no instrument will reach; at $a \gtrsim 0.9$ they differ by a fraction of order one. High-spin ringdown is therefore where the structural difference between them becomes an observational one.

Appendix A.14 supplies the quasinormal-mode calculation, and it changes the arithmetic. The linear response of the fundamental $\ell = 2$ Schwarzschild mode to a horizon reflectivity is $M d\omega/d\mathcal{R}_h = 0.00735845 - 0.0646771 i$, and since $\tau = -1/\text{Im}\omega$ that is a fractional change of $-0.727 \mathcal{R}_h$ in the damping time against $+0.0197 \mathcal{R}_h$ in the real frequency. The damping answers thirty-seven times harder than the pitch, so a ringdown search for a reflecting horizon should be pushed on the decay time.

Two things follow that a magnitude alone could not give. The damping response is linear in the reflectivity and carries a sign, so a reflecting horizon can shorten a ringdown as readily as lengthen one, and which it does turns on the phase of the reflection rather than its size. And the absorption model used above, $\tau \rightarrow \tau/(1 - |\mathcal{R}|^2)$, is quadratic and signless, so it is wrong at leading order: at $|\mathcal{R}| = 0.1$ the linear term is 7.2 times the quadratic one and at 0.01 it is 73. Reproducing $\delta\hat{\tau}_{220} = 0.14$ therefore needs $|\mathcal{R}_h| = 0.19$ at linear order rather than the 0.35 the quadratic model asks for, a factor of 1.8 in reflectivity, which moves the horizon temperature the excess would imply.

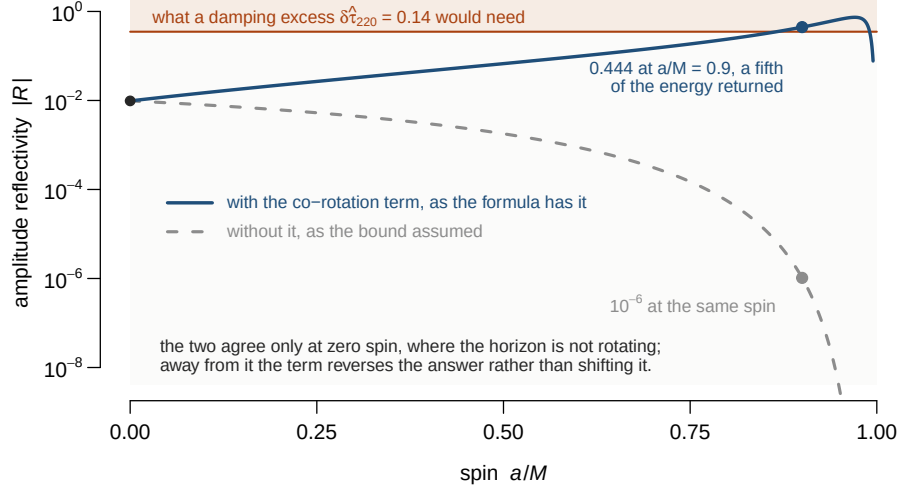


Figure 6: What the co-rotation term does. Solid, the generalised Boltzmann factor as written, $|R| = \exp[-|\omega - m\Omega_H|/2T_H]$; dashed, the same expression with $m\Omega_H$ dropped. Both use the Berti-Cardoso-Will fit [14] for the fundamental $l = m = 2$ mode. The two agree only at zero spin, where Ω_H vanishes, and thereafter run in opposite directions: ω and $m\Omega_H$ both grow with spin and their difference shrinks, so the reflectivity rises. At $a/M = 0.9$ the horizon returns about a fifth of the incident energy where the dropped-term expression gives 10^{-6} . The shaded band is what a damping-time excess $\delta\hat{\tau}_{220} = 0.14$ would need under the most generous model, in which all ringdown damping is horizon absorption; the solid curve enters it above $a/M \simeq 0.87$. No exclusion follows either way, for the reason given below: a reflectivity magnitude is not a quasinormal spectrum.

Two limits on that, and the first is the one that matters. The calculation is Schwarzschild at $\ell = 2$ while the discussion above is at $a \gtrsim 0.9$, where the coefficient will differ and is not computed here; what carries over is the structure, that the leading term is linear and signed, and not the number. And $\mathcal{R}_h$ is the reflectivity in the power-coefficient normalisation A.14 states, which is not the same as the amplitude reflectivity quoted against unit waves in the tortoise coordinate; the two differ by $e^{2\rho}$, and a response quoted without its convention is not a number.

Betzios, Gaddam and Papadoulaki’s near-horizon scattering model is a further concrete alternative [15]. Its small corrections are compatible with present echo bounds, so null searches do not yet distinguish the two implementations. Reflection alone also does not imply a resonance pole; a prompt constant reflection coefficient is a counterexample.

LIGO-Virgo-KAGRA’s GWTC-3 analysis reports a damping-time deviation $\delta\hat{\tau}_{220} = 0.14^{+0.11}_{-0.11}$ when multiplying posteriors and $0.13^{+0.21}_{-0.22}$ under hierarchical combination [16]. These are 90% credible intervals. The first excludes zero; the second includes it. The analysis discusses positive bias and possible prior effects, and does not claim a violation of GR. We do not convert the quoted interval width into a one-standard-deviation error.

Stated the other way round, this is the paper’s most definite near-term commitment, and it is a null. Every gravitational-wave signature the fold could produce, a horizon reflectivity, the $\ell \rightarrow \ell \pm 1$ mixing of A.13, the $\omega \rightarrow \omega^*$ spectral action of A.14, requires a nonzero seam weight, and this implementation sets that weight to zero. So it predicts an ordinary Kerr ringdown, no echo, and a damping-time deviation of exactly zero. Zero lies outside the posterior-multiplication interval and inside the hierarchical one, so the prediction is neither excluded nor comfortable. Holding the central value fixed, shrinking the interval far enough to separate the two cases needs roughly twice the current exposure, which places the decision inside a single observing run. If the excess firms up, the transparent seam adopted here is excluded and the reflecting branch set out above takes its place, with the spin dependence that distinguishes them. If the ringdown settles onto Kerr, the

reflecting proposals are the ones in difficulty.

A robust intrinsic departure from the predicted Kerr response, after the relevant astrophysical and waveform effects are accounted for, would challenge the adopted minimal implementation. It would not identify reflection as the cause or prove a particular quotient geometry. Echo-like propagation effects must likewise be distinguished from a new horizon reflection law.

Setting the seam weight to zero is a choice within this action, and A.11 makes it explicitly; the structure a nonzero weight would carry is worked out separately anyway, because a seam action carrying a scale is not excluded and its reader should be shown the shape a reflecting implementation would have. A.10, A.13 and A.14 do that in three steps. A.10 asks what the kernel ratio of A.6’s worked instance becomes at a general bifurcate Killing horizon, and how much of it survives without de Sitter’s maximal symmetry. A.13 asks what a latitude-dependent horizon response does to the multipoles. For a *scalar* field that is an $\ell \rightarrow \ell \pm 2$ mixing at fixed m , forced by the evenness of the transverse parity and switched on by spin alone; it does not carry over to gravitational perturbations, whose spin-weighted harmonics admit $\ell \rightarrow \ell \pm 1$. A.14 asks what the antilinear lift’s action $\omega \rightarrow \omega^*$ requires of a reflecting horizon’s spectrum: for a constant reflectivity, a complex value breaks the $\omega \rightarrow -\omega^*$ mirror symmetry and a real value preserves it. None of the three supplies an amplitude; the matching law above would have to do that. They constrain the shape a reflecting implementation must have, and the minimal implementation adopted here sets all three to zero.

5. Where the two sheets can touch

Section 4 adopted a transparent seam and predicted an ordinary Kerr ringdown. That is what the fold does at a horizon it is given. This section asks the prior question, where in a spacetime can a point influence its own fold image at all? That one has an exact answer, it needs no seam coefficient, and nothing in it was adopted.

Section 1 separated two readings of “touch”, and this is where the difference bites. We mean the diagonal: x causally connected to Θx , which is what a local cross-sheet term $\int \phi(x)\phi(\Theta x)$ would need. A.15 has the derivation, the charged and higher dimensional cases, and the limits; what follows is the result and what it means.

5.1 The answer, and what charge does to it

Outside a horizon, nothing. A point and its image are spacelike separated for every such pair, which is Section 1’s theorem and the reason its silence is a theorem. Inside one the radial part of the separation turns timelike, so what is left to settle is the transverse sphere. There the fold’s map is the antipode, which puts a point half a turn of angle from its own image, and causality caps how far round a falling curve can get. The cap works out to

$$\Delta\phi_{\max}(r) = 2\pi - 4 \arcsin \sqrt{\frac{r}{2M}},$$

which reaches π at $r = M$. So the two sheets can reach each other in the inner half of a Schwarzschild interior and nowhere else in that spacetime, and the bound is attained, by a null geodesic sitting at $X \equiv 0$ in Kruskal coordinates and spiralling, which is what makes the statement an equivalence.

That derivation and the dimension count of 5.2 are set out for a general reader in a separate Letter [38], and A.15 has the full working, the charged and higher-dimensional cases and the limits. What the rest of this section does is go where the Letter does not, to compact extra dimensions, and to the other horizon.

Charge says which half of that sentence is robust. A charged hole has two horizons, and the same calculation gives $r \leq (r_+ + r_-)/2$, at every charge, always exactly half way across the interior band. **The result is about the inner half of the interior; half the horizon radius is the uncharged special case.**

The contact region, and the angle that decides it

Colour is the angle itself. The spectrum starts exactly where contact does.

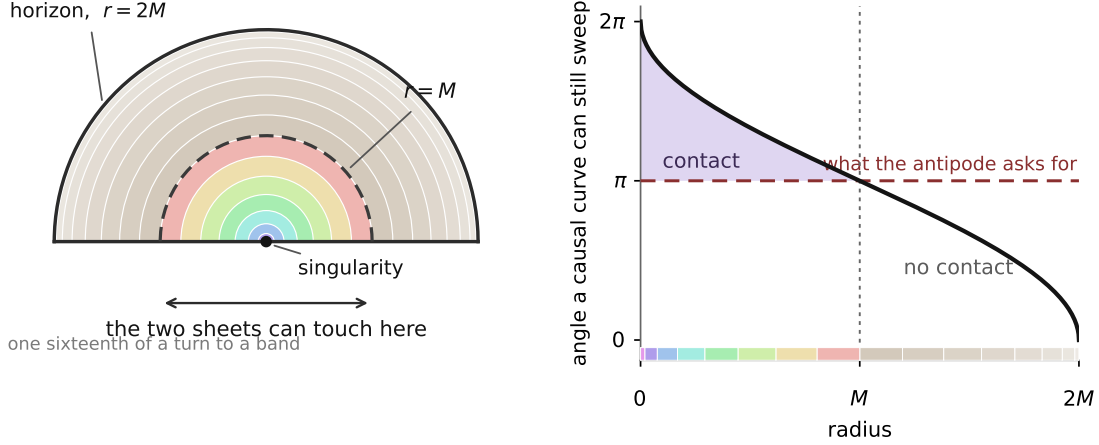


Figure 7: The contact region drawn through the interior. Each band is one sixteenth of the angle $\Delta\phi_{\max}(r) = 2\pi - 4\arcsin \sqrt{r/2M}$ a causal curve starting at radius r can still sweep, so a band is narrow where that angle accumulates fastest. They are narrowest at the horizon and at the singularity and widest at $r = M$ itself. The eighth contour is $r = M$, where the running total first reaches the half turn the antipode asks for. Bands inside it carry the spectrum and bands outside it do not, so the colour starts exactly where contact does. Right, the same quantity drawn as a curve.

The half that is not robust is the exclusion. Reaching the inner half is proved at every charge, by an explicit $E = 0$ null geodesic that attains π there, but ruling out everything outside it needs the angle-maximising curve to sit at $X \equiv 0$, and A.15 shows the condition for that, $C(r) = r(2\kappa - f') + 2f < 0$, holds throughout a Schwarzschild interior and fails at every nonzero charge in a shell outside the Cauchy horizon, with no threshold below which it is safe. So at charge the inner half is a floor on the contact region rather than the whole of it, and whether the region reaches into (M, r_+) is open. The gap runs the forgiving way, since a maximiser that leaves $X \equiv 0$ delivers more angle and can only enlarge the region, so nothing claimed here shrinks if it is closed against us.

A.15 traces both to one algebraic fact about $r^2|f|$ being a quadratic whose roots bound the region, and shows what happens when it is not. A positive cosmological constant pushes the boundary outward, reaching the whole interior at the Nariai point. Hold on to that endpoint. Section 6 arrives at it from an unrelated direction.

5.2 Why four dimensions

Run the same calculation in D dimensions and it permits contact in four alone, which is the one place the fold says anything about how many there are. A Schwarzschild interior supplies exactly $\pi/(D-3)$ of turning to any causal curve crossing it, horizon to singularity, whatever the mass, while the antipode asks for π in every dimension, because that is fixed by the map being a free involution of a sphere. Two legs give $2\pi/(D-3)$ against it, so four dimensions clears the distance with a factor of two to spare, five falls short by exactly nothing, and six and above fall short outright. The comparison is drawn in the right panel of this paper's own opening figure, and set out for a general reader in the Letter [38].

That is a statement about large dimensions, since it runs on Tangherlini holes. If the extra dimensions are instead compactified at a scale R , the sweep depends on the hole. For $r_h \gg R$ the hole is a uniform black string, and that product is not an approximation: a four-dimensional Ricci-flat metric times a flat direction is Ricci-flat, so the string is an exact vacuum solution, checked numerically to the finite-difference

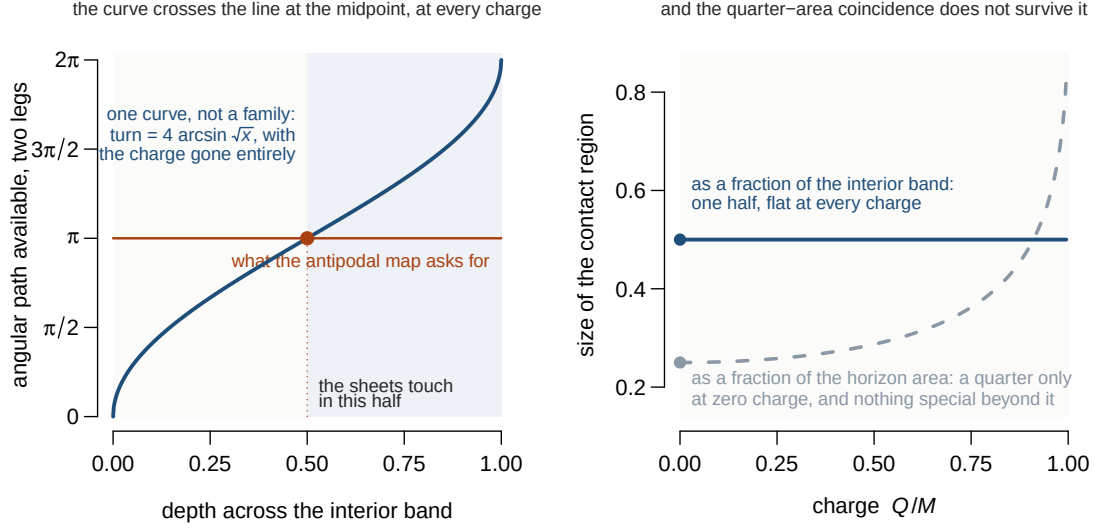


Figure 8: Left, the contact condition once the interior is measured in its own units. One leg from radius r out to the horizon integrates to $2\arcsin \sqrt{(r_+ - r)/(r_+ - r_-)}$, so with x for that depth fraction the two-leg sweep is $4\arcsin \sqrt{x}$ and the charge has gone entirely: one curve, not a family. It crosses the π required at $x = 1/2$ because $\sqrt{x} \geq \sin(\pi/4)$ is the condition, which is why the answer is the inner half of the band at every charge, exactly. Right, the same result stated two ways. As a fraction of the band it is one half and does not move; as a fraction of the horizon area it is a quarter at zero charge and better than three quarters near extremality. Anything thermodynamic would have to survive charge, and the quarter does not, so it is a coincidence of $r_- = 0$.

floor against a warped control four orders larger. The compact directions then enter the causal bound only as $-dK^2$ under the square root, so moving in them subtracts from the angular progress and never adds to it, the maximiser sits still in K and the sweep is the four-dimensional 2π exactly.

For $r_h \ll R$ the hole does not see the compactification, it is Tangherlini in $D = 4 + n$, and its sweep is $2\pi/(n+1)$ against the same π as ever, which one compact dimension already reduces to equality and two put out of reach. So small extra dimensions are not excluded by any of this. What follows is that the fold can only reach inside holes large compared with them, and that its contact switches off below that scale. Which solution describes the hole settles the crossover, and that is the Gregory-Laflamme question [34]. A uniform string is linearly unstable once the compact circumference is long compared with the horizon: for the neutral five-dimensional string the threshold wavenumber is $k_{\text{GL}} r_h \simeq 0.876$ [35], so a periodic unstable mode fits once the circumference passes $2\pi/0.876 \simeq 7.17$ horizon radii, which is $r_h/R \simeq 0.876$.

That threshold locates a linear instability, and where the phase boundary sits is a separate question. Gregory and Laflamme offer the fragmentation picture as a conjecture, and Lehner and Pretorius, evolving the instability numerically [36], find a cascade of necks and bulges rather than a clean handover to a localised hole at one radius. So this section claims only what the numerics carry, and that is enough. A crossover exists, it sits at r_h/R of order one, and above it the fold has its four-dimensional sweep while below it the hole is genuinely higher-dimensional and the sweep falls short. Where the handover finishes is a question for that literature and nothing here turns on the answer. The threshold quoted is theirs and we have not re-solved the eigenvalue problem. Whether the fold acts on the compact factor is separate and would add to the distance required rather than to the sweep.

The same sweep answers a different question at the other horizon. Beyond a cosmological horizon $f = 1 - r^2/L^2$ is negative and the turning available is $\int_L^\infty dr/(r\sqrt{|f|}) = \pi/2$ per leg, so two legs supply exactly the π the antipode asks for and supply it only as $r \rightarrow \infty$. De Sitter sits on the threshold, and it sits there in every dimension, since f does not know D . At finite radius the sweep falls short by L/r , so a point in a Λ -dominated universe never reaches its own image and the image stress vanishes in exact de Sitter. Adding

a mass changes that answer. The horizon comes in to $r_c < L$, the integration begins where $|f|$ is smallest, and the sweep per leg becomes $\pi/2 + 2M/L$, which opens contact at $r_* = L^2/2M$. The expansion carries a comoving worldline out from the horizon to that radius in $L \ln(L/2M)$ of cosmic time. For holes between 6.6×10^{10} and ten solar masses that is 27.47 to 50.08 Hubble times in the future, so nothing in the observable universe or its past is in contact. What remains there is the de Sitter scale $1/16\pi^2 L^2$, 2×10^{-71} of the dark energy.

That places the argument in a small and old genre, and A.15 says where. Most entries assume something about matter or observers. The closest relative assumes neither and is a century old: Weyl's observation that the Maxwell action is the simplest integral invariant that exists at all, and that no such invariant exists in any other dimension. His selection rests on the degree of a form and this one on a causal sweep, and we claim no more than that they are independent.

5.3 What the region is

The same result read another way needs nothing new. The fold pairs every point with a distinct one, so it is a free involution, the quotient is a manifold, and on it x and Θx are the same point. Then "a point can influence its own image" says there is a closed causal curve through that point. What the inner half of the interior locates is the causality-violating core of the folded hole: closed timelike curves strictly inside it, the closed null geodesic exactly on its boundary, nothing outside, and the whole of it sealed behind a horizon and then behind half of what is left.

The sign of the image stress does not rest on a single Hadamard slot. The calibration was made by moving the coupling off conformal at zero mass, and the same Hadamard coefficient carries m^2 , so the test available is a mass at conformal coupling where the coupling slot is empty. Summing the massive conformally coupled modes on the same geometry, with the fold first checked against the massless closed forms to 6×10^{-4} , the null-null component is negative at every mass and close to proportional to m^2 . So A.15's self-censoring reading holds for a massive field whatever its coupling. The two slots do not diverge at the same rate as the caustic is approached, so the proportionality to $-V_0$, the Hadamard coefficient A.19 writes as $\Delta^{1/2}[m^2 + (\xi - \frac{1}{6})R]/2$, calibrated on one is not a constant shared by both and what transfers is the sign. Those rates are accounted for in A.15. Read in the distance to the caustic rather than in $1 + \cos \tau$, which is its square, the coupling slot goes as $\delta^{-4.010}$, exactly what the caustic-order rule and two derivatives predict, and the mass slot as $\delta^{-2.037}$. That is one power softer because a mass carries one power of the world function, and one more because the conformal coupling is, on this geometry, the value at which the leading mass parts cancel.

Two things travel with that and A.15 states both. First, the fold reverses time orientation, so the quotient is not time-orientable and the loop closes with the transported arrow flipped, which makes the phrase "closed timelike curve" a convention that has to be argued for. That reversal is a theorem of the fold. The modular flow of a wedge is the boost, whose generator gives $dX_0/ds = X_1$. Sampling four thousand points of the right static patch and four thousand of its antipode returns a positive value at every point of the first and a negative one at every point of the second, while a rotation generator returns no sign and points outside both patches return both. Its two limits are checked in a finite-dimensional model where each modular operator is built from $S(a\Psi) = a^\dagger \Psi$ rather than written down: the entropy is constant along the flow to twelve figures, and J commutes with the flow, so what runs the two patches oppositely is $\Delta_{A'} = \Delta_A^{-1}$ and not anything J does. Second, surfaces of constant r are spacelike inside, so this chronology horizon is spacelike rather than the null kind Hawking's theorems produce, and those assume the time-orientability this quotient lacks.

Hawking's chronology protection conjecture [37] holds that the laws of physics prevent closed timelike curves from forming, and it has stood without a general proof. The theorems that support it assume a null chronology horizon on an orientable spacetime. This case is neither. The horizon is spacelike, because surfaces of constant r are spacelike inside, and the quotient is not time-orientable, because the fold reverses time orientation. So those theorems reach no verdict here in either direction, and whatever the interior does is a fresh case.

What the fold does here is support the conjecture from a geometry it was not framed for. The null-null contraction is negative at the caustic and that sign carries to a hole, so the image stress defocuses the

congruence along which the two sheets would touch, and the region works against the closed curves it contains. That is evidence for the conjecture and not a proof of it, and the distance between those is the radial contraction, which is still held by conservation rather than by a sum done in the interior. Should that step go the other way the fold would instead predict a causality-violating core sealed behind a horizon and then behind half of what remains, and would say so.

A.16 identifies two loci at which the fold joins the sheets structurally, the bifurcation surface and the singularity, since the map exchanges the future and past branches of the curvature hyperbola. That appendix works out what a matter transit there would require. That is a different sense of joining from the one this section uses, and the two should not be run together.

There is also a reason to think the fold polices itself. The world function between a point and its image vanishes linearly at the boundary, calculated from the connecting geodesics in A.15, and a simple zero of it makes the image term of a free field's two-point function diverge there as an inverse power. If that survives a full calculation the stress tensor blows up on the contact sphere and the fold censors its own closed curves the way a chronology horizon does. It is tempting to read a power off that: σ linear in the distance s from the contact sphere, the parametrix image term going as $\Delta^{1/2}/\sigma$, a stress tensor quadratic in derivatives, therefore s^{-3} . That counting needs the Van Vleck factor $\Delta^{1/2}$ to be finite on the contact sphere, and it is not. The fold's transverse factor is the antipodal map on a round S^2 , and the antipode is a conjugate point of every geodesic leaving a point, because the whole one-parameter family of great circles refocuses there. On Nariai this is explicit, $\Delta = \varphi^2/(\sin \varphi \sinh \varphi)$ with its first caustic at $\varphi = \pi$; on the Schwarzschild contact orbit the first angular zero is at contact and not before it. The image point does not sit near a caustic, it sits on one, so the parametrix is being expanded about the one place the expansion does not hold.

So the power cannot be read off the parametrix, and the question stays open. A caustic is where the approximation fails, and the exact two-point function has to take its place, which asks nothing of a mode sum because no Van Vleck factor appears in one. Two things are needed before the sum can be done. The state is the first. It is not an image sum over a $\mathbb{Z}_2$ quotient, since the fold is antilinear and sends (q, p) to $(Dq, -Dp)$, so each parity sector of the quotient carries an identically zero symplectic form; what two exteriors related by an antiunitary map carry is a two-sheet state, and A.8's propagation fixes which one. At a bifurcate Killing horizon it is the thermofield double at $\tanh r = e^{-\beta\omega/2}$, carrying the transverse parity that separates $J \circ P_\perp$ from J , and that parity is the insertion of $(-1)^\ell$ on the cross-sheet term. The insertion and the antipodal map are one operation. The second thing is a geometry in which the sum closes, and A.18 supplies one.

A.18 uses the smallest geometry carrying the feature at issue, which is not Schwarzschild and is not offered as it. Null geodesics leaving a point spread over every great circle and refocus together after exactly π of transverse angle, the same one-parameter connecting family the contact sphere has. A generic null pair in it returns -1.0095 against the known -1 , which is the control that could have failed. The image pair returns $-3/2$, sharpening on that value as the window closes on the caustic, and removing the antipodal parity by hand removes the divergence outright. So the caustic steepens the divergence by half a power: the image term goes as $s^{-3/2}$ and the stress tensor as $s^{-7/2}$, against the parametrix count's s^{-3} .

So the self-censoring reading holds, with a steeper exponent. The sign, which is what decides whether the divergence closes the region or merely marks it, takes a second model, and A.19 supplies one. Point-split against the image kernel, the fold's own pullbacks assemble the energy density into the wave operator, which annihilates the two-point function, so in A.18's geometry the density vanishes identically. That zero belongs to the constant transverse radius, let the radius vary and there is nothing for the cancellation to be. Flat space is that case and is exact, its fold being the time reflection composed with the spatial point reflection, and the image energy density there is $(8r^2 + 6\xi^2)/4\pi^2 D^3$ with $D = 4r^2 - \xi^2$, positive everywhere and diverging positively at contact. The exponent is the generic -3 rather than A.18's $-7/2$, because a straight line is the only geodesic joining a point to its image and there is no caustic.

So for the energy density one model has the power and the other has the sign, and a black hole has both and is neither. Each result is right in a geometry this paper is not about, and that is the limit to carry away from both, for the density.

The contraction the defocusing argument runs on is not the density and is in better shape. The null-null

component T_{kk} comes out negative at the caustic by two routes that share no assumption, moving the mass at conformal coupling and moving the coupling at zero mass, and the transfer of that sign to a hole is not a signature argument. A.18's mode sum and A.19's amplitude rule compute the same image two-point function, one by a Bessel tower with an $i\epsilon$ and no Van Vleck factor anywhere in it, the other by geometric optics with a Van Vleck factor and a proper-time integral, and they agree to the last digit where both apply. A relative phase between them would have to be a multiple of π , since a Wightman function at spacelike separation is real, and they agree, so it is zero and the signs of the stress components carry. What still comes from A.15's conservation chain rather than from a sum done in the interior is the radial contraction, and that is the one step this section does not close. What would settle it on Schwarzschild is radial: A.19 carries the fold through for any static spherically symmetric metric and finds the transverse sphere supplying the angular part of the potential exactly, at every one of them, so no ℓ survives in the density and what is left is a two-dimensional object in r alone.

5.4 How far a point is from its own image

Section 5 has used the separation between a point and its own image only at its zero, and the whole field is calculated here, since that is where any local effect of the fold would have to live. Write $s(x)$ for it: a scalar determined by the geometry and the fold with nothing chosen, whose vanishing at $r = M$ is the contact result restated. Away from there it has a profile. On the $t = 0$ slice the geometry is the Einstein-Rosen bridge, the fold carries the point at (r, θ) to the mirror point at $(r, \pi - \theta)$ at the same Killing time, and the connecting curve must both cross the bridge and turn through π on the sphere. The pairing is of points and not of observers, so the field exists along any worldline through them, null ones included. That is a geodesic on a surface of revolution, so Clairaut's relation applies and the answer is a one-dimensional integral. In units $2M = 1$ the separation is π at the horizon, 6.054 at $r = 2$, 8.519 at $r = 3$ and 23.80 at $r = 10$, monotone outward, with $|\nabla s| \rightarrow 2$ far away because leaving the hole moves you away from your image twice over.

Two features follow, and the first carries a qualifier. On this slice the floor is πr_h exactly, half the horizon circumference, because the angular half-turn has to be covered and the sphere is smallest at the throat. That is a bound on *spacelike* separation across the bridge and not on separation in general: 5.1's whole point is that inside $r = M$ a point and its image are causally connected, so the separation there is null or timelike and the exterior floor says nothing about it. The two statements are about different regions and different signs. And the whole structure is switched on by mass. In empty de Sitter the same invariant is $Z(x, \mathcal{A}x) = -1$ at every point, so the separation is the same everywhere and its gradient vanishes identically; there is no local fold structure in an empty universe at all.

How the separation reaches that floor takes one more line, and ties the field to something familiar. For a static observer the redshift factor is $\sqrt{f}$ with $f = 1 - r_h/r$, so f is the square of the local clock rate, and near the horizon

$$s - \pi r_h \longrightarrow 2\sqrt{2} \coth\left(\frac{\pi}{2\sqrt{2}}\right) r_h f = 3.51656 r_h f.$$

The clock is a reading of f and not a restriction on it: f is a property of the point, so the law holds at every point whatever passes through it, and a null geodesic crossing r meets the same separation field a hovering mass does.

The coefficient is a pure number with no mass in it, and it is a limit. The residual against it falls linearly in $r/r_h - 1$ over four decades, while a near-miss such as $9\pi/8$ leaves a residual that does not go to zero at all. So the fold's reach above its floor and the local clock rate are not two facts about the geometry. They are one quantity, and the horizon is where both arrive, the clock stops exactly where the separation to one's image can fall no further.

That is a geometric fact and not yet a force. For the field to act it has to enter an effective action, and the only place it does is the image term of a two-point function, whose coefficient A.8 drives to zero. Exactly what that zero covers needs seeing, because it is both broader and narrower than it first reads. Broader:

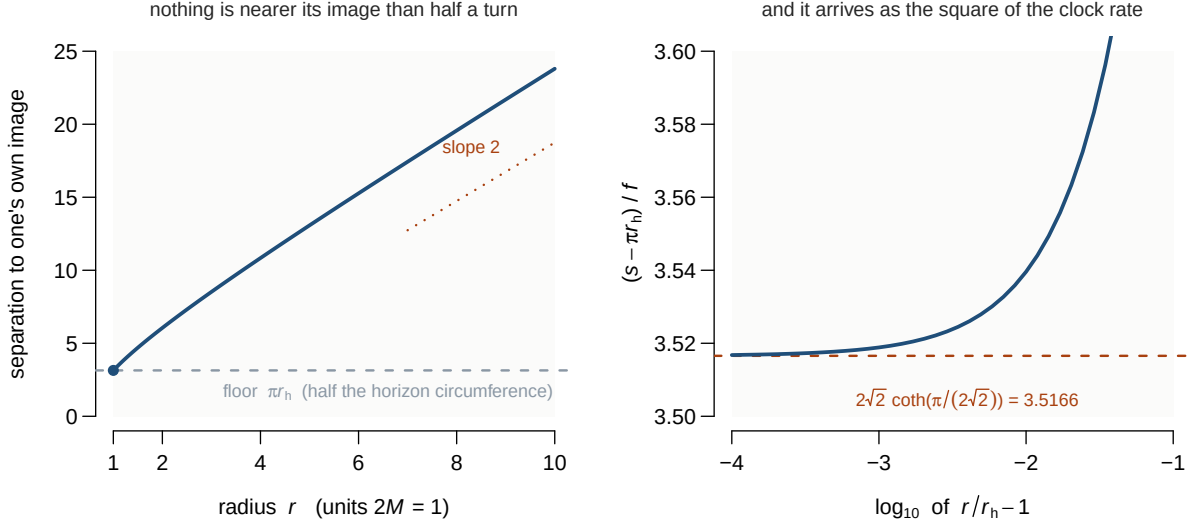


Figure 9: How far a point is from its own fold image. Left: the separation across the exterior on the $t = 0$ slice, against the floor πr_h that the angular half-turn requires and which nothing on that slice goes below. Far from the hole the curve runs at slope two, because leaving the hole moves a point away from its image twice over. Right: the approach to the floor divided by $f = 1 - r_h/r$, the square of the local clock rate, flattening onto $2\sqrt{2} \coth(\pi/(2\sqrt{2})) = 3.5166$. The two integrals are redone in R for the figure and reproduce the values quoted here to four decimals, which makes the figure a check on its own numbers as well as a picture of them.

the exclusion of a constant, unsmoothed image kernel is not a de Sitter statement at all. Such a kernel is singular wherever a point is null-separated from another point's *image*. The Hadamard condition is not a prohibition on singularities away from coincidence, since it carries the ordinary lightcone singularity at $\sigma(x, y) = 0$; what it does not allow is a singularity at a pair that is not null-related in the first place, and the image-null locus is full of such pairs, including spacelike ones. That argument is local and survives away from maximal symmetry.

Narrower, nothing in that argument reaches a smooth image correlation, and A.8 records why. Squeeze a single smooth mode, $v = \cosh r u + \sinh r \bar{u}$, and the two-point function changes by $2 \sinh^2 r \operatorname{Re}[u(x)u(y)] + \sinh 2r \operatorname{Re}[u(x)u(y)]$. If the mode is fold-real, so that $u(\Theta y) = \bar{u}(y)$, the second term is an image correlation and nothing was inserted to make it one. The Bogoliubov map is canonical, so positivity and the commutator survive; the change is a finite sum of products of smooth mode functions, so the short-distance singularity survives and the state stays Hadamard; and the local stress moves. So positivity and Hadamard do not forbid the fold a local effect, and the silence of Section 1 does not extend from commutators to the stress tensor on those grounds.

A calculation is what is missing here. A.8 fixes the squeeze at a bifurcate Killing horizon to $\tanh r = e^{-\beta\omega/2}$, so the amplitude is not free; what stays open is the choice of mode and the phase the bang leaves per mode, and the stress shift has been evaluated for neither. The argument for silence at this level has been removed, and no argument for an effect has replaced it.

6. Two classes of black hole, and whether the second one exists

Section 5 puts the contact region inside a hole possessing a past, and collapse does not make such a hole. A star that falls in produces one exterior and one horizon, and the region that would hold the mirror sheet's matter never exists. So the fold does not modify every black hole. It sorts them into two classes: one-sided holes, which is everything we have ever observed, and two-sided holes, which have both sides in place.

All of it rests on A.15's hypothesis (H), which that appendix reduces to a single statement, the fold maps the hole to itself. That is a commitment, and a programme that replaces the interior with structure at the horizon denies exactly it. A hole with no interior has no past region for a mirror sheet to occupy and no bifurcate Killing horizon for an involution to fix, so the second class would be empty and Sections 5 and 6 would describe nothing. Sections 2 and 3 are untouched either way, since they never needed a black hole. This paper does not settle which picture is right and does not claim to. What it does is put the two on the same fork: the minimal implementation predicts an ordinary Kerr ringdown with no echo and a damping-time deviation of exactly zero, which is the observable where a horizon carrying structure and a horizon with an interior behind it part company. Everything else follows, the second exterior included, so the sorting is not a question about formation histories at all. A.15 then counts how many holes satisfy it, and 6.2 reports the count.

Nothing in the rest of this section uses the depth of the contact region, which Section 5 has already settled. It matters here only for what it rules out: no local cross-sheet coupling can act anywhere an outside observer could ever look.

The order here is consequences first and the count second, because the count is a short answer and the consequences are what it is an answer about.

6.1 What a two-sided hole does

Everything here is worked at arbitrary mass. That is how the consequences come out, and the first of them says nothing except that the mass does not matter. Section 6.2 then finds that the one family in which the count can be done holds a single two-sided hole, and it is the maximal one, where an accretion picture does not apply. Whether sub-maximal ones exist outside that family is open. So read what follows as the structure's consequences. No such hole is known to be there.

That leaves a question to answer, since the overmassive holes are not hypothetical. They exist, so a cosmology that is right has to be consistent with them. This one is, and by having nothing to say, they are one-sided; one-sided holes in a folded universe obey ordinary general relativity, and the fold modifies none of it. Consistency is not explanation, and the two routes by which the fold might have explained them are both shut. The formation channel fails in 6.2 below, for reasons that are about structure rather than about counting. And the other route, an enhanced primordial tail, is closed in the cosmology paper. The linear fold map adds no connected three-point function to the Gaussian free state, so the fold supplies no bispectrum to steepen a tail with. What the fold does offer here is a constraint rather than a mechanism, and it runs from the data to the theory.

Such a hole is fed from both sheets. Since the fold reverses charge, our side supplies matter and the mirror supplies antimatter, and the two streams meet inside. That meeting is a weaker condition than the contact of A.15 and should not be confused with it. Both exteriors of a two-sided hole share one interior, so infalling material from either arrives in the same region and can meet anywhere in it; A.15's $r \leq M$ is the stronger statement that a point is causally connected to its own fold image, which is what a cross-sheet interaction would need. The annihilation below uses only the first. Whether they finish annihilating before the singularity has an answer that does not depend on the mass, which is unusual for a black-hole calculation. Eddington-limited accretion gives $\dot{M} \propto M$, the infall time from horizon to singularity goes as M , and the interior volume as M^3 , so the interior density falls as $1/M$, the reaction rate falls as $1/M$, and the rate multiplied by the time is flat. The computed slope of optical depth against mass is 9.6×10^{-17} across ten decades, and breaking the volume scaling on purpose moves it to $+0.50$, so the flatness belongs to the geometry and survives the arithmetic being broken.

The optical depth is flat at roughly 46.7, and that level is not robust: it rides on an annihilation cross-section and an accretion efficiency, and moving either by a decade runs it from single digits to the high hundreds. What survives the whole range is the conclusion rather than the figure. The optical depth exceeds unity everywhere in the box, so the streams annihilate before the singularity at any mass and any plausible cross-section, and the interior of such a hole is radiation rather than baryons. So the independence is what to quote.

A hole fed from both sheets would also grow faster, and the change would land in the exponent rather than the rate. Eddington-limited growth is exponential with e-folding time t_S , so a supply multiplier multiplies the e-fold count over a fixed window while leaving the rate alone. Between $z = 20$ and $z = 7$, a window of 0.582 Gyr in the Planck cosmology, a hole at radiative efficiency 0.1 and Salpeter time 50.1 Myr gets 11.63 e-folds one-sided; double the supply and it gets 23.3, ending 1.1×10^5 times heavier from the same seed. A factor that size, sitting next to the overmassive holes the James Webb telescope has found at one to two orders above the local relation, warrants checking rather than adopting, and it would be natural to carry as a free parameter with observation left to bound it. The fold does not deliver it. It forbids it twice, and both arguments are short enough to give here.

The first is the time orientation. Take a null shell falling in from our side at Kruskal $V = V_0 > 0$. Ingoing Vaidya has $8\pi T_{vv} = 2\dot{m}/r^2$ with every other component of the Einstein tensor vanishing and Misner-Sharp mass $m(v)$ exactly, so a stream of positive energy is one whose mass function rises toward the future, and the shell leaves $M + \delta$ at $V > V_0$ with M behind it. The fold sends that shell to $V = -V_0$ and reverses time orientation with it, so the mirror sheet's future is our decreasing V and their shell leaves $M + \delta$ at $V < -V_0$. Our exterior is the whole of $U < 0, V > 0$, and never meets that region. Both sheets end at $M + \delta$, each from its own infall. Turn the second shell round, so our exterior gains what they drop, and that needs $\dot{m} < 0$ in the mirror sheet's own advanced time and so a negative energy density on their clock; one sign has to serve both sheets, so a fold-invariant state cannot have it both ways. The step from "their infall is the image of ours" to a doubled exponent counts one shell twice.

That argument assumes the sheets are disjoint images glued along a bifurcation surface, which is the one thing Section 5 is at length denying, so it is not the end of the matter. Inside the contact radius a point and its fold image are joined by a causal curve, and matter crossing there reaches our hole without passing through our exterior. Let p be the fraction that crosses. Here the second argument bites, and it is only counting. Fold invariance makes the two sheets' infall equal, and a crossing is a swap, so our hole receives $(1 - p)$ of our own stream and p of theirs, which is $(1 - p)F + pF = F$ for every p . The version that gains, in which we keep all of our own inflow and add p of theirs, turns (F, F) into $((1 + p)F, (1 + p)F)$ and creates $2pF$ from nothing. The multiplier is one whatever the contact geometry turns out to be, and the transmitted-fraction reading fails for the same reason the symmetric reading did, it counts one stream twice.

What, if anything, the mirror sheet does double is the next question, because a whole extra exterior ought to show up somewhere. Section 3 tests three implementations of the copy reading and all three fail, which constrains those three choices and leaves other quotients untouched. Read as a place carrying its own content, the total content of the two-sheet geometry genuinely is twice what either sheet accounts for. It shows up nowhere, and the reason is the same one every time. Our future horizon is $U = 0$ and theirs is $V = 0$: two different null surfaces, each of area $16\pi M^2$. Their infall lands in their future interior, which is our past interior, different events. Their radiation contributes at events we do not occupy, which is why N_{eff} does not move. Nothing is shared except the bifurcation surface, and a two-surface carries no energy. The doubling is real and it is invisible, and those two facts are the same fact.

So accretion cannot tell a two-sided hole from a one-sided one, and it cannot do so for reasons that survive every quantity this paper has left uncomputed. Section 4 reached the same verdict through the other channel, where a transparent seam rings down as Kerr whatever the hole's class, which is the null prediction stated there against GWTC-3. Both of the observational routes we can construct return nothing. What the section keeps is the classification itself, which none of this touches: the fold sorts holes into those with a past region and those without, and 6.2's one derived member of the second class is the maximal hole. The little red dots are not evidence for it, and this is why.

A doubled exponent over this window is a factor of 10^5 , and the two arguments that remove it are a sign and a sum. Neither needed a measurement. Section 6.2 goes further. Every member of the class this paper can derive is maximal, and far too large for any accretion argument to touch, so even had the count come out the other way there would have been no object to apply it to.

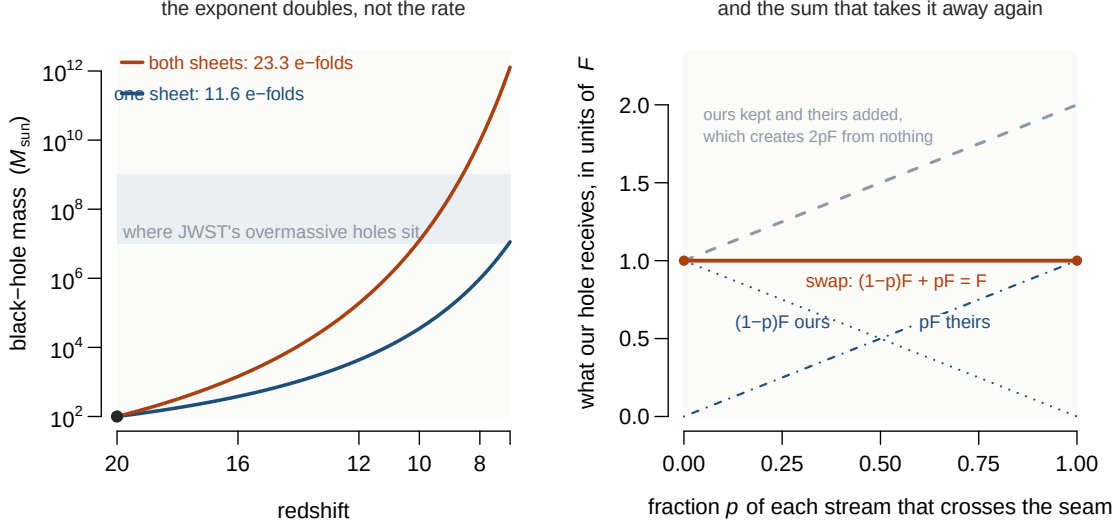


Figure 10: What a second supply would have given, and the count that removes it. Left: growth from a $100 M_{\odot}$ seed between $z = 20$ and $z = 7$, Eddington-limited at radiative efficiency 0.1. A supply multiplier does not add to the mass, it multiplies the exponent, so 11.6 e-folds become 23.3 and the final hole is 10^5 times heavier. That is why the section checks it. Right: the two-channel count that forbids it. Fold invariance makes the two sheets’ infall equal, and a crossing inside the contact radius is a swap, so our hole receives $(1 - p)F$ of our own stream and pF of theirs for a total of F at every p . The alternative, keeping our own and adding theirs, is the dashed line, and it creates $2pF$ from nothing.

6.2 Counting the second class

None of the above has an object to describe unless the second class is populated. This subsection counts it. The count is small, and the answer has a shape an abundance argument does not: it is a uniqueness statement, and it is reached from the fold’s own condition rather than from a rate.

Formation through a throat rather than by collapse has a literature and a name. Kopp, Hofmann and Weller [17] showed that a large enough perturbation closes its overdense region off into what that field calls a separate universe, and Uehara, Escrivà, Harada, Saito and Yoo [18,19] have since simulated the collapse numerically and classified the outcome by whether the trapping horizon bifurcates, which is their type II-B. Type II-B reads like the channel that populates the second class, and it is not that channel, for two independent reasons, so nothing here counts it as a mechanism.

The first reason is structural. A type II-B hole evolves from regular data on a spacelike slice, and such a spacetime has no white-hole region. What the throat opens onto is reached to the future, an inflating baby universe, and matter falls from our side into it. That is not a second exterior feeding a shared interior, which is what 6.1 uses, and it is not the past interior P that Section 5’s contact needs. The terminology differs too. A bifurcating trapping horizon is an apparent horizon splitting into branches in a dynamical spacetime, while A.10’s reduction and A.15’s boost gauge both need a bifurcate *Killing* horizon, with a Killing field vanishing on a two-surface. Sharing the word is the whole of the resemblance.

The type II-B arithmetic is kept because it answers a different question correctly. Normalised to the observed rate of ordinary primordial black hole formation, and separating the two thresholds by the twenty per cent that arXiv:2401.06329’s profiles imply, a Gaussian curvature field leaves four type II-B holes in every hundred thousand, rising to 7×10^{-4} at a stretched-exponential index $p = 1.5$, one per cent at an exponential tail $p = 1$, and eleven per cent at $p = 0.5$. The threshold for type II-B sits above the threshold for ordinary formation, so softening the tail raises both rates and the ratio gains far less than a rare-tail intuition suggests: ordinary holes per type II-B one run from about five at $p = 2$ to five hundred at $p = 0.5$. What none of those numbers measure is the population of this section.

What does populate it is a question about the fold and not about a tail. A.15 reduces the requirement to one statement, that the fold maps the hole to itself, and shows that statement carries the rest. The map is then forced, and the second exterior, the past interior and the bifurcation surface arrive with it. No formation channel has to supply a white-hole region, so no formation channel settles this.

A.15 then counts the class, and the count is one. Schwarzschild-de Sitter is the one exactly solvable background carrying a black-hole horizon and a cosmological one together, so it is where the question can be settled. Below the Nariai limit it admits no single involution serving both horizons: the two wedge reflections compose to a timelike translation, so identifying them closes timelike curves, and the surface gravities differ as well. At Nariai the geometry is the product $dS_2 \times S^2$, both bifurcation surfaces sit in one dS_2 at $X_0 = X_1 = 0$, and the single linear map $(X_0, X_1, X_2) \mapsto (-X_0, -X_1, X_2)$ fixes both while reversing time orientation and carrying our static patch to the other one.

So the second class has exactly one member in the uncharged family, and it is the largest hole that family holds: $M = 1/(3\sqrt{\Lambda})$, about 2.2×10^{22} solar masses at the observed Λ , with a horizon ten billion light years in radius. Charge and spin both widen the point to a branch without changing its character. In Reissner-Nordström-de Sitter the fold-invariant set is the whole Nariai branch, from the uncharged case to the ultracold point at $Q = 1/(2\sqrt{\Lambda})$ where all three horizons meet; the extremal branch is excluded because a zero-surface-gravity horizon has no bifurcation surface for a wedge reflection to act on. Rotation does the same. The transverse map is an exact isometry of Kerr-de Sitter at every spin and every Λ , for the same reason it works in Kerr, and the merger of the outer pair runs from $a = 0$ with $\Delta_r'' < 0$ along all of it, so the near-horizon radial factor is dS_2 and never AdS_2 . That branch ends where Δ_r acquires a triple root, where the quartic factorises as $-(\Lambda/3)(r - r_0)^3(r + 3r_0)$ and $\Lambda r_0^2 = 2\sqrt{3} - 3$, giving

$$\frac{a}{M} = \frac{3}{4\sqrt{2\sqrt{3} - 3}} = 1.1009173688,$$

the same number in every de Sitter background and past the Kerr extremal bound, which a positive Λ permits.

The branch between those two ends is a single curve and can be written down. In units where $r = M\rho$ and $a = M\alpha$, imposing $\Delta_r = \Delta_r' = 0$ and eliminating Λ leaves one quartic in ρ alone,

$$\rho^4 - 3\rho^3 + 2\alpha^2\rho^2 - \alpha^2\rho + \alpha^4 = 0, \quad 9\Lambda M^2 = \frac{27(\rho - \alpha^2)}{\rho^4},$$

whose largest real root traces the class. At $\alpha = 0$ it is $\rho^3(\rho - 3)$, so $\rho = 3$ and $9\Lambda M^2 = 1$, which is Nariai; the root disappears at $\alpha = 1.1009173688$, and the numerical branch meets the closed form there to 7×10^{-16} . Solving it this way matters, because hunting roots of Δ_r directly loses two thirds of its digits exactly where three of them coincide, which is the end of the branch and the only place the endpoint can be read off.

So the class is better described as the boundary of the parameter space than as a point: maximal holes, black-hole horizon coincident with cosmological, carrying charge and spin up to that value. Every member is still maximal. That is not a population of objects inside a universe. It is a statement about a universe, and as a population count it is thin.

Followed across the family rather than inspected at its endpoint, that stops looking like an isolated accident. The fold is present throughout, because the cosmological horizon always has a wedge reflection, and that is the map Section 2 uses at our own horizon. What changes with mass is whether the same map also fixes the black hole, and A.15 shows it cannot while the two horizons have distinct bifurcation surfaces. So a hole is two-sided exactly when its horizon is the cosmological horizon, which happens at the maximum mass and nowhere else. The second class is not a separate population. Sections 2 and 6 are looking at a single structure from its two ends, and Section 5's contact radius climbing from half the horizon radius to all of it is the parameter between them made visible.

Two things follow from that. The first is that a measurement of the primordial curvature tail does not falsify the fold: it measures the wrong quantity, and no test of the fold can be built on it. Sections 2 and 3 are

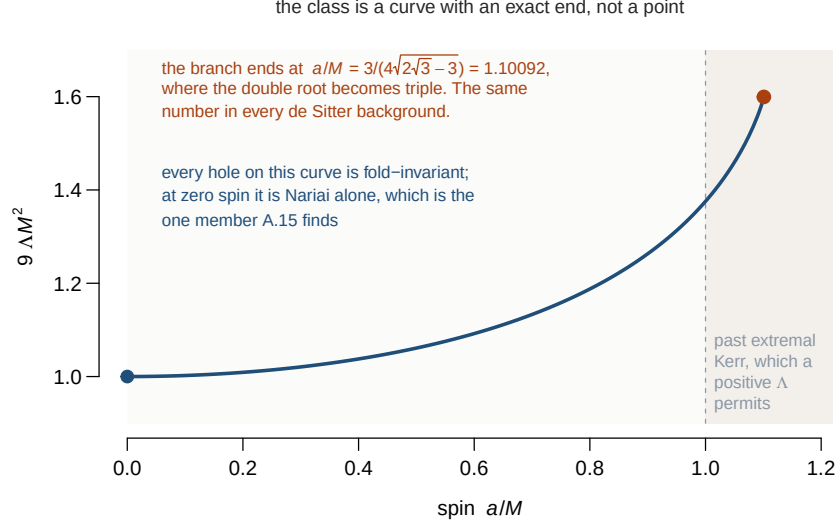


Figure 11: The fold-invariant class across spin. Each point is a Kerr-de Sitter hole whose black-hole and cosmological horizons coincide, which is what a single involution serving both requires. At zero spin the curve starts at Nariai, $9\Lambda M^2 = 1$, the one member A.15 finds in the uncharged family. It ends at $a/M = 3/(4\sqrt{2\sqrt{3}-3}) = 1.10092$, where the double root becomes triple, and that ratio is the same in every de Sitter background because both a/M and $9\Lambda M^2$ are fixed numbers there. The shaded strip is past the Kerr extremal bound, which a positive Λ permits: the branch crosses $a/M = 1$ at $9\Lambda M^2 = 1.3755$ and keeps going.

untouched because they never needed a black hole, and Section 5’s contact condition is untouched. What changes is that the object it describes is one configuration rather than a class with an abundance.

The second is that two other calculations, run for different reasons and neither looking for the other, single out the same configuration. Section 5’s contact radius, swept against Λ , runs from half the horizon radius at $\Lambda = 0$ to 0.991 of it at $9\Lambda M^2 = 0.9999$, so the one hole the fold can fix is also the one whose interior is entirely in contact, with no outer shell left where the sheets cannot reach.

And the objection a maximal configuration invites, that it does not last, turns the same way, though not as far as it first appears to. Unfolded Nariai is unstable. The sphere-radius perturbation has exactly three modes and two of them run away, which is the known decay toward Schwarzschild-de Sitter. Both runaways are odd under the fold map and neither descends to the quotient, so the folded solution does not carry them. That result is narrow in three ways. The mode that does survive is bounded inside a static patch and unbounded outside one. The non-spherical sectors had no instability to remove, their potentials being nonnegative at every $\ell \geq 1$. And the Euclidean negative mode is a constant on both spheres of $S^2 \times S^2$, invariant under every isometry, so the semiclassical decay is untouched. Two runaway modes go, in one sector, inside one patch. That is a real result and it is not the same thing as making Nariai stable.

What is open now is whether anything outside this family qualifies, and whether the embedding argument carries into the warped rotating near-horizon geometry. Neither is a question about a tail.

7. Matter rule, neutrino masses and the conditional decay scale

The fold also commits the matter sector, conditionally on the production model and state selection of the cosmology paper, and nothing in Sections 2 to 6 depends on it. With the stabilising rule of Boyle, Finn and Turok exact, one light neutrino is massless and the two measured splittings fix the sum of the masses. With the rule weakly broken by a small Yukawa column, the sterile neutrino decays to $h\nu$, $Z\nu$ and $W\ell$ in the ratio $1 : 1 : 2$, with a hard neutrino line near half its mass, and the light-neutrino mass the same column induces

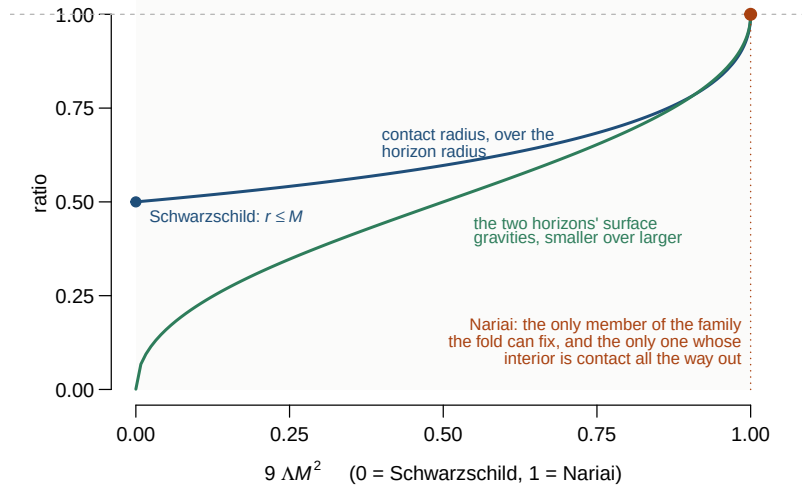


Figure 12: Two quantities that were computed for unrelated reasons, across the Schwarzschild-de Sitter family from Schwarzschild at the left to Nariai at the right. In blue, the contact radius of Section 5 over the black-hole horizon radius: it is exactly one half at $\Lambda = 0$, which is $r \leq M$, and climbs to one, so the region where a point can reach its own fold image grows until it fills the interior. In green, the ratio of the two horizons' surface gravities, which decides whether one involution can serve both and so whether the fold can fix the hole at all. The first is a causality calculation about angular sweep; the second is a statement about horizon structure. Neither was derived from the other and both reach one only at the endpoint.

is at most 2.1×10^{-55} eV at an example lifetime of 10^{28} s. Flavour cannot separate that decay component from an astrophysical population, and the fixed ratio of line to continuum can. A.20 has the working and its conditions.

8. What the mirror sheet is, and why counting the radiation does not decide it

Everything so far concerns what the fold does, at a horizon and then to the matter sector. This section concerns what the fold *is*. The obvious measurement does not reach it, for a reason followed below, and the answer left standing is more specific than either of the two names on offer.

The fold keeps two sheets and relates them by an antilinear map. What the cosmology paper leaves open is whether the mirror sheet is a **copy**, a relabelling of the same degrees of freedom seen under the involution, or a **place**, a second region carrying its own stress-energy. Two results in this paper bear on it and neither is in that paper.

Section 3 tests three readings that make the mirror sheet a single copy of our own space, and all three fail: on positivity, on the first law, and against the orbital decay of a binary pulsar. That constrains those three implementations and leaves every other quotient untouched. And A.8 says what kind of place it is, which turns out to be a third option neither name covers. At a bifurcate horizon in equilibrium the two-sheet state is a thermofield double: the mirror sheet carries its own algebra, which is the commutant of ours, and every expectation in it is fixed by the state on our side. A place, then, and a purification rather than an independent world. Whatever it holds is determined by what we hold, without being the same degrees of freedom relabelled.

None of that changes the radiation arithmetic below, and the reason it does not is the point of the rest of this section. The two readings disagree about how much radiation the universe contains, and that quantity has been measured.

the test that looks decisive, and the reason it does not apply

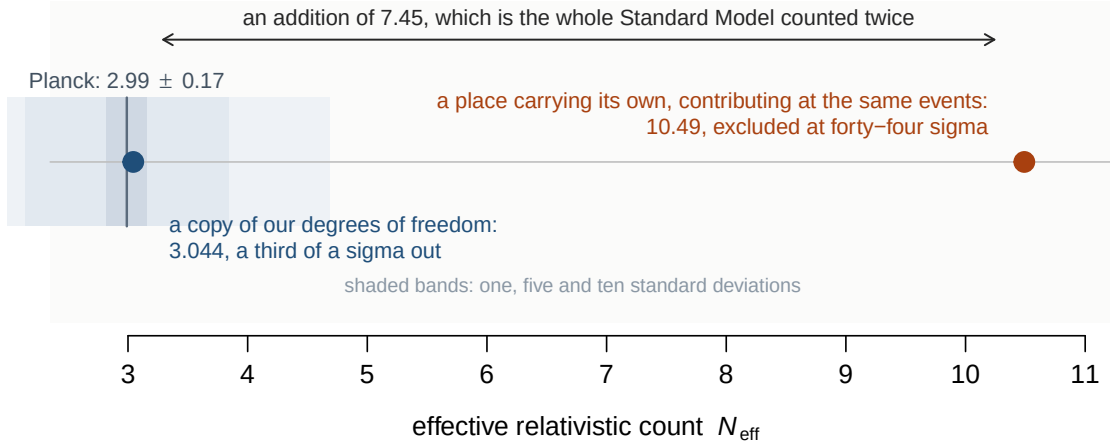


Figure 13: The arithmetic that looks decisive. Writing $\rho_{\text{rad}} = \rho_\gamma(1 + 0.2271N_{\text{eff}})$ with $0.2271 = (7/8)(4/11)^{4/3}$, the Standard Model's $1.691\rho_\gamma$ duplicated is an addition of 7.45 in N_{eff} units, which is the whole Standard Model counted twice. Against Planck's 2.99 ± 0.17 the copy reading sits a third of a standard deviation away and the place reading forty-four. **The paper does not claim that exclusion.** Duplicating the density requires the two sheets' stress-energy to add at the same events, and the fold puts them at different ones, which is the same fact that makes the silence of Section 1 a theorem. Both readings therefore predict the left-hand point and Planck does not separate them. The figure is here to show how large a gap the argument would have carried had it applied, and how little it gives without co-location.

A copy contributes nothing to the expansion rate. The mirror sheet is the same content described twice, so the effective relativistic count is the Standard Model's, $g_* = 106.75$, which is what the cosmology paper uses. A place contributes its own. Recombination tests that more cleanly than the bang does, since the radiation there is measured directly as N_{eff} , the effective number of neutrino species. A mirror sheet carrying the same content adds its entire radiation density while adding no photons we can see, and that is exactly the quantity N_{eff} counts. Writing the radiation density as $\rho_{\text{rad}} = \rho_\gamma(1 + 0.2271N_{\text{eff}})$, the Standard Model's $1.691\rho_\gamma$ is duplicated, which in N_{eff} units is an addition of 7.45,

$$N_{\text{eff}}^{\text{copy}} = 3.044, \quad N_{\text{eff}}^{\text{place}} = 10.49.$$

Planck measures $N_{\text{eff}} = 2.99 \pm 0.17$. The copy reading sits 0.3 standard deviations from that; the place reading is excluded at forty-four.

That arithmetic is correct and it does not apply here, which is the point of the rest of this section. The step from “a place carries its own stress-energy” to “the radiation density at our recombination is doubled” needs the two sheets' stress-energy to add *at the same events*. Nothing in the fold puts them there. Everything else in this paper insists on the opposite: Section 1's silence theorem, its proof that one point in each wedge is always spacelike separated, and A.15 entirely. On de Sitter the map sends $X^0 \mapsto -X^0$, so a point and its image are not even contemporaneous. The Friedmann equation at our last-scattering surface is sourced by the stress tensor at our events, and a sheet occupying other events contributes nothing to it whichever reading is taken. So $N_{\text{eff}} = 3.044$ under both, and the forty-four standard deviations exclude a co-located mirror sector, which is a model this paper does not propose. The caveat travels with it: what is tested is silence at recombination, while the g_* commitment is about the bang.

An obvious caveat, that N_{eff} constrains recombination while the g_* commitment is about the bang, is about *when* and not about *whether*, and the second is the question that bites. Are the sheets ever co-located at all? Geometry answers no, everywhere except the bang itself, which is the one surface they share.

What survives runs the other way, and counts for more than the test. If the geometry forbids the place reading from contributing locally, then the cosmology paper's $g_* = 106.75$ follows from the fold instead of being chosen, and its dark-matter ceiling of 491.6 PeV does not carry the factor-of-two ambiguity that this section was written to resolve. That question dissolves rather than being decided by data. That is a smaller claim than a passed measurement and a defensible one.

Two things would put a real test back. An argument that the sheets are co-located at the bang in the sense the production calculation needs, which would make g_* the observable and the abundance the measurement, and which we have not constructed. Or a mechanism by which the mirror sheet gravitates non-locally, which the fold as built supplies no route to. Until one of those exists, this paper's empirical weight sits on Section 4's ringdown null, which a running instrument is already testing.

The first of those two routes is the interesting one and it is where the surviving test lives, so it is followed one step here. The bang is the single surface the two sheets share, and g_* enters the abundance calculation there rather than at recombination. So if co-location at the bang can be argued, the ambiguity returns as a shift in the produced abundance and therefore in the mass, and that shift is already quantified. It moves the dark-matter ceiling from 491.6 to 526.9 PeV and the two-body neutrino line from 245.8 to 263.4 PeV, a fixed

$$\frac{\Delta E_\nu}{E_\nu} = 2^{1/10} - 1 = 7.18 \times 10^{-2}$$

of wherever the line sits. Quoting a shift like that in units of the propagated theory width, which is the natural thing to do, describes the theory error and not what an experiment faces. The limiting uncertainty on a cascade line position at these energies is the absolute energy scale, which carries contributions from the inelasticity distribution at $x \sim 10^{-6}$, the ice absorption model, photomultiplier saturation and the shape of the background, and is a common-mode systematic of order fifteen to twenty-five per cent. A common-mode systematic does not average down with exposure, so the significance saturates below one standard deviation at any event count. So the line test requires absolute calibration better than seven per cent to reach one standard deviation at all, and better than two and a half per cent for three. No existing detector is close, so the test should be quoted as deferred to a detector that is, and never as a sensitivity. It is now doubly conditional, on that calibration and on the bang co-location argument nobody has made, and we would rather say so than carry it as an asset.

9. Conclusions

A horizon surrounds every observer. Ours is a cosmological one, and the mathematics sealing its far side is the mathematics that seals a black hole. This paper asked what a mirror-symmetric universe does at such a surface, and the answer divides into what the symmetry determines and what it only permits.

At our own horizon nothing is chosen. The reflection is fixed by theorems and the transverse part is the antipodal parity their composition leaves behind, and a reciprocal law follows: the correlation across a reflected separation over the direct one, times the same ratio reflected, is exactly one, for every state, every metric on the surface and every free involution. Its logarithm is odd about ninety degrees, so all even multipoles vanish. The fold's map is a half-period shift, the square root of the thermal transformation, and primitivity picks the fundamental period. Three readings of the mirror sheet as a single copy of our own space fail, by algebra, by thermodynamics, and against the orbital decay of a binary pulsar.

Where the two sheets can touch is settled by causality alone, with no seam, no coefficient and no choice. Outside any horizon they cannot. Inside a hole possessing a past, contact needs half a turn of the sphere as well as a fall, and causality caps the angular path of any causal curve at $2\pi - 4 \arcsin \sqrt{r/2M}$, which reaches π at the midpoint of the interior, $r = M$ for an uncharged hole and $(r_+ + r_-)/2$ for a charged one. So the sheets touch in the inner half of the interior, and nowhere else when the hole is uncharged. That region is where the quotient carries closed causal curves. In D dimensions the interior supplies $2\pi/(D-3)$ against the same π , so four large dimensions is the only case in which the two sheets reach each other at all.

At the contact radius the image term diverges as $\delta^{-3/2}$, and inside it the term turns imaginary. An imaginary energy is a decay rate and a vacuum decay rate is pair creation, so the region where the sheets touch makes matter and antimatter from the causal structure alone. Of everything in this paper that is the result we would most like to see someone find a way to test, and we see none. The stress the image term carries defocuses the curves that would make contact and leaves the hypothesis of Penrose's theorem where it was, with the radial half of that sign still resting on conservation.

One thing is not supplied. Carrying the fold to a black hole requires a seam coefficient the theory does not fix, and no symmetry can pick one out. Exactly one member of the family absorbs no momentum, and it is adopted on that premise. It predicts ordinary Kerr ringdown and a damping-time deviation of exactly zero. Zero sits outside GWTC-3's posterior-multiplied interval and inside its hierarchical one, and roughly twice the present exposure separates those two readings.

One by-product holds whatever the fold turns out to be. A.14 computes $M d\omega/d\mathcal{R}_h = 0.00735845 - 0.0646771 i$ for the fundamental $\ell = 2$ Schwarzschild mode, so a horizon reflectivity moves the damping time by $-0.727 \mathcal{R}_h$ and the real frequency by $+0.0197 \mathcal{R}_h$, thirty-seven times harder on the decay than on the note. The response is linear and signed, so the absorption model in common use, $\tau \rightarrow \tau/(1 - |\mathcal{R}|^2)$, is quadratic and signless and therefore wrong at leading order.

A hole with a past would take matter from our side and antimatter from the mirror, and it would grow no faster than a one-sided hole. Within the Kerr-Newman-de Sitter family the fold maps a hole to itself only when it is maximal, 2.2×10^{22} solar masses at the observed Λ , out to a spin $a/M = 1.10092$. The contact region fills 0.991 of such a hole's interior, and two of its three runaway modes are odd under the fold inside one static patch, which does not make Nariai stable. No such hole is known to exist. Counting the radiation does not decide whether the mirror sheet is a copy of our degrees of freedom, since N_{eff} is 3.044 under either reading.

What stays open is named where it arises. The seam coefficient is adopted. The reduction A.10 rests on is not proved there. Section 6's count is made inside one family of solutions. The contact radius is a conjecture at spin. And the interior algebra, the one place the two sheets can reach each other, is not built here.

Two of these results are one result read at different distances. The fold's parity is the classical-quantum split, so the classical world is the part of a two-sheet field that survives the quotient, and no limit in $\hbar$ has to be taken to reach it; at a horizon the weight of the part that does not survive is a temperature. That is why the cosmology paper gets the field equations out of a symmetry with the metric left as a background, and it is why asking where the two sheets can touch has an answer at all. They touch inside a hole with a past, across the inner half of its interior, in four dimensions and in no others, and the nearest test of that is an exactly Kerr ringdown with zero damping-time deviation, decidable at about twice present exposure. What the two papers share is the shape of the answer: gravity did not have to be quantised for any of it, and the symmetry was doing the work.

Appendices

The appendices are numbered from A.6 because A.1 to A.5 belong to the cosmology paper.

A.6 A worked instance of the in-in dictionary. §1's closed-time-path dictionary, worked on one pair. Take the massless conformally coupled scalar on dS_4 and put the observer on the central worldline, $r = 0$. There the transverse sphere degenerates, the transverse parity acts trivially, and $A = J$ exactly, so the fold's image kernel and the thermal contour's cross-leg propagator are the same function of time:

$$G_{\text{img}}(t) = \frac{H^2}{16\pi^2} \text{sech}^2\left(\frac{Ht}{2}\right) = W(t - i\beta/2) = G_{+-}^{\sigma=\beta/2}(t), \quad \beta = \frac{2\pi}{H}.$$

The three expressions agree in the stated Bunch-Davies conformal-scalar example. Define the analytic half-period continuation $F_{\text{half}}(t) = F_W(t - i\beta/2)$. With the convention $F(E) = \int dt e^{-iEt} F(t)$, strip

analyticity gives $F_{\text{half}}(E) = e^{\beta E/2} F_W(E)$. Identifying this analytic continuation with a geometric image $W(x, \alpha y)$ additionally requires the state's geometric reflection and parity implementation, including index transport for fields with spin; KMS alone does not establish that identification for every state, mass or spin. With the spectral density $\rho(E) = E/4\pi^2$, fixed by the Gibbons–Hawking detector response, the reconstruction $\int_0^\infty dE \rho(E) \cos(Et)/\sinh(\beta E/2)$ returns the same sech^2 , and a five-energy numerical check reproduces $F_{\text{img}} = e^{\beta E/2} F_{\text{BD}}$. These are implementation checks of the stated analytic conformal-scalar identities. No claim of physical precision attaches to them. The half-period identity itself is checked to 4×10^{-17} at four times, and fails by 0.115 ten per cent away in β .

The superscript on G_{+-} is not a convention here. A thermal contour may be displaced downward by any σ in $[0, \beta]$ without changing a physical correlator, and it carries two cross kernels, one for each ordering of the legs. For one mode their difference is $i \sin \omega t \sinh(\omega(\beta/2 - \sigma))/\omega \sinh(\beta\omega/2)$, reproduced against the mode sum to 6×10^{-16} and vanishing at every separation only at $\sigma = \beta/2$. With $\omega = 1.3$ and $\beta\omega = 2 \ln 3$, so that $1/\omega$ is the unit for the three numbers that follow, at $\sigma = \beta/3$ the two kernels still differ by 0.216, and at $\sigma = 0$ by the whole commutator $i \sin \omega t/\omega$ whatever the temperature.

Since Θ is antilinear with $\Theta^2 = 1$, exchanging the sheets must leave the cross kernel alone, so the fold admits one member of the family. The selection is the Bose factor doing the work, through $(1+n)e^{-\omega\sigma} = ne^{\omega\sigma}$: with the thermal occupation dropped and only the vacuum piece kept, the two kernels differ at $\beta/2$ as much as anywhere else, by 0.256 against 0.496 at $\sigma = 0.2\beta$, a ratio near a half where the thermal case drops to machine zero. No mode is indifferent, and modes with $\beta\omega$ near two are the most sensitive, losing 0.0259 of the coincident correlator for a displacement of one per cent of β . The equal-time two-sheet matrix at $\sigma = \beta/2$ has the fold-even and fold-odd combinations as eigenvectors, with eigenvalues $\coth(\beta\omega/4)$ and $\tanh(\beta\omega/4)$ in units of $1/2\omega$ whose product is 1 to twelve figures at every temperature, giving the cosmology paper's ratio $4 \tanh^2(\beta\omega/4)$, which is 0.2872 at the crossing ceiling $\beta\omega = \ln 3$ and 1 at $\beta\omega = 2 \ln 3$.

One caution belongs with this. The quantum-quantum correlator of the symmetric contour does not vanish: it is $\tanh(\beta\omega/4) \cos \omega t/\omega$, which is 0.905 in units of $1/\omega$ at coincidence for $\beta\omega = 6$, and the familiar statement that it vanishes belongs to $\sigma = 0$, where the computation returns zero and falls to it monotonically. What vanishes at $\beta/2$ is the quantum-quantum commutator, which is the statement §1 makes. Off the central worldline the two maps part company. For two points at radii r_1 and r_2 subtending an angle γ at the centre, with time separation Δt , the embedding invariants differ by

$$Z_\alpha - Z_J = -2r_1 r_2 \cos \gamma,$$

independent of Δt , which reduces to $-2r^2$ at equal radii and $\gamma = 0$. A relative discrepancy needs a denominator, and the two choices behave differently. Referred to the fold kernel it is $r^2/[(1-r^2) \cosh^2(\Delta t/2)]$, which is 9.9% at $r = 0.3$ and 178% at $r = 0.8$ for $\Delta t = 0$; referred to the thermal kernel it is $-r^2$, bounded by unity. Both the invariants above and these two figures are reproduced from the kernels. The two differ by $(1-Z_J)/(1-Z_\alpha)$ and that factor carries the apparent growth.

Only the ratio is invariant. At $\Delta t = 0$ and $\gamma = 0$ the invariants are $Z_J = -1 + 2r^2$ and $Z_\alpha = -1$, so $1 - Z_\alpha = 2$ identically and

$$G_\alpha = \frac{1}{2}, \quad G_J = \frac{1}{2(1-r^2)}, \quad G_\alpha/G_J = 1-r^2,$$

in units where the prefactor is dropped. The fold's image kernel is constant in radius. Divergence at the de Sitter horizon belongs to the thermal contour's cross-leg kernel, and at $r = 1/\sqrt{2}$ the fold kernel is half of it. The transverse parity flips the sign of the r^2 term, which is what cancels the radial dependence. This is a statement about $\gamma = 0$ and does not generalise as it stands. At the horizon $1 - Z_J \rightarrow 1 - \cos \gamma$ and $1 - Z_\alpha \rightarrow 1 + \cos \gamma$, so each kernel is singular exactly where its own map brings the two points together, the thermal contour's at coincident directions, the fold's at antipodal ones. Neither is the better behaved overall. The general statement is the horizon ratio,

$$G_\alpha/G_J \rightarrow \tan^2(\gamma/2),$$

independent of everything else and reproduced to 4×10^{-7} . Written with unit direction vectors this is

$$\frac{G_\alpha}{G_J} \rightarrow \frac{|\hat{n}_1 - \hat{n}_2|^2}{|\hat{n}_1 - (-\hat{n}_2)|^2},$$

the squared chord between the two points over the squared chord from one to the other's antipode. J leaves the transverse direction alone, while α carries it to the antipodal one, so each kernel is controlled by the separation of its own pair of arguments. The three features the cosmology paper's A.1 states then need no calculation here. Each is the ordinary short-distance singularity of a two-point function, appearing at the angle its own identification selects. The $1 - Z_\alpha = 2$ identity is specific to $\gamma = 0$ and $\Delta t = 0$; the $\tan^2(\gamma/2)$ ratio holds at every angle. Both are statements about the conformally coupled massless scalar in the Bunch-Davies state, and nothing is claimed for spin two or for other states. The $\cosh^{-2}$ confines the effect to coincident times, decaying as $4e^{-|\Delta t|}$ once $|\Delta t| \gg 1$.

Angular dependence needs care here. The invariant difference is exactly $P_1(\cos \gamma)$, but the discrepancy is not, because $\cos \gamma$ also enters its denominator. Writing $b = r_1 r_2$ and $a = 1 + \sqrt{(1 - r_1^2)(1 - r_2^2)} \cosh \Delta t$, the fold-referred discrepancy $-2b \cos \gamma / (a - b \cos \gamma)$ expands as $-(2b/a) \sum_{n \geq 0} (b/a)^n \cos^{n+1} \gamma$, with no alternation because the denominator carries $-b \cos \gamma$; the thermal-referred form $-2b \cos \gamma / (a + b \cos \gamma)$ gives the same series with $(-1)^n$ inserted, so every multipole appears, suppressed by powers of b/a , which at equal radii and $\Delta t = 0$ is $r^2/(2 - r^2)$.

The dipole dominates only while that is small: the quadrupole is 3.1% of it at $r = 0.3$, 23% at $r = 1/\sqrt{2}$ and 53% at $r = 0.9$. The angular zero holds exactly at every radius and every time separation, and the numerator alone carries it: the discrepancy vanishes at $\gamma = \pi/2$ and reverses sign across it whatever the higher multipoles do. That zero does not weaken with radius and is the sharp feature.

Contamination by higher multipoles belongs to the variable. The fold does not produce it. The discrepancy is a difference of kernels; the ratio is not, and by the reciprocal law of A.10 $\log(G_\alpha/G_J)$ is odd about $\gamma = \pi/2$ for any Hadamard state, so its even multipoles vanish *identically* at every radius and every time separation, rather than only while b/a is small. That is algebra. A projection at $r = 0.9$, where the quadrupole of the discrepancy is 53% of its dipole, returns even coefficients below 10^{-15} for four unrelated profiles, which tests the quadrature and not the claim. The quantity that matters is the truncation error, since the logarithmic endpoints make the series converge in L^2 but not pointwise there, and a truncated reconstruction rings within $O(1/\ell_{\max})$ of $\gamma = 0$ and π . The dipole dominance that has to be argued for in the difference is exact in the logarithm, away from the two endpoints.

Two readings have to be excluded. The discrepancy depends on r_1 , r_2 , γ and Δt alone, so it is invariant under a rigid rotation of both points and selects no axis; P_1 is a parity and parity commutes with rotations. The fold cannot generate a directional dipole of the kind reported in radio and quasar number counts, at any amplitude. Neither is this a correction to ordinary clustering: the objects compared are the cross-leg kernels G_{12} of the cosmology paper's §2.1, while the galaxy two-point function is built from the same-leg propagator. Read as a fractional correction to $w(\gamma)$ in a narrow redshift slice it would give tens of per cent by $z \simeq 1$ and more than unity by $z \simeq 2$, which observed clustering excludes outright, and that rules out the identification rather than the fold. The cross-leg kernel enters the noise and dissipation sector, which is where the force-noise ports below place it.

For scale, r is the areal radius in units of the de Sitter radius, so in the flat slicing it is physical distance in horizon radii. Taking $H_\Lambda = H_0 \sqrt{\Omega_\Lambda}$ and a Λ CDM distance-redshift relation for orientation only, $r = 0.3$ is $z \simeq 0.40$, $r = 0.8$ is $z \simeq 1.4$, and the horizon $r = 1$ is $z \simeq 2.0$. The largest angular scales of the microwave background lie well outside this range: a 60° separation on the last-scattering surface corresponds to $r \simeq 2.6$, so nothing in this comparison bears on the large-angle correlation anomalies.

That same structure governs a quantity which is not a consistency check. For a comoving scalar charge the local and image force-noise kernels form a two-port matrix with fold-even and fold-odd eigenvalues

$A_0 \pm A_I$. On the central worldline the image kernel is exactly minus the half-period shift of the local one, so $A_I = -A_0 = -1/12\pi^2$ and the even port vanishes identically. The $-2r^2$ breaks that identity. Carrying the discrepancy into the kernel and integrating, with $\int dt \operatorname{sech}^6(t/2) = 32/15$, gives an even-port lift against an unchanged odd port $1/6\pi^2$. Referred to the fold kernel the lift is $-r^2/[15\pi^2(1-r^2)]$; referred to the thermal kernel it saturates, approaching $-1/12\pi^2$ as $r \rightarrow 1$, exactly half the odd port. Under that thermal referencing the even channel is lifted off zero at every $r > 0$ and stays the smaller of the two at all radii; referred to the fold kernel it instead reaches the odd port at $r = \sqrt{5/7}$, which is the same normalisation artefact again, with no physical crossover behind it. The degeneracy is exact only on axis, the misalignment enters at second order, and the sign is fixed. Carrying a propagator-level discrepancy into the force-noise kernel is a leading-order identification rather than a spin-2 derivation; the lifting of the zero and its order in r do not depend on that step, but the coefficients do.

A.7 The first-tick sum and its asymptote. The closed form quoted in §2.1 and its large- A limit, derived. Closed de Sitter in conformal time has $a = 1/(H \cos \eta)$, $aH = \sec \eta$ and $\dot{a} = \tan \eta$. Write $\chi = a\varphi$, and take scalar harmonics $-\nabla^2 Y_n = (n^2 - 1)Y_n$ of multiplicity n^2 . The environment is then $n \geq 2$, since $n = 1$ is the homogeneous mode and belongs to minisuperspace, and

$$\begin{aligned} \chi_n'' + (n^2 - 2 \sec^2 \eta) \chi_n &= 0, \\ v_n &= C_n e^{i n \eta} (\tan \eta + i n), \\ |C_n|^2 &= \frac{1}{2n(n^2 - 1)}, \\ \psi_n &\propto \exp(-\Omega_n \chi^2/2), \\ \Omega_n &= -i \frac{v_n'}{v_n}, \quad \Omega_n' = i(\omega^2 - \Omega_n^2). \\ \Omega_n &= \frac{n(n^2 - 1) - i \tan \eta \sec^2 \eta}{\tan^2 \eta + n^2}, \quad \Omega_n(0) = \frac{n^2 - 1}{n} \in \mathbb{R}, \\ \operatorname{Re} \Omega_n &= \frac{n(n^2 - 1)}{\dot{a}^2 + n^2}, \quad \operatorname{Im} \Omega_n = -\frac{\dot{a} (aH)^2}{\dot{a}^2 + n^2}, \\ \tan \theta_n &= -\frac{\dot{a}(1 + \dot{a}^2)}{n(n^2 - 1)}. \end{aligned}$$

$$\psi_- = \bar{\psi}_+ \implies D_n = |\langle \bar{\psi} | \psi \rangle| = \left(\frac{\operatorname{Re} \Omega_n}{|\Omega_n|} \right)^{1/2} = (1 + \tan^2 \theta_n)^{-1/4} = (1 - r_n^2)^{1/4}.$$

$$\log |\langle E_- | E_+ \rangle| = -\frac{N_f}{4} \sum_{n \geq 2} n^2 \log \left[1 + \frac{A^4 (A^2 - 1)}{n^2 (n^2 - 1)^2} \right], \quad A \equiv aH.$$

With $u = n/A$, the large- A limit is

$$-\frac{N_f}{4} A^3 \int_0^\infty u^2 \ln(1 + u^{-6}) du = -\frac{\pi}{12} N_f A^3.$$

Each environment mode is a Gaussian whose frequency function solves the Riccati equation; the no-boundary condition is that Ω_n be real at the throat, which the exact mode function delivers, and because the Wheeler–DeWitt operator is real the two branch-conditioned environment states are complex conjugates, so the branch overlap is the Gaussian integral $\int \psi^2$ and not $\int |\psi|^2$. The squeezing phase is exactly odd in $\dot{a}$ mode by mode, which is the time-odd order parameter, while $|D|$ depends on $\dot{a}^2$ alone, as time reversal requires; summing the mode overlaps over the S^3 tower with degeneracy n^2 gives the closed form. At large A the summand depends on n only through $u = n/A$, so the sum becomes $A^3 \int_0^\infty u^2 \ln(1 + u^{-6}) du = \pi A^3/3$, and the asymptote is already good to 0.07 per cent at $A = 10$, the sum is -261.627 against -261.799 , a gap of 0.066 per cent. The integral is $\pi/3$ to 2×10^{-16} and the sum is converged in the truncation.

A.8 The image term: what positivity and Hadamard forbid, what they allow, and what sets its size. §3.1's two inequalities, derived. Let $\mathcal{A}$ be the antipodal map (A is reserved here for the amplitude), take the conformally coupled massless scalar on dS_4 , where the Bunch–Davies kernel is elementary, and consider the following real, de Sitter-invariant two-amplitude ansatz on one static patch:

$$W(x, y) = \frac{1}{2}i\Delta(x, y) + A G_{\text{sym}}(Z) + B G(-Z),$$

$$G(Z) = \frac{H^2}{8\pi^2(1-Z)}, \quad Z(x, \mathcal{A}y) = -Z(x, y).$$

The antipode pairs each cover mode with itself, antilinearly:

$$y_n \circ \mathcal{A} = -\bar{y}_n,$$

$$\langle a^\dagger a \rangle = \nu, \quad \langle aa \rangle = \langle a^\dagger a^\dagger \rangle = \kappa, \quad A = 2\nu + 1, \quad B = -2\kappa,$$

$$\sigma = \begin{pmatrix} \nu + \frac{1}{2} + \kappa & 0 \\ 0 & \nu + \frac{1}{2} - \kappa \end{pmatrix}, \quad \Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

$$\sigma + \frac{i}{2}\Omega \succeq 0 \iff \nu \geq 0 \text{ and } \nu(\nu + 1) \geq \kappa^2 \iff A \geq 1 \text{ and } A^2 - B^2 \geq 1.$$

Since $G(-Z) = H^2/[8\pi^2(1+Z)]$ is regular as $Z \rightarrow 1$, Hadamard normalisation gives $A = 1$ and hence $B = 0$.

Within the displayed de Sitter-invariant two-amplitude ansatz, the mode blocks may be chosen diagonal in this basis and are described by the 2×2 covariance σ . Positivity, $\sigma + (i/2)\Omega \succeq 0$, then says that the spectral weight of the symmetric part must dominate the commutator $\frac{1}{2}i\Delta$ mode by mode, which in the amplitudes is $A^2 - B^2 \geq 1$.

That equivalence is checked against direct diagonalisation on a 301×601 grid in (ν, κ) . Criterion and eigenvalues agree at all 180901 points, both classes are populated, 112865 against 68036, and three wrong criteria each produce tens of thousands of mismatches. At three points the grid lands exactly on the boundary, and at one of them the two sides round opposite ways, so the count without a matched tolerance is 180900.

One thing that argument does not do is forbid the fold an image correlation altogether, and the gap is named here. The ansatz tested above is a constant amplitude multiplying the whole spectrum, which is why the ultraviolet can refuse it: it puts a singularity on the image lightcone, at pairs that are not null-related in the first place. A finite, smooth image correlation is a different object, because it adds only smooth terms and leaves every singularity where it was. Squeeze a single mode, $v = \cosh r u + \sinh r \bar{u}$, and the two-point function changes by

$$\Delta W(x, y) = 2 \sinh^2 r \operatorname{Re} [u(x) \overline{u(y)}] + \sinh 2r \operatorname{Re} [u(x) u(y)],$$

and if the mode is fold-real, $u(\Theta y) = \overline{u(y)}$, the second term is an image correlation without anything having been inserted to make it one. The Bogoliubov map is canonical, $\cosh^2 r - \sinh^2 r = 1$, so the commutator and positivity are untouched; ΔW is a finite sum of products of smooth mode functions, so the short-distance singularity is untouched and the state remains Hadamard; and the local stress moves. Constancy across the spectrum is what the argument above excludes, not image correlation as such. Nor is the fold confined to a parity-blind spherical mode. A transversely odd $\ell = 1$ mode, for which $u(\Theta x) = -u(x)$, flips the sign of the image coefficient and still gives a nonzero stress shift.

Nor does it need a mass. Running the same construction at $m = 0$ on the Nariai background leaves the image correlation intact, with a stress shift whose sign varies with position on the sphere as well as with the squeeze. What does fail at $m \rightarrow 0$ is something else, and the distinction is the useful part: the Euclidean reference state's homogeneous mode has variance going as Λ/m^2 and does not survive the limit. A genuine

massless canonical pair does exist. On the dS_2 factor the homogeneous equation $\ddot{q} + \tanh t \dot{q} = 0$ has the constant and the Gudermannian $\text{gd } t = \arctan \sinh t$ for solutions, and $u_\alpha = \sqrt{\Lambda/16\pi^2\alpha}(1 - i\alpha \text{gd } t)$ carries unit Klein-Gordon norm for every $\alpha > 0$, the $\cosh t$ of the circle cancelling against the $\text{sech } t$ of the current. So the zero mode is a state to be chosen and not a barrier to having one, and what the massless limit kills is the reference, not the field. A massless scalar is not a photon and nothing here is a Maxwell calculation.

That family is where the fold's remaining freedom sits, and the fold does not reduce it. Since gd is odd and Θ acts on a mode as $(\Theta u)(t) = \overline{u(-t)}$, the pair u_α is fold-real at every α , so Θ -invariance selects no member. The ambiguity is the familiar infrared one of a minimally coupled massless field in de Sitter, in this model's form, and it does not touch the conformally coupled case worked above, where ξR acts as a mass and there is no zero mode to argue about.

What that leaves is a selection problem rather than an exclusion, and we should be exact about which kind, because the fold is not silent on selection in general. For an antiunitary Θ with $\Theta^2 = 1$, a Θ -invariant state and a Hermitian A , $\langle \Theta A \Theta \rangle = \overline{\langle A \rangle} = \langle A \rangle$. That is antilinearity and hermiticity and nothing else. Where Θ exchanges two things it therefore forces their expectations equal, which is how the cosmology paper's contact condition $\langle N_+ \rangle = \langle N_- \rangle$ arises at the bang, a consequence, and with it a definite occupation.

Where Θ fixes the object instead, the same identity constrains nothing, and a fold-real mode at a horizon is that case. A squeezed vacuum with real r has real Fock amplitudes, so it is Θ -invariant at every r while its occupation $\sinh^2 r$ runs freely; giving the amplitudes a phase makes the invariance bite again, so the check is not vacuous. The fold selects where it exchanges and not where it fixes.

So the squeeze parameter is not a foundational gap waiting for a new principle. The state at a horizon is whatever the selected bang state evolves into, which makes this a propagation question and not a question about what selects. What survives as genuine freedom is the phase the bang leaves per mode, and that is a smaller and better-posed object than a free amplitude. It is also bounded. A squeeze is measured by its occupation, $\langle N \rangle = \sinh^2 r$, so the occupation the bang selects is a squeeze it selects, $r = \text{arcsinh} \sqrt{n_*}$ with $n_* = (1 - \sqrt{1 - P})/2$ the cosmology paper's floor; and since $n_* \leq \frac{1}{2}$, $r \leq \text{arcsinh}(1/\sqrt{2}) = 0.65848$, a pure number. Reading that as a ceiling here needs two things neither paper establishes, that the crossing's invariance reaches the mode and that its squeeze is inherited rather than regenerated, and both are stated as assumptions. That propagation is short, because §3's horizon law, derived in A.10, has already done its hard half. The fold's map is the square root of the thermal transformation at the primitive period, so for a state in equilibrium at a horizon the cross-sheet correlator is the direct one shifted by half a period, which is a consequence. Carrying it through for one mode of frequency ω at inverse temperature β ,

$$W\left(t - \frac{i\beta}{2}\right) = \frac{\cos \omega t}{2\omega \sinh(\beta\omega/2)},$$

real and even, which is the fold-real property the fold needs; the two exponentials collapse to a single coefficient because $(1 + n)e^{-\beta\omega/2}$ and $ne^{\beta\omega/2}$ are equal. A two-mode state with that cross correlator is the thermofield double, and the thermofield double is a two-mode squeezed vacuum at

$$\tanh r(\omega) = e^{-\beta\omega/2}, \quad \langle N \rangle = \sinh^2 r = \frac{1}{e^{\beta\omega} - 1},$$

the second identity exact. So the squeeze the fold carries at a horizon is fixed by the mode frequency and the surface gravity and nothing else. The amplitude left open above is not open.

Two things travel with that and both are stated. The fold is $J \circ P_\perp$ and not J , and the embedding settles that: on the de Sitter hyperboloid in $\mathbb{R}^{1,4}$ the two factors compose to $-\text{Id}$ to machine zero, J alone has a three-dimensional $+1$ eigenspace whose restricted form is positive definite and therefore a whole two-sphere of fixed points, $P_\perp$ alone fixes a hyperbola, and a π rotation put in place of the parity fixes two points. A fold pairs two sheets, so a fixed point is a point that is its own mirror and none is allowed; total inversion is the only one of the four with no $+1$ eigenvalue, and its differential on the tangent space has trace -4 . Scaling one transverse direction by 1.1 inside the reflection breaks the isometry by 0.210, so the enumeration is over isometries and not over maps. What travels with the choice being forced is that the state is the

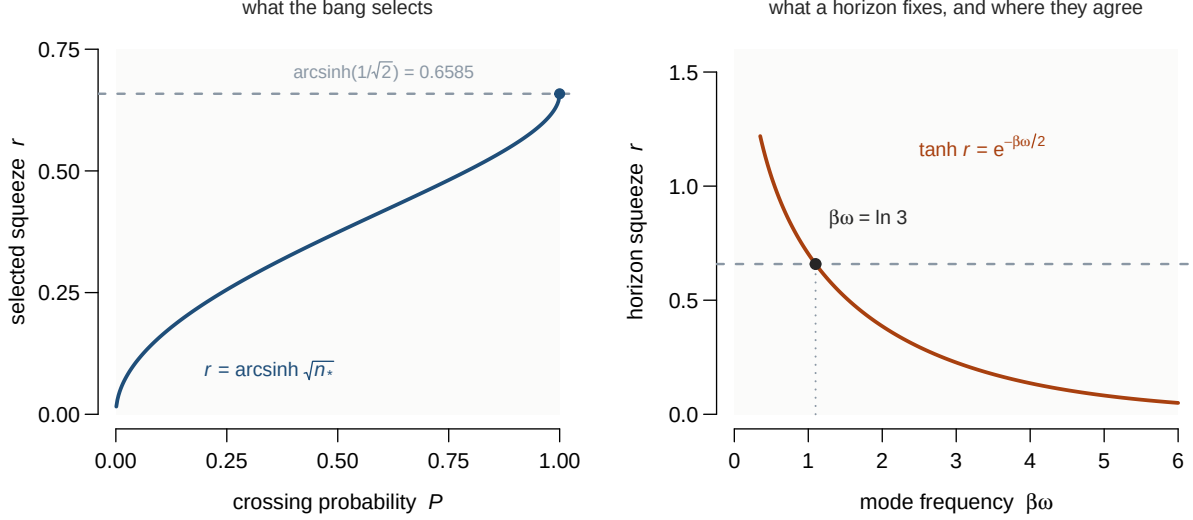


Figure 14: The two ends of the fold, and the one frequency where they agree. Left: what the crossing at the bang selects. Fold-invariance forces half-and-half occupancy, so the pair-block floor is $n_*(P) = (1 - \sqrt{1 - P})/2$, and since a squeeze carries $\langle N \rangle = \sinh^2 r$ the selected occupation is a selected squeeze $r = \text{arcsinh} \sqrt{n_*}$, topping out at $\text{arcsinh}(1/\sqrt{2}) = 0.6585$ when the crossing is complete. Right: what equilibrium at a bifurcate horizon fixes, the half-period shift making the state a thermofield double with $\tanh r = e^{-\beta\omega/2}$. The dashed line is the same in both panels, and the horizon curve crosses it at $\beta\omega = \ln 3$ exactly, to machine zero. The reason is that both ends are statements about one half: Θ -invariance caps the crossing at half occupancy, a horizon mode is half occupied where the Bose factor is one half, and that is $e^{\beta\omega} = 3$. The curves meet because they say the same thing.

thermofield double carrying an extra transverse parity. The marginals on either sheet are unchanged while the cross-sheet term picks up $(-1)^\ell$ per multipole, which is why a stress evaluated at a single squeeze comes out with opposite signs at the pole and at the equator. And equilibrium at the horizon is not a theorem about every state, though it is a weak assumption, as the next paragraph shows.

The Hadamard condition, imposed on every state used here, makes the singular part of a two-point function a matter of geometry and the state-dependent part smooth, and the temperature comes from the singular part. On a patch of proper size r the state-dependent part reaches the ratio $W_{\text{reg}}/W_{\text{sing}}$ only at order r^2 : a thermal state at temperature T gives $\pi^2 r^2 T^2/3$, with the coefficient 3.2898681 against $\pi^2/3$ and the exponent 2.000000 fitted over five decades, and a classical background gives $4\pi^2 r^2 \phi_{\text{cl}}^2$ at 1.999972. Since the balance is already first order in the patch, an error of relative order $(rT)^2$ in the temperature falls one order below the equation returned.

Alter the singular coefficient, which is what leaving the Hadamard class does, and the ratio goes to the alteration rather than to zero, at exponent 10^{-4} or smaller. Two limits belong with that. It is a statement about a shrinking patch and not about a whole horizon, a stellar-mass hole in the present microwave background having rT near 3.57×10^6 ; and it quantifies the deviation rather than removing it, so a state with no KMS property at the horizon remains a state the fold has nothing to say about. A third qualifier belongs with those two and is a restriction on *which* horizons the result describes. The half-period shift is A.10's statement at a bifurcate Killing horizon, and a hole formed by collapse has none, no past singularity, no second exterior, no bifurcation surface. So the squeeze computed here is the squeeze at the cosmological horizon and at a two-sided hole, and Section 6 counts that class at one member. What the result does not do is assign a two-sheet correlation to the horizon of an ordinary astrophysical black hole, where the fold still pairs our sheet with the mirror but pairs our hole with a mirror hole rather than with a second exterior of its own.

One number joins this to the other end of the fold. The cosmology paper's crossing bounds the squeeze at the bang by $\text{arcsinh}(1/\sqrt{2}) = 0.65848$, and that is exactly the horizon's own value at $\beta\omega = \ln 3$, since $\sinh r = 1/\sqrt{2}$ gives $\cosh r = \sqrt{3/2}$ and therefore $\tanh r = 1/\sqrt{3} = e^{-\ln 3/2}$; the two agree to 8×10^{-22} . That agreement is no coincidence: the ceiling is half occupancy, the Bose factor reaches half at $e^{\beta\omega} = 3$, and the two curves meet there because they are the same statement. The ceiling is exactly one half, which is what Θ -invariance gives and nothing else chose. Modes softer than $T \ln 3$ sit above the bang's ceiling and harder ones below it. The two ends of the fold meet at one frequency, and the meeting is arithmetic.

The Hadamard step is short-distance, the image term is regular as $Z \rightarrow 1$; its only pole sits at $Z = -1$, which a pair of points reaches only on the horizon; so the whole coincidence singularity of W is A times the cover's, and fixing it to the cover's geometric value is $A = 1$, whence $B = 0$. Against that, the tested projected image prescription, $(A, B) = (1, 1)$ or $(\nu, \kappa) = (0, -\frac{1}{2})$, misses positivity by exactly $\frac{1}{4}$; it assigns zero variance to one quadrature, $\sigma_{\underline{x}} = 0$, minimum eigenvalue $\frac{1}{2} - 1/\sqrt{2} = -0.2071067812$ exactly, and by Reeh–Schlieder the offending test function can be taken supported inside one static patch, so the failure is already local. The Mottola–Allen α -vacua give the positive comparison family, with $A = (1+|c|^2)/(1-|c|^2)$ and $B = 2|c|\cos \beta/(1-|c|^2)$. A nonzero image coefficient requires $A > 1$, contrary to the Hadamard normalisation $A = 1$.

A.9 A selected time-symmetric response has zero net dissipative work. §3.3's vanishing, derived. The antipode is an isometry that reverses time orientation, so it exchanges the past and future light cones and therefore $G_{\text{ret}} \circ (A \times A) = G_{\text{adv}}$. The Wheeler–Feynman exchange [27] is also checked numerically, $\max |G_{\text{ret}}(Ax, Ay) - G_{\text{adv}}(x, y)| = 0.000\text{e}+00$ for four masses; hence **within the retarded/advanced affine family** the unique invariant combination is the even one. This does not make the complete quotient Green problem unique without boundary data. For two sources at separation R coupled through a stationary reciprocal mutual kernel K ,

$$W = \int dt [\dot{q}_2(t)\Phi_1(t) + \dot{q}_1(t)\Phi_2(t)], \quad \Phi_i = \frac{K * q_i}{4\pi R}.$$

For an even kernel, $K(t-t') = K(t'-t)$, and the integrand is $(\partial_t + \partial_{t'})[q_2(t)q_1(t')]K(t-t')$. Because $(\partial_t + \partial_{t'})K(t-t') = 0$, that integrand is the total derivative $(\partial_t + \partial_{t'})\{q_2(t)q_1(t')K(t-t')\}$; with matching steady-state periods or vanishing all-history boundary terms it integrates to $W = 0$. What vanishes is the symmetric derivative acting on the kernel. The derivative acting on the product does not. In frequency space, $\tilde{G}_{\text{sym}} = \cos(\omega R)$ is real and $\max |\text{Im } \tilde{G}_{\text{sym}}| = 0$; meanwhile $\frac{1}{2}\tilde{\Delta} = i \sin(\omega R)$ carries the dissipative part.

$$\begin{aligned} W(G_{\text{ret}}) &= -w, & W(G_{\text{adv}}) &= +w, \\ W(G_{\text{sym}}) &= 0, & W(\tfrac{1}{2}\Delta) &= -w = W(G_{\text{ret}}), \end{aligned}$$

for a single positive w fixed by the source parameters. The value of w carries no meaning on its own. What is exact is the zero, and that the two nonzero entries differ only in sign. Both are reproduced here for Gaussian sources at seven parameter choices, with $|W(G_{\text{sym}})|$ and $|W(G_{\text{ret}}) + W(G_{\text{adv}})|$ at 10^{-16} , and a kernel skewed away from even gives a nonzero answer, so the check is not vacuous. The retarded value alone is quoted to no digits here, because a number for it means nothing without the source width, separation and R it was calculated at; the structural statement is what the argument uses and it is exact. The exact zero for the selected symmetric kernel comes from the integration-by-parts argument. Nothing was rounded to reach it.

For a complete steady-state response that is this reciprocal even combination, the absorptive operator vanishes and so does **total cycle-averaged secular work**. This does not set instantaneous work, individual-source transfers or outgoing-wave amplitude to zero; nor does it include an independent homogeneous or environmental response. The commutator $G_{\text{ret}} - G_{\text{adv}}$ identifies the part the selected classical response omits, but symmetrising one kernel does not prove the full field theory classical. Weisberg and Huang report the observed-to-GR orbital-decay ratio for PSR B1913+16 as 0.9983 ± 0.0016 [8]; the zero-damping complete-response implementation is excluded by it.

A.10 The law is the KMS condition off the surface, and how far the horizon ratio carries. §3's law, derived, and how far it carries. The objection to meet first is that the reciprocal law is definitional. It holds for any involution and any positive kernel, so on B it constrains nothing. That objection is right, and it is right off B as well. With $R(x, y) = W(x, JP_\perp y)/W(x, Jy)$ and $P_\perp^2 = \mathbb{1}$, substitution alone gives $R(x, P_\perp y) = R(x, y)^{-1}$ at every separation, so the identity carries no information anywhere and no temperature enters it. What carries the argument off B is a match and never the identity, and the boost separation that has not yet dropped out is what leaves room for one. A.6 derives the image kernel $G_{\text{img}} \propto \text{sech}^2(Ht/2)$ from the geometry of the antipodal map on the central worldline, with no temperature anywhere in that derivation; the form is quoted here because what follows uses only the form. Requiring that geometric object to be a thermal half-period shift $W(t - i\beta/2)$ of the Bunch-Davies kernel then fixes β . The match holds to 4×10^{-17} at $\beta = 2\pi/H$ and fails by 0.115 ten per cent away on either side, a factor 2.6×10^{15} worse and symmetric in the two directions. Minimising the match itself over β , with $2\pi/H$ nowhere in the search, returns 6.2831853072; minimising the $\alpha^2 = 1$ condition instead returns 6.2832000828. Two conditions, two minimisations, one temperature. But it holds equally at every odd multiple $\beta = (2j + 1)2\pi/H$, because $\sinh(z - i(2j + 1)\pi/2) = (-1)^{j+1}i \cosh z$ and squaring removes the sign, so the match alone determines β only modulo odd multiples; checked to 3×10^{-16} at $j = 0, 1, 2, 3$ and failing at 25.6 at every even multiple. What picks the fundamental is primitivity, which the KMS condition already contains: a function of period $2\pi/H$ has period $3 \times 2\pi/H$ as well, and the KMS temperature is the primitive period. No match occurs below the fundamental, which is what primitivity requires and is checked. The temperature is returned by that match and is not inserted. Only then does $\alpha^2 = 1$ read $W(t - i\beta) = W(t)$, which is the KMS condition; and this last step is a consistency check rather than a derivation, since csch^2 has period $2\pi i/H$ however it was obtained. In the frequency domain the same statement is that $e^{\beta E/2}$ squares to the KMS factor $e^{\beta E}$: the fold's map is the square root of the KMS transformation, and the horizon temperature is returned.

None of that is new physics and it is not offered as any. α contains the wedge reflection, which is the boost through imaginary angle π , so $\alpha^2 = 1$ says a 2π Euclidean rotation is the identity, and obtaining a horizon temperature from smoothness is standard. The fold's $\mathbb{Z}_2$ is that condition, so the law's triviality on B is a constraint degenerating rather than a constraint missing, and the ninety-degree equality is the boundary value of something that fixes a temperature. What none of it touches is the reduction to W_B itself, which is the step this appendix rests on and does not prove. A.12 carries the rest of what the law reaches.

How far the horizon ratio carries. A.6 obtains $G_\alpha/G_J \rightarrow \tan^2(\gamma/2)$ at the de Sitter horizon for the conformally coupled scalar in the Bunch-Davies state. This section separates the part of that result which belongs to horizons from the part which belongs to that spacetime and that state, because the two are easily conflated.

Let B be the bifurcation surface of a bifurcate Killing horizon. The wedge reflection J fixes B pointwise; $P_\perp$ is a free involutive isometry of B and moves it; $\alpha = J \circ P_\perp$. The status of $P_\perp$ differs between the two settings and the difference matters. On de Sitter nothing is chosen: α is the antipodal involution of the cover, J is fixed by the wedge-reflection theorems, and $P_\perp$ is whatever their composition makes it, the $(-1)^\ell$ parity of the cosmology paper's §2.1. At a black hole the geometry selects it as well.

A free involutive isometry of a round B is an element of $O(3)$ with no $+1$ eigenvalue, so it is $-\text{Id}$ and nothing else. Kerr's B is not round. It depends on θ only through $\cos^2 \theta$, and its latitudes are separated by the pair of Gaussian curvature and circumference at every spin up to $a/M = 0.998$, which fixes its isometry group at $O(2) \times \mathbb{Z}_2$; of that group's eight involutions exactly one is free. Those two cases exhaust the stationary vacuum holes in four dimensions, and the count stays one beyond them: charge moves r_+ and nothing the argument uses, and on the round S^{D-2} the eigenvalue argument runs unchanged at every dimension, which is the range these contact results are stated over. A B with no equatorial symmetry admits no free involutive isometry at all, so the failure mode is absence and never ambiguity, and $P_\perp$ is an input nowhere the fold is applied. With both arguments at B the boost separation drops out of both kernels, and

$$\frac{G_\alpha}{G_J} \rightarrow \frac{W_B(x, P_\perp y)}{W_B(x, y)},$$

the state's own correlator restricted to B , at the reflected separation over the direct one. This uses no metric

on B , no choice of state and nothing about the ambient spacetime beyond the near-horizon form. It is why each kernel is singular exactly where its own map brings the arguments together, and that much is general.

This reduction rests on three properties and nothing else, and the rest of this appendix leans on it. Expand both kernels in transverse modes on B , carrying two labels rather than one, since assuming the modes diagonalise the state's kernel would be a symmetry assumption about the state and we have just said there is none:

$$G_J = \sum_{I,K} R_{IK}(\Delta\tau) Y_I(x) Y_K(y)^*, \quad G_\alpha = \sum_{I,K} R_{IK}(\Delta\tau) Y_I(x) Y_K(P_\perp y)^*.$$

Since $P_\perp$ is an isometry of B it acts on the transverse labels alone, and since it commutes with the boost, $R_{IK}(\Delta\tau)$ is one common matrix for both kernels, whatever it is. At $\Delta\tau = 0$, which is both points on B , $R_{IK}(0)$ is exactly what defines the restriction W_B , so the ratio is $W_B(x, P_\perp y)/W_B(x, y)$ and no diagonality was used to get there. Checked on a dense random kernel with more than half its weight off the diagonal: the reciprocal law holds to 2×10^{-16} on it, while the diagonal truncation of the same kernel gives a visibly different ratio, so the two expansions are not the same object and the narrower one was never needed. Specialise to a round B and the statement becomes visible by hand. There the antipodal insertion is $(-1)^\ell$, and the Legendre polynomials obey $P_\ell(-x) = (-1)^\ell P_\ell(x)$, so the two signs are the same sign and cancel term by term: the reflected sum at γ is the direct sum at $\pi - \gamma$ for every boost profile whatever, and three well-chosen ones prove less than that. Four profiles, one of them random, confirm it to 9×10^{-15} , which is roundoff. A round horizon is only the easy illustration. The argument above used no maximal symmetry, no round B , no particular state and no closed form.

The same check makes plain how little the identity itself demands. It holds on modes of the wrong parity and on involutions of B that are not free, because it follows from $P_\perp^2 = 1$ and nothing else. Freeness and parity carry the geometry downstream. The law itself needs neither. Of the three properties the third carries the content, namely that the $\Delta\tau \rightarrow 0$ restriction exists at all; for a Hadamard state it does away from transverse coincidence, which is where the ratio is evaluated, but a restriction to a codimension-two surface is not automatic and we do not prove it. The first two are geometry.

It is a limit and not an identity, and the boost dependence does not cancel away from B . At $\gamma = 110^\circ$ and $rH = 0.9999$ the B -form is already off by 3.1×10^{-4} at zero boost, and the error grows as $\cosh \Delta t$ from there, reaching a per cent by $\Delta t \simeq 4.8$.

That reduction has a rate, which matters because any horizon calculation is done at small but nonzero lapse. The boost separation enters both kernels only through the coefficient $\sqrt{(1-r_1^2)(1-r_2^2)} = N_1 N_2$, so it does not cancel *between* them; it vanishes in each separately as B is approached. What is left is then first order in N^2 , and its constant has a closed form,

$$\frac{G_\alpha/G_J}{\tan^2(\gamma/2)} - 1 = \frac{2 \cos \gamma (\cosh \Delta t + 1)}{\sin^2 \gamma} N^2 + O(N^4),$$

with Δt the static-patch time separation. The measured constants at $rH = 0.9999$ are what that formula says they should be: 12.1 at 60° and the same 12.1 at 120° , because $|\cos \gamma|/\sin^2 \gamma$ is even about a right angle; 61.8 at 150° ; 517 at 170° , where $\sin^2 \gamma$ is running out; and exactly zero at 90° . Fixing the one free boost from the 60° entry alone reproduces the other three to better than three per cent, with nothing left to adjust. With $N^2 \simeq d^2/16M^2$ for a proper distance d above a Schwarzschild horizon, a per cent needs $d = 0.4M/\sqrt{c}$ in terms of the N^2 coefficient c , which is 0.058 of a Schwarzschild radius at the moderate-angle coefficient, 0.025 at 150° and 0.009 at 170° where the fold kernel's own pole is being approached. Moderate angles are the forgiving case and the near-antipodal ones the tight case, by about a factor of two and then again by three. So the reduction is a controlled limit with a calculable correction, and the ninety-degree equality is exact at every radius and at every boost. A right angle sends $\cos \gamma$ to zero, which removes the transverse term from Z_J and Z_α at once, so the two kernels are the same function and their ratio is one identically. That angle alone carries no truncation error, which is a second reason to lean on it.

One more statement is general. Write $R(x, y) = W_B(x, P_\perp y)/W_B(x, y)$. Replacing y by $P_\perp y$ and using $P_\perp^2 = 1$,

$$R(x, y) R(x, P_\perp y) = 1,$$

identically. The proof is that displayed line and nothing else enters it, so the law holds for every state, every correlator, every metric on B and every free involution, since it uses only that $P_\perp$ is one. On a round B with an isotropic state $P_\perp$ carries γ to $\pi - \gamma$ and the law reads $R(\pi - \gamma)R(\gamma) = 1$, so $\log R$ is odd about $\gamma = \pi/2$ whenever $R > 0$. The law is exact by the displayed line, so the numerics that accompany it confirm an implementation and not a mathematical fact; they are reported at better than 2×10^{-14} across ten positive decreasing profiles at four radii and four hundred angles, and four thousand random configurations at unequal radii and nonzero time separation, in that spirit. The oddness needs $\log R$ to be a function it makes sense to expand: R has a zero at one coincidence and a pole at the other, so $\log R$ diverges at both ends, and it lies in $L^2(\sin \gamma d\gamma)$ exactly when those divergences are logarithmic. Hadamard states give power-law singularities and therefore $\log R \sim \log \gamma$, which is square-integrable; a state with an essential singularity there would not be, and the multipole statement below would not apply to it.

Three consequences follow that the pointwise statement did not supply. The unit value at $\gamma = \pi/2$ is the fixed point of the law, so it needs no isotropy argument and no state. $\tan^2(\gamma/2)$ satisfies the law, but that form is not universal: what is universal is the functional equation, and the conformal scalar's closed form is one solution of it. And the law fixes a sign across the whole surface rather than at one angle. For any correlator decreasing in separation, $R < 1$ wherever x lies nearer y than $P_\perp y$ and $R > 1$ where it does not, so the fold suppresses correlations on the near side of the involution's equidistant locus and enhances them on the far side, with the crossing exactly on that locus. Magnitude belongs to the state and the spacetime; the sign and the locus do not.

How far the law reaches, what it measures when a state falls short of its own fold image, and what it says about the closed-de Sitter overlap's branch weights are taken up separately in A.12, so that this appendix can return to the question it set, how much of that value survives.

The value at a given angle is not universal. On a round B with an isotropic state the ratio is $G(\pi - \gamma)/G(\gamma)$, and three features of it hold for every such state: it vanishes like γ^2 , it equals exactly one at $\gamma = \pi/2$, and it has a $(\pi - \gamma)^{-2}$ pole. The first and third are the Hadamard singularity at each coincidence; the second needs only that a point at $\pi/2$ from x has its antipode at $\pi/2$ as well. Nothing beyond those three is universal. Three profiles with the same short-distance behaviour at both coincidences give 3.00, 4.00 and 2.75 at $\gamma = 2\pi/3$. They are the chordal form $1/(1 - \cos \gamma)$, the arclength form $1/\gamma^2$, and a chordal form carrying an asymmetric finite-separation factor (log-slopes +2.000 and -2.000 at the two ends for all three, and unity at $\pi/2$ to twelve digits). The first of them gives $\tan^2(\gamma/2)$, exact on de Sitter because that state's two-point function is exactly $1/(1 - Z)$ at every separation, which is a consequence of maximal symmetry and not of the horizon.

It is therefore not established for the Hartle-Hawking state on Schwarzschild. The near-horizon route obtains it there by expanding about $r = 2M$ to Rindler $\times S^2$ and using a flat transverse chord; the chord approximates the separation on a curved sphere only at small angle, so that route does not reach the angles the law is read at, and it is not a derivation. What survives on Schwarzschild is the general ratio above and the three universal features, all of which $\tan^2(\gamma/2)$ happens to satisfy.

Kerr keeps the structure and loses the closed form. Its bifurcation surface carries

$$ds^2 = \rho_+^2 d\theta^2 + \frac{(r_+^2 + a^2)^2 \sin^2 \theta}{\rho_+^2} d\phi^2, \quad \rho_+^2 = r_+^2 + a^2 \cos^2 \theta,$$

which is axisymmetric and even in $\cos \theta$, so $P_\perp : (\theta, \phi) \mapsto (\pi - \theta, \phi + \pi)$ is an exact isometry, free and involutive, checked to 1.8×10^{-15} . The general ratio therefore applies. But B is squashed, with $g_{\theta\theta}(\text{pole})/g_{\theta\theta}(\text{equator}) = 1 + a^2/r_+^2$, equal to 1.024, 1.072, 1.167, 1.393 and 1.753 at $a/M = 0.3, 0.5, 0.7, 0.9$ and 0.99 and to unity at $a = 0$. Isotropy fails, and with it any closed form and the unit value *at that angle*. The reciprocal law survives,

because it needs only the involution and Kerr keeps that, and it still holds to 6×10^{-15} at $a/M = 0.3, 0.7$ and 0.95 . What the law then locates is the set where $W_B(x, P_\perp y) = W_B(x, y)$, and identifying that set with the curve equidistant from y and $P_\perp y$ requires the state's restriction to B to depend on those points only through geodesic distance there. Given that, the locus is a deformed curve in place of a great circle, and six unrelated distance-only profiles return $R = 1$ to twelve digits at the same point on it. That check confirms the geometry and not the state.

State dependence demands less than it might seem. $P_\perp$ factors as a rotation by π about the axis composed with the equatorial reflection $\theta \mapsto \pi - \theta$, and Kerr has both symmetries, so an axisymmetric equatorially symmetric restriction is $P_\perp$ -invariant without that having to be assumed; kernels built from the invariants such a state may depend on are invariant to 10^{-15} over 2×10^4 random pairs. Then $W_B(x, P_\perp y) = W_B(P_\perp x, y)$ and the locus where $R = 1$ exists and is well defined, it is the set on which $P_\perp x$ and x correlate equally with y .

What does fail is its identification with the *geodesic* equidistant curve, and the reason is a count rather than a computation. On a round sphere $SO(3)$ acts transitively on pairs at a fixed separation, so the space of pairs modulo isometry is one-dimensional and every invariant is necessarily a function of distance. Spin breaks that group to the rotations about the axis and the equatorial reflection, the quotient becomes three-dimensional in $(\theta_1, \theta_2, \Delta\phi)$, and distance is one function on it whose level sets are surfaces. An invariant kernel is another such function, and nothing makes it constant on those surfaces. So spin moves the locus off the equidistant curve, and computing where it goes needs the state and not only the metric. That is the gap that remains. The departure is switched on by spin alone and vanishes identically for a non-rotating hole. A.13 shows it is not a separate fact.

A.11 Transparent matching is adopted, and at a codimension-one interface the corner term is the only seam carrying no scale. §4's boundary condition, and why it is adopted. One thing holds and it is narrow. Among quadratic boundary terms at the specified codimension-one interface, the corner term is the one that can carry a dimensionless coefficient and still act. That is not a classification of actions on B itself, which is codimension two, where $\int_B d^2x \phi^2$ already has a dimensionless coefficient for a four-dimensional scalar. Within that action r cannot depend on frequency, and unitarity fixes the matching to $r(\kappa) = (1 - \kappa^2)/(1 + \kappa^2)$, $t(\kappa) = 2\kappa/(1 + \kappa^2)$.

No derivation of $\kappa = 1$ is offered. No symmetry of the fold fixes it and no regularity condition does. What selects the transparent point is a flux condition on one stated premise, and a variational criterion that is soft: the transparent seam maximises the entanglement across it, and a five per cent reflection retains 99.7 per cent of that. So $\kappa = 1$ is adopted, with the corner term's uniqueness among scale-free seams behind it and a maximum rather than an exclusion.

Parametrise the seam by κ in the corner term $(\kappa/2)q^T J \dot{q}$, where $q = (q_1, q_2)$ are the two sheets' field values at the join and J is the antisymmetric unit. The reflection and transmission that term produces are not quoted from anywhere; they follow from it. Varying $S_c = (\kappa/2) \int dt (q_1 \dot{q}_2 - q_2 \dot{q}_1)$ and integrating by parts in t leaves $\kappa(\dot{q}_2 \delta q_1 - \dot{q}_1 \delta q_2)$, which against the bulk variation $-\phi' \delta \phi$ gives the matching conditions $\phi'_L(0) = \pm \kappa \dot{q}_2$ and $\phi'_R(0) = -\kappa \dot{q}_1$. With $\phi = e^{ikx} + r e^{-ikx}$ on the left and $t e^{ikx}$ on the right at frequency $\omega = k$, these read $1 - r = \kappa t$ and $t = \kappa(1 + r)$, whose unique solution is

$$r(\kappa) = \frac{1 - \kappa^2}{1 + \kappa^2}, \quad t(\kappa) = \frac{2\kappa}{1 + \kappa^2}, \quad r^2 + t^2 = 1.$$

The other sign gives $r = (1 + \kappa^2)/(1 - \kappa^2)$, which is a different branch altogether and fails unitarity: at $\kappa = \frac{1}{2}$ it returns $r = 5/3$ and $t = 4/3$, so $r^2 + t^2 = 41/9$, and the violation grows with κ . Unitarity therefore selects the sign, where a convention would have left it free. A full two-channel scattering matrix with both bulk actions, normals and incoming channels specified is not constructed here. Both expressions and the unitarity are verified against the matching condition at $\kappa = 0, \frac{1}{4}, \frac{1}{2}, 1, 2, 4$ to machine precision. So $\kappa = 1$ is the transparent point, $r = 0$ and $t = 1$, and $\kappa = 0$ and $\kappa \rightarrow \infty$ are the two perfect mirrors.

Because κ is dimensionless and the corner term carries no other scale, r and t cannot depend on frequency. That is the whole problem. A seam reflecting with the same amplitude at every frequency, including arbi-

trarily short wavelengths, is not a physical interface, real interfaces become transparent in the ultraviolet. The corner term as written therefore needs a scale it does not have.

No symmetry of the fold fixes κ . Demand that the fold be a symmetry of the seam, on the ground that the seam is the fold locus and cannot be less symmetric than the map defining it. That demand holds at every κ and therefore selects nothing, and here is why in one line. The fold does two things: it exchanges the sheets, and it reverses time orientation. The corner term is antisymmetric in the sheet labels, so the exchange flips its sign; it is first order in a time derivative, so the reversal flips it back. Those two flips cancel for any coefficient, which is checked directly on profiles with no parity of their own, with each half of the fold applied alone as the control and each half flipping the sign as it must.

What the same action says about the fluxes does select. Vary the two bulk actions and the corner term together at the join, with ϕ_L on $x < 0$ and ϕ_R on $x > 0$ and opposite outward normals. The matching conditions are $\partial_x \phi_L(0) = \kappa \dot{q}_2$ and $\partial_x \phi_R(0) = \kappa \dot{q}_1$, and the scattering matrix that follows is

$$S = \begin{pmatrix} r & t \\ -t & r \end{pmatrix}, \quad r = \frac{1 - \kappa^2}{1 + \kappa^2}, \quad t = \frac{2\kappa}{1 + \kappa^2},$$

a rotation with $\det S = 1$. Under the fold law $E \bar{S} E = S^\dagger$, with E the exchange of sheets and the dagger carrying the exchange of incoming and outgoing channels, that matrix is invariant at every κ , which is the same no-selection result seen on the matrix.

Now take the two fluxes across the join, from the same matching conditions. The energy defect $T_{tx}(0^-) - T_{tx}(0^+)$ is identically zero at every κ , the seam conserves energy for free. The momentum defect is not,

$$T_{xx}(0^-) - T_{xx}(0^+) = \frac{1}{2} (1 - \kappa^2) (\dot{q}_1^2 - \dot{q}_2^2),$$

so for general data the seam pushes, and the push vanishes for arbitrary data at exactly $|\kappa| = 1$. That is the transparent member. The coefficient is $3/8$ at $\kappa = \frac{1}{2}$ and $-3/2$ at $\kappa = 2$, and a self-coupled matching in place of the cross-coupled one gives $\frac{1}{2}(1 + \kappa^2)(\dot{q}_1^2 - \dot{q}_2^2)$, which never vanishes, so the selection belongs to the cross coupling rather than to the counting.

What that momentum is, written in the variables the cosmology paper's §2.1 splits fields into, takes a line, because it is not what the raw expression looks like. Putting $q_1 = \Phi_c + \Phi_q/2$ and $q_2 = \Phi_c - \Phi_q/2$ into the defect gives

$$T_{xx}(0^-) - T_{xx}(0^+) = (1 - \kappa^2) \dot{\Phi}_c \dot{\Phi}_q,$$

the classical velocity times the quantum one. So the seam's force is a classical-quantum cross term. It vanishes when the two sheets agree, because then $\Phi_q = 0$; it vanishes when the classical field is static; it vanishes identically at the transparent point; and it has no long-range tail, since it exists only where the sheets meet. Whatever else the fold does, this is the one interaction in the fold that is not gravity and does not reach.

An interface that absorbs no momentum is one whose stress tensor is continuous across it, which is what distinguishes a topological interface from a material one. So the family has exactly one topological member and the paper adopts it. That is not a derivation of $\kappa = 1$ from the fold, and the difference needs stating exactly, the fold does not imply that its seam is topological. What is established is that if the seam is topological then $\kappa = 1$ uniquely, so the adoption rests on a stated physical premise with a name rather than on a choice of number. Reject the premise and one sentence goes; the rest stands.

No symmetry of the fold can fix the coefficient, so a derivation would have to come from dynamics or from a completion of the corner term, which is ultraviolet-incomplete.

Regularity does not fix the coefficient either. One might argue that a constant reflectivity puts an image term in the two-point function which diverges between mirror-placed points, that no Hadamard state is singular

at spacelike separation, and so that r must vanish. The mode sum can be counted two ways and neither supports it. Sum both incident channels of the two-port, as completeness of the mode basis requires, and the cross-seam image coefficient is $rt + tr'$; unitarity of $S = \begin{pmatrix} r & t \\ t & r' \end{pmatrix}$ forces $r' = -r$ whenever t is nonzero, so that coefficient is identically zero at every κ , checked by reading it off the mode functions at $\kappa = \frac{1}{4}, \frac{1}{2}, 1, 2, 4$ and returning 10^{-17} each time. The alternative $r' = +r$ leaves $2rt$ behind, which is the plant that makes the zero mean something. Project the mode space instead, as a $\mathbb{Z}_2$ quotient does, and the coefficient becomes $(t^2 - r^2)/2$: nonzero, and *largest at the transparent point*, where it is exactly $\frac{1}{2}$ against -0.2785 at $\kappa = \frac{1}{4}$ and zero at $|r| = |t|$. So counted the first way nothing is singular at any κ , and in the second the transparent seam is the worst offender, where the argument needs it to be the only clean one. The conclusion does not depend on which count is right.

One criterion does select the transparent point, and it is variational. A free-fermion chain with a defect bond is the same two-port scattering problem, with transmission $T = 4t_d^2/(1 + t_d^2)^2$, in a setting where entanglement is computable exactly. Total entropy across the defect is *not* a function of T . A strong bond builds a dimer across the cut whose local contribution swamps the transmission-dependent part, and $t_d = 0.5$ and $t_d = 2$ give entropies differing by more than a factor of two at identical T .

The coefficient of $\log L$ is a function of T and of nothing else. A bond defect and a site defect tuned to the same transmission perturb the chain differently and return the same coefficient to within 0.63 per cent, at $T = 0.95, 0.88, 0.64$ and 0.30 ; the $T = 0.88$ pair is 0.15363 against 0.15460 . Defects at *different* transmissions differ by tens of per cent, so the agreement is not an artefact of the fit. The absolute coefficient depends on the chain's boundary conditions and on how many cut points the entropy counts; what is comparable between two defects is the percentage, and that is what the claim uses. These are finite-size fits and not a proof of universality. That coefficient is maximal at $T = 1$, with both neighbours below it. Its symmetry under $t_d \rightarrow 1/t_d$ is inherited from the symmetry of T under that map, so it is the seam's rapidity inversion seen again. The transparent seam is therefore the one that maximises the entanglement between the two sides, which for a framework claiming the sheets are joined is the natural way to say how strongly.

Selection here is soft and should not be oversold: a five per cent reflection retains 99.7 per cent of the coefficient, so this identifies the maximum without excluding its neighbourhood, and nothing above excludes that neighbourhood either. And the lattice shares only the scattering problem; the dynamics differ, and the entropy calculated is the chain's.

The classification reaches further than the constant case, which needs showing because the obvious objection is that it does not. A seam might be non-linear, with a reflectivity that vanishes at small field amplitude and switches on above some A_0 ; such a seam is transparent wherever the field is weak, and it looks like a way of having a reflecting seam without introducing a scale. It is not, and the reason is that short distance is not small amplitude. Vacuum fluctuations diverge as the separation closes, the amplitude a probe at separation s samples running as $1/2\pi s$, so for any finite A_0 the seam is switched on below $s_* = 1/2\pi A_0$ and its reflectivity tends to one as the separation shrinks further, 0.99 at $s_*/10$ and 0.9999 at $s_*/100$, independently of A_0 . So A_0 is a scale and the seam is a scaled seam whatever the field-dependence looks like at weak field, which is the case already covered, it falls outside the corner term and needs its own matching law.

Scale dependence is real and runs opposite to the intuition that suggests it. A seam reflecting at large amplitude is transparent in the infrared and reflecting in the ultraviolet. It is not a seam that is transparent where we can look and active only near a singularity. That is the expected sign for a boundary coupling carrying a dimensionful scale, which is irrelevant in the infrared and grows toward the ultraviolet, and it is why an ultraviolet condition catches it. Ringdown sits at $0.374 c^3/2\pi GM$, which is 41 orders of magnitude below a Planck seam scale, 37 below a grand-unified one, 30 below the 492 PeV mass of the cosmology paper's §3.1 and 24 below a TeV.

Write the response as $r(\omega/\Lambda_P)$, with Λ_P the Planck scale or any other microscopic one in its place; the letter is subscripted throughout this appendix to keep it clear of the cosmological constant, which is Λ everywhere else in this paper. Astrophysical ringdown sits at $\omega/\Lambda_P \simeq 4 \times 10^{-40}$ for $10M_\odot$ and 4×10^{-48} for $10^9 M_\odot$, which is the deep *infrared* of that function and not its ultraviolet. A condition forcing $r \rightarrow 0$ at large argument constrains nothing there, $r(x) = e^{-x}$ obeys it and equals one to forty significant figures at these arguments. A microscopic scale therefore does not make a seam invisible at ringdown, and no statement

about astrophysical waveforms follows from an ultraviolet condition alone.

The implementation adopted here is transparent at every frequency by choice, which is a different matter and is unaffected.

The corner term is not one choice among many. The action supplies a constant reflectivity by carrying no scale of its own, so the question is which boundary terms on B can carry a dimensionless coefficient at all. At quadratic order there is one. A scalar has dimension $(d-2)/2$ and a boundary density must have dimension $d-1$, so at $d=4$ only terms with exactly one derivative qualify, namely $q\dot{q}$ and $q\nabla q$. The second is $\nabla(q^2)/2$ and integrates to zero on a closed bifurcation surface. The symmetric part of the first is $\frac{1}{2}d(q^2)/dt$ and drops out of the matching condition, which we check directly; the antisymmetric pairing does not and is the corner term. So on a codimension-one matching surface “a seam carrying no scale of its own” and “the corner term” name the same object at this order. On B itself, which is codimension two, the count runs the other way: $g\int_B d^2x\phi^2$ has a dimensionless g , so the counting does not extend there: a codimension-two defect does not scatter like a delta in one normal coordinate, needs its own regularisation, and can generate logarithmic scales of its own. Relatedly, $r(\omega) \rightarrow 0$ is not by itself Hadamard regularity, since removing a pointwise divergence does not remove a wavefront singularity, and a proper microlocal treatment is not attempted here.

So the statement to make is narrow and is stated narrowly. The dimensional counting above concerns a specified codimension-one interface and does not classify every action that could live on the bifurcation surface.

Seams compose. Writing $\kappa = e^u$, the family’s transfer matrix is $\begin{pmatrix} \cosh u & \sinh u \\ \sinh u & \cosh u \end{pmatrix}$ to 1.8×10^{-15} , so seams in series add rapidities, $\kappa \rightarrow 1/\kappa$ is the group inverse, and reversing the crossing sends $u \rightarrow -u$. Requiring the seam to look the same from either sheet does not pick $u = 0$. The two channels are the sheets, so the exchange is $E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which *anticommutes* with the corner term’s complex structure $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Under the exchange together with the time reversal the fold’s map carries, $q \rightarrow Eq$ and $\dot{q} \rightarrow -E\dot{q}$, so $(\kappa/2)q^T J\dot{q}$ returns to itself for every κ , verified on 2000 random $(q, \dot{q}, \kappa)$.

The group has an identity, and it is the transparent seam. The composition law carries a consequence. A one-parameter group has an identity element, and here it is $u = 0$, which is $\kappa = 1$. The transparent seam is the member that composes with any other seam and leaves it where it was, and it is the only member that does. That is what having no seam means as an operation on the theory. So adopting $\kappa = 1$ adds no boundary term to the action, which is the choice the minimal implementation makes wherever else the fold leaves room, in taking a linear fold map and a Gaussian free state. A reflecting seam is an addition needing a matching law of its own, and the question to press is what would require one. None of this derives $\kappa = 1$. A theory may add a boundary term, and an identity element is privileged by minimality and by nothing else. What it settles is where the burden sits. Adopting the identity is adopting nothing, and that is a lighter thing to have to defend than a choice among a continuum.

If the seam carries the horizon’s own scale. One alternative remains. r cannot depend on frequency *because the corner term supplies no scale*, and a horizon supplies several. A seam matched at the horizon may carry T_H , and then $r = r(\omega/T_H)$ with $r(\infty) = 0$ satisfies Hadamard regularity without being transparent. That is a different action needing its own matching law, but it is not free, because two predictions follow before any amplitude is chosen. Both $M\omega$ and MT_H are dimensionless functions of a/M alone, so ω/T_H depends on spin and not on mass: such a seam reflects identically for a $10 M_\odot$ and a $10^9 M_\odot$ hole at the same spin, which is a null test across eight orders of magnitude available from ringdowns in matching spin bins. And with

$$\frac{\omega}{T_H} = 4\pi (M\omega) \frac{1 + \sqrt{1 - a^2}}{\sqrt{1 - a^2}},$$

equal to $8\pi M\omega$ at $a = 0$ and diverging at extremality, spinning the hole up pushes the ringdown to larger argument, where r is smaller. Horizon reflectivity effects are therefore largest at low spin and vanish as extremality is approached *for a response depending on ω/T_H alone*, which §4’s co-rotating case is not, since there the amplitude rises from 9.8×10^{-3} at zero spin to 0.444 at $a = 0.9$. An asymptotic falloff does not

imply global monotonicity, and a rotating response can depend on $(\omega - m\Omega_H)/T_H$. What follows is therefore a property of the assumed response law, for every fall-off; three unrelated fall-off laws give the same monotone ordering. Echo models with a reflecting wall at fixed proper distance predict the opposite trend, since the light-crossing delay grows as the horizon is approached. The two are separated by the sign of the trend with spin, with no amplitude required from either. The quasinormal frequencies themselves are not computed here; the ratio above is carried by T_H , which varies by a factor 8.4 between $a = 0$ and $a = 0.998$ while $M\omega$ varies by a factor of a few, so the trend does not depend on values we have not calculated.

Those two predictions are not equally robust, and the difference matters. Mass independence survives any choice of borrowed scale, because every scale a horizon can lend, the temperature, the horizon radius, the square root of the area, the angular velocity, the curvature, is a power of the mass times a function of spin, so the mass cancels identically from the dimensionless combination whichever is taken. The vanishing toward extremality does not: it belongs to the temperature, which vanishes at extremality and changes by a factor of twelve across the spins sampled, where the radius and the area change by less than two and give a nearly flat trend instead. Attaching the extremality behaviour to any horizon scale is therefore wrong. So the spin dependence discriminates among reflecting implementations built from horizon scales, while the mass independence is common to all of them.

We do not establish that the scale must be the horizon's, and the obvious reply is one word long, Planck. A seam could carry a scale of external origin, and such a seam is neither transparent nor built from the horizon. It does not escape the statement, but it changes it from a prohibition into a classification. An external scale far above the horizon's puts the ringdown at ω/Λ_P between 4×10^{-40} and 4×10^{-48} across astrophysical masses. The reciprocal quantity $M\Lambda_P$ runs from 10^{38} to 10^{47} and is not the argument of the response. These arguments are the deep infrared, an ultraviolet falloff condition does not constrain a response there. An external scale *comparable* to the horizon's does show up, as a dependence on mass at fixed spin, and it would then be measuring that scale rather than refuting anything; for that to be visible the scale must sit between about 10^{-11} and 10^{-20} eV, where nothing in particle physics currently sits.

So what the seam analysis gives is a sorting of cases. Reflectivity mapped across mass and spin separates three cases. No variation at any mass or spin means a scale-free seam, or any scale far above the horizon's, and does not distinguish between them. Variation with spin alone means a horizon scale, which is the fold's own branch, with the trend fixed by which scale it is. Variation with mass at fixed spin means a scale of external origin near the horizon's. That is a measurement programme, and it requires no amplitude.

A.12 How far the reciprocal law reaches. A.10 establishes the law and its immediate consequences. This appendix collects what it reaches beyond the bifurcation surface. Its algebraic form, the measure it supplies for a state falling short of its own fold image, and what that measure says about the branch weights of §2.1.

The law does not need a manifold, and saying so settles how much of this belongs to horizons and how much to the fold. For a state ω on an algebra carrying an involutive automorphism σ , write $R(A, B) = \omega(A\sigma(B))/\omega(AB)$; the same substitution gives $R(A, \sigma B) = 1/R(A, B)$, needing only $\sigma^2 = \text{id}$. Checked on random density matrices with random involutive automorphisms in eight dimensions over four thousand operator pairs, to 4×10^{-14} , and the check is validated by substituting a generic unitary conjugation for the involution, which breaks it at order unity. The geometric statement above is one instance.

Once the law is algebraic, the natural question is how far a state falls short of being its own fold image, and the measure is the relative entropy $S(\omega \| \omega \circ \sigma)$. Two properties of it belong to the involution rather than to relative entropy generally. It is symmetric: relative entropy is invariant under a common automorphism, so $S(\omega \circ \sigma \| \omega) = S(\omega \circ \sigma \| (\omega \circ \sigma) \circ \sigma) = S(\omega \| \omega \circ \sigma)$ once $\sigma^2 = \text{id}$, and four hundred random state-and-involution pairs return an asymmetry below 3×10^{-12} against a median divergence of 1.07. And it is second order in the misalignment: mixing a fold-symmetric state toward a generic one by a fraction ϵ , the divergence grows as ϵ^2 , the log-log slope running 1.913, 1.953, 1.980, 1.993 and 1.998 as ϵ falls from 10^{-1} to 3×10^{-4} . It vanishes only at exact fold symmetry. Fold-symmetric states are therefore a stationary point of this divergence and not merely its zero set.

Second-order behaviour is the part with consequences, because it fixes how misalignment accumulates. Independent departures of size ϵ over N events sum to order $N\epsilon^2$ rather than $(N\epsilon)^2$, so the accumulation is

diffusive, and a universe-wide tally comes out far smaller than a naive addition of amplitudes suggests. We do not compute ϵ for any process and nothing in the fold supplies it, so this is a scaling only; turning it into one means calculating the misalignment a definite process produces, for which the branch decoherence of §2.1's closed-de Sitter overlap is the natural first case.

Applying it to that case gives a definite answer. The two branches of §2.1 carry environment states of overlap s ; writing the branch-label state as $a|+\rangle|E_+\rangle + b|-\rangle|E_-\rangle$ and tracing out the environment gives a two-by-two density matrix with diagonal $(|a|^2, |b|^2)$ and off-diagonal carrying s . Because Θ is antilinear *and* exchanges the branches, $\Theta\rho\Theta$ is ρ with the two weights swapped. The misalignment therefore turns on the branch weight and not on the coherence, and vanishes identically at equal weight, at every overlap.

Decoherence regulates the divergence without causing it. At unequal weight a coherent superposition is pure, and a pure state has infinite relative entropy against anything off its support, while the fully decohered value is finite and closed form, $S = (2p - 1) \log[p/(1 - p)]$, which is $8(p - \frac{1}{2})^2$ to leading order and reproduces the coefficient the general argument gave. And the timing is early: using §2.1's own overlap with $aH = \cosh Ht$, which returns its quoted $Ht = 1.28984$ at $aH = 1.95376$, the misalignment has settled onto its decohered value by the threshold.

Equal weight at the fold point does not propagate. Θ is antiunitary, and CPT covariance gives $\Theta U \Theta^{-1} = U^{-1}$, which is not $[\Theta, U] = 0$, so a state that is Θ -invariant at the fold point is not Θ -invariant afterwards, $\Theta|\psi(t)\rangle = |\psi(-t)\rangle$. The branch imbalance $\langle N \rangle$ is therefore an odd function of time, zero at the fold point and nonzero away from it, checked to 2×10^{-15} on a Hamiltonian built to satisfy $\Theta H \Theta^{-1} = H$.

What the oddness means is more useful than what it removes. $\langle N \rangle$ is neither conserved nor zero on either sheet; what vanishes exactly is the sum over the pair, since the mirror sheet's value at its time t is ours at $-t$. The two sheets carry equal and opposite branch imbalance at every time and the pair carries none. That is a joining of a kind not otherwise in this paper. Not a force between the sheets and not a shared boundary condition, but a conserved quantity that exists only when both are counted.

The divergence should be read accordingly. $S(\omega \| \omega \circ \sigma)$ is not a defect accumulating with cosmic time; the two sheets are meant to differ. What is constrained is the antisymmetry, and that is exact. No rate is delivered: the growth depends on the Hamiltonian, and the one used here is a random CPT-covariant matrix. The closed-de Sitter overlap's is a different Hamiltonian. The shape is established; the number is not.

That last point needs a qualification, and it is the one that applies to §2.1's closed-de Sitter overlap. The calculation assumed a state evolving under a CPT-covariant Hamiltonian. Wheeler-DeWitt supplies a constraint solution rather than an evolving state, and the no-boundary wavefunction is real, $\psi \sim \cos S = \frac{1}{2}(e^{iS} + e^{-iS})$, so the two branch amplitudes are complex conjugates, their moduli are equal identically at every aH , and there is no time dependence to track. In the setting of §2.1 the imbalance vanishes for that reason and not because of CPT.

Those two conditions differ, and the stronger one is imported. CPT covariance gives an imbalance odd in time and generally nonzero; reality of the wavefunction gives zero identically. Reality is not, however, only a choice of cosmological boundary condition: the paragraphs below obtain it from Θ -invariance instead, and the cosmology paper's A.4 states the commitment that makes that available. Preferring a tunnelling wavefunction therefore gives up the no-boundary route to reality. Reality itself survives, provided they keep Θ -invariance of physical states. What survives either way is the cross-sheet statement, since $\langle N \rangle(t) + \langle N \rangle(-t) = 0$ holds trivially when both vanish. The conservation law is therefore the more robust of the two claims, and the equal weight on a single sheet is the weaker.

That weaker claim can be improved, though not to a theorem. Reality of the wavefunction need not be borrowed from the no-boundary proposal, because Θ is antilinear and imposing $\Theta|\psi\rangle = |\psi\rangle$ is a reality condition. Writing $\Theta = K \circ S$ with K complex conjugation and S linear, $\Theta^2 = 1$ requires $SS = 1$, and in the branch basis with S the branch swap the condition reads $\psi_- = \bar{\psi}_+$, which is the no-boundary form and forces $|\psi_-| = |\psi_+|$. Equal branch weight and Θ -invariance are the same condition. The restriction is well defined. The Wheeler-DeWitt operator is real, so its solution space is closed under conjugation, checked directly, and the invariant part carries exactly half the real dimension, measured as the rank of $(1 + \Theta)/2$ applied to a complex basis at $d = 1, 2, 3, 5$. That is the same halving A.16 obtains at the singularity through $(1 + JU)/2$, two loci, one mechanism, an involution keeping its even sector.

What this changes is where the assumption comes from. Its status is unaffected. It need not be borrowed from Hartle and Hawking and can be stated in the fold's own terms. Two things it does not do. It still needs the physical state to solve the constraint instead of evolving, since Θ -invariance is not preserved under evolution, and timelessness is a feature of the formulation, which no theorem supplies. And the cosmology paper's C.2 gauges the *linear* involution $P_\perp$, following Harlow and Numasawa; gauging an antilinear one is a different operation which we do not perform. So Θ -invariance is a condition the fold may adopt naturally, and not one the fold already carries.

That reach matters for the horizons A.10 cannot otherwise touch, and it also marks where a natural repair fails. One might hope to move from B to a late-time cross-section, since the reduction needed only $\Delta\tau = 0$ and a cross-section supplies that by construction. It does not work: J fixes B pointwise and fixes nothing else on the horizon, carrying every cross-section of H^+ to a cross-section of the mirror sheet's H^- , so the two arguments no longer share a surface and the ratio is a correlator between two horizon branches rather than one restricted to a single surface. The reciprocal law itself is untouched by that, since both kernels send their second argument to the same place and differ only by the transverse reflection. The obstruction was never the absence of B ; it is the absence of the mirror sheet. So the algebraic law transfers to a collapse hole, whose settled exterior is stationary and axisymmetric and whose transverse antipodal map is an automorphism of the exterior algebra, while $\alpha = J \circ P_\perp$ as the second Keldysh leg does not, because no algebra supplies a mirror sheet.

An obvious objection is that the law is definitional, it holds for any involution and any positive kernel, so it constrains nothing. That is correct on B and, as A.10 records, off it too, the identity being algebraic at every separation; and the two are one equation in different limits. Off B the boost separation has not dropped out, and A.6's identification makes α act as the half-period continuation $W(t - i\beta/2)$, which reproduces the image kernel there to 4×10^{-17} . Then $\alpha^2 = 1$ reads $W(t - i\beta) = W(t)$, and that is not free: sweeping the shift period, the condition fails by $O(25)$ at every value tried and holds only at $\beta = 2\pi/H$ and its multiples, while minimising the residual returns 6.2831853072 against $2\pi/H = 6.2831853072$ from three of four sample times, the fourth a boundary artefact of the search window carrying a residual of 45. In the frequency domain the same statement is that $e^{\beta E/2}$ squares to the full KMS factor $e^{\beta E}$, so the fold's map is the square root of the KMS transformation.

A.10 sets out why none of that is new physics and what it leaves untouched: the reduction to W_B itself, which is the step A.10 rests on and does not prove. That limit carries to everything collected here.

A.13 A latitude-dependent horizon response mixes ℓ , by two for a scalar and by one for a graviton. §4's second reflecting signature. The squashing of A.10 is one function, and it has a second consequence that is easier to look for. Write the local anisotropy of the bifurcation surface, azimuthal proper length per unit coordinate over meridional, as

$$A(\theta) = \frac{\sqrt{g_{\phi\phi}}}{\sin\theta \sqrt{g_{\theta\theta}}} = \frac{r_+^2 + a^2}{r_+^2 + a^2 \cos^2\theta},$$

verified against the induced metric to 4×10^{-16} at four spins up to $a = 0.9M$. It is unity at every latitude only when $a = 0$, and $A(\pi/2)/A(0)$ is exactly the squashing ratio $1 + a^2/r_+^2$ of A.10, at every spin tested. The two statements are one function seen twice.

Teukolsky's separation puts the horizon condition on the radial function alone, which works because that condition is the same at every latitude. A response weighted by A is not, so it cannot be imposed mode by mode, it couples different ℓ at fixed m . Because A is even in $\cos\theta$, its Legendre expansion carries even terms only, so on a *scalar* field the coupling is $\ell \rightarrow \ell \pm 2, \ell \pm 4$ and never $\ell \pm 1$. Computed on a 200-node Gauss-Legendre rule, every odd coefficient of $\langle P_\ell | A | P_{\ell'} \rangle$ vanishes. The largest such entry sits at 4×10^{-15} and does not move whether the cut-off is $\ell = 6$ or $\ell = 12$, which is the rule's own floor and not a residual, since its Legendre orthogonality holds only to the same figure. Validated by planting an anisotropy odd in $\cos\theta$, which lifts the odd entries to 0.21.

That rule does not survive the move to gravitational perturbations. Teukolsky modes are spin-weighted, $s = -2$, and spin-weighted harmonics do not carry the parity the scalar argument uses. The decisive matrix element is non-zero: $\langle {}_{-2}Y_{22} | \cos^2 \theta | {}_{-2}Y_{32} \rangle = \sqrt{35}/21 \simeq 0.281718$, reproduced to 2×10^{-16} on the same rule with the harmonics checked orthonormal first, against the quadrature floor for the same element built from Legendre functions. An even response can therefore mix $\ell = 2$ with $\ell = 3$ at fixed $m = 2$. The scalar coefficients below are correct; what fails is carrying their selection rule into the gravitational case.

The strength is the quadrupole of the anisotropy relative to its mean, c_2/c_0 , where $c_\ell = \frac{2\ell+1}{2} \int_{-1}^1 A(x) P_\ell(x) dx$ with $x = \cos \theta$. It is $-0.00167, -0.01552, -0.04614, -0.21875$ and -0.36711 at $a/M = 0.1, 0.3, 0.5, 0.9$ and 0.99 , and identically zero at $a = 0$, with every odd coefficient vanishing by the same parity.

What is useful here is the selection rule, and it has limits. It supplies no amplitude: the law that would fix how strongly a horizon response is weighted by A is the matching law for a general rotating-horizon seam, which §4 still records as unconstructed. For the corner term the weight is adopted as zero rather than derived. So the rule is not a falsifier of this paper. Read as a falsifier it is empty, because the implementation defended here predicts no such response to begin with.

It is not a discriminator on somebody's positive detection either. The multipole structure of a ringdown anomaly does not decide whether the fold could be its cause: a parity-even transverse map can produce $\ell \pm 1$ mixing for spin-weighted modes, by the element above, so an $\ell \pm 1$ admixture rules nothing out and its absence confirms nothing. What remains is the scalar statement and the general one. A response that depends on latitude couples different ℓ at fixed m , and the size of that coupling still has no law to fix it.

A selection rule that says nothing about detectability is not yet a prediction, so here is an estimate. Take the seam weight w to be bounded by the echo searches, since a reflecting seam is what those searches look for, and write the fractional power going to $\ell = 4$ as $w^2 |c_2/c_0|$. A three-sigma detection of that contamination then needs a fundamental-mode ringdown signal-to-noise of roughly $3/(w\sqrt{|c_2/c_0|})$, which at $a = 0.9$ and $w = 10^{-2}$ is about 600 and at $w = 10^{-3}$ about 6000. Current instruments reach a few tens on loud events, so confirmation waits on next-generation detectors and, at the lower weight, on nothing yet planned. These are order-of-magnitude figures and the detector numbers are nominal.

No asymmetry between refuting and confirming survives. The fold does not forbid $\ell \pm 1$ identically for gravitational modes, so a target amplitude has to be reached on both sides, and telling a real odd-parity admixture from noise, from waveform systematics, or from an $\ell = 2$ overtone is itself amplitude-dependent. We have no matching formula for that side and quote no figure for it.

A.14 The antilinear lift acts on the quasinormal spectrum as $\omega \rightarrow \omega^*$, and the outgoing branch quantifies a reflecting horizon. §4's third reflecting signature. This follows from the definitions in the cosmology paper's §2.1 with no further input. Acting on a mode $e^{-i\omega t}$, the pullback by α reverses t , giving $e^{+i\omega t}$, and antilinearity then conjugates the coefficient, giving $e^{-i\omega^* t}$. A mode damped in our time therefore pairs with one growing in our time, which is the content of the CPT relation and nothing is wrong with it: the mirror sheet carries the opposite time orientation, so each sheet sees ordinary damping on its own clock, and the growth exists only when one sheet is described in the other's.

That action has a consequence for a reflecting horizon. The Schwarzschild spectrum has the mirror symmetry $\omega \rightarrow -\omega^*$, if ω is a quasinormal frequency then so is $-\omega^*$. The fold sends $\omega \rightarrow \omega^*$, so composing the two gives $\omega \rightarrow -\omega$, and whether the fold's action is compatible with the spectrum's own symmetry is therefore a question about the boundary condition. It has a two-line answer. Leaver's coefficients are real polynomials in $\rho = -i\omega$, so the continued fraction obeys $\overline{F(\omega)} = F(-\omega^*)$, and the same argument gives it for G . Checked for F at six frequencies, real and complex, with a defect of exactly zero, since the two sides run the identical arithmetic; the check is validated by planting an imaginary part in β_n , which breaks it. The G half is the argument and not a computation, because G is not built anywhere here. Conjugating the quasinormal condition $F(\omega) + \mathcal{R}G(\omega) = 0$ then gives $F(-\omega^*) + \overline{\mathcal{R}}G(-\omega^*) = 0$, so $-\omega^*$ solves the same equation exactly when $\mathcal{R}(-\omega^*) = \overline{\mathcal{R}(\omega)}$. For a *constant* reflectivity that condition is $\overline{\mathcal{R}} = \mathcal{R}$, and everything below is that case. It is not the general condition. A delayed response $\mathcal{R}(\omega) = r_0 e^{i\omega\tau}$ with real r_0 and τ is complex at every real frequency and satisfies it exactly, as does any $\mathcal{R}$ with real Taylor coefficients in $i\omega$. Reality is sufficient and not necessary.

To test this, write the horizon behaviour in powers of $x = r - 1$, with $2M = 1$ and $\rho = -i\omega$. The two Frobenius solutions there go as x^ρ and $x^{-\rho}$, the first carrying the wave transmitted into the horizon and the second the wave reflected off it, so a reflecting horizon imposes

$$\psi \sim x^\rho + \mathcal{R}_h x^{-\rho}$$

with unit leading coefficients on both. That normalisation is part of the statement. Measured instead against unit-amplitude waves in $r_* = r + \log(r - 1)$, the same physical reflectivity reads $\mathcal{R}_{\text{wave}} = e^{2\rho} \mathcal{R}_h$, because $x^\rho = e^{-\rho} e^{\rho r_*}$, a factor of $0.1085 + 1.4233i$ at the fundamental, and shifting the additive origin of r_* moves it again. A response quoted without its convention is not a number, so the convention comes first here.

Leaver's continued fraction is built on the ingoing branch, so the outgoing one has to be constructed, and its ansatz is fixed before any recursion is written. A trial $x^A r^B e^{-\rho x}$ carries r -power $A + B$ at infinity, where outgoing means $e^{-\rho r_*} = r^{-\rho} e^{-\rho r}$, so $A + B = -\rho$. Leaver's branch takes $A = +\rho$ and needs $B = -2\rho$; the outgoing branch takes $A = -\rho$ and needs $B = 0$, no power of r at all. Over six decades of r the ratio to $e^{-\rho r_*}$ drifts by 10^{-10} for that choice and by 1.9 for either $B = \pm 2\rho$, and at the horizon its ratio to $x^{-\rho}$ is one to eight figures while Leaver's winds through 3.7 radians. Under that prefactor the three-term recursion reads

$$\alpha_n = (n + 1)(n + 1 - 2\rho), \quad \beta_n = -(2n^2 + 2n + \ell(\ell + 1) - s^2 + 1), \quad \gamma_n = n^2 - s^2,$$

which is Leaver's confluent reduction redone rather than his coefficients with signs moved. The series it generates satisfies the original Regge-Wheeler equation to 5×10^{-41} at forty digits, and a part-per-million nudge to any one of the three families raises it to between 2×10^{-7} and 7×10^{-6} , while Leaver's own coefficients under this prefactor fail at 0.8.

With both branches in hand the reflecting condition is $\mathcal{S}(\omega) = \mathcal{R}_h$, where $\mathcal{S} = -F/G$ is the ratio of the two horizon amplitudes carried by the solution that is outgoing at infinity, so the linear response is $d\omega/d\mathcal{R}_h = 1/\mathcal{S}'(\omega_0)$. Normalising the two Frobenius solutions is what makes that a number: F and G can be rescaled independently without moving their zeros, and $-G/F'$ moves with them, but a ratio of two solutions each carrying unit leading coefficient cannot be rescaled at all. Continuing the equation in from large r along a ray rotated off the real axis, so that the wanted solution is the dominant one going inward, and projecting onto the two Frobenius solutions at the match point, gives

$$M \frac{d\omega}{d\mathcal{R}_h} = 0.00735845 - 0.0646771 i,$$

or $0.0928510 + 0.0034534 i$ in the tortoise normalisation. The same machinery returns $M\omega_0 = 0.3736716844 - 0.0889623157i$ as the root of $\mathcal{S}$, which is Leaver's fundamental with no continued fraction anywhere in the calculation, and the independent Leaver solver used to seed that root returns the standard values for five other modes it was not built around. Neither number moves by more than 5×10^{-41} when the contour is changed from a ray at 45° reaching $r = 300$ to one at 60° reaching $r = 120$. Planting a scalar potential in place of the gravitational one, or $\ell = 3$ in place of $\ell = 2$, or a flipped sign in the $1/r^3$ term, moves $|\mathcal{S}(\omega_0)|$ from 5×10^{-48} to of order one.

What that response says is that a reflecting horizon shows up in the damping long before it shows up in the pitch. A reflectivity of one per cent moves the real frequency by two parts in ten thousand and the damping rate by seven parts in a thousand, a factor of 37 between them, so a ringdown search for a reflecting horizon should be pushed on the decay time.

The mirror symmetry then behaves as the algebra said it would. Solving $\mathcal{S}(\omega) = \mathcal{R}$ for the mode near ω_0 and for the one near $-\omega_0^*$, the defect $|\omega_+ + \bar{\omega}_-|$ is exactly zero at $\mathcal{R} = 0.001$ and at $\mathcal{R} = 0.01$, zero rather than small because the mirrored contour runs the identical arithmetic, and it is 2.604×10^{-4} at $\mathcal{R} = 0.001i$ and 2.604×10^{-3} at $\mathcal{R} = 0.01i$, linear in the imaginary part and proportional to the response above. Within the constant-reflectivity case, then, reality preserves the mirror symmetry and an imaginary part breaks it. Nothing here extends that to a frequency-dependent seam.

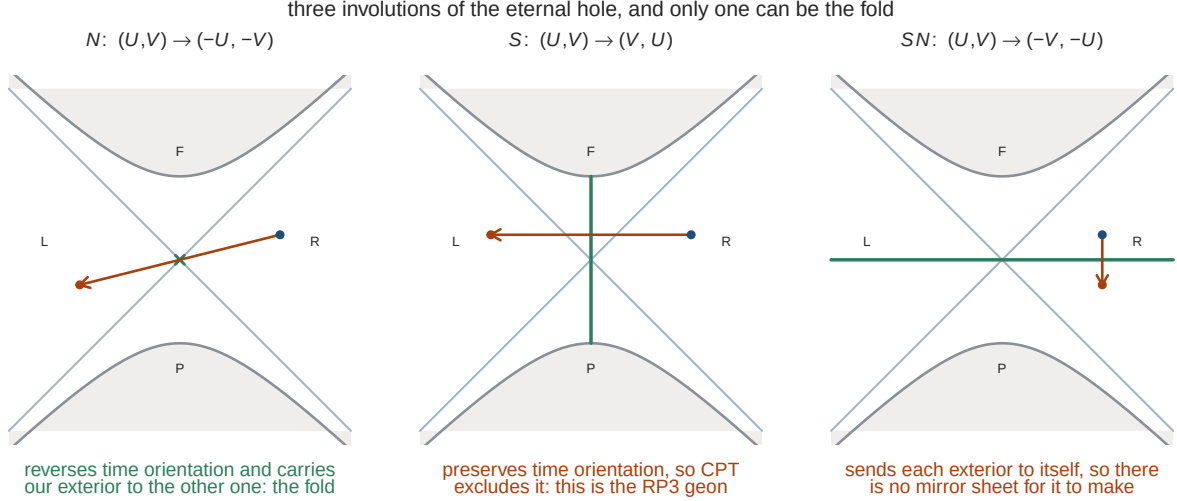


Figure 15: The classification, and what separates the three. In Kruskal coordinates $T = (U + V)/2$ and $X = (V - U)/2$ the survivors of the squaring are the point reflection through the origin, the mirror in the vertical axis and the mirror in the horizontal one. Shaded, the singularities; thin diagonals, the horizons; green, each map's fixed set, which is a point for N and a line for the other two. The blue dot is a point of our exterior R and the orange one is its image. S carries R to L but preserves time orientation, so a CPT map cannot be it; it is the $\mathbb{RP}^3$ geon. SN reverses time orientation but sends each exterior to itself, so it offers no mirror sheet at all, and the contact it would describe would sit where we can look. N does both things the fold needs and, unlike the swaps, admits no boost freedom to fix afterwards. The script checks that all three preserve UV and square to the identity, and that a fourth map does neither.

Two limits bound what this shows. It is a check of internal consistency, the prediction and the test are both ours, and no measurement enters. And it does not favour a reflecting horizon, which the minimal implementation of §4 does not adopt; it says what a reflecting implementation would have to look like, namely that a complex reflectivity carries a spectral signature a real one cannot, so the two are distinguishable in principle by the spectrum alone.

A.15 Where the two sheets can meet, and what that requires. §5's condition and §6's count, both derived. *One hypothesis is assumed throughout, and it is stated first. (H) The fold maps the hole to itself as a set.*

That is the whole of it. It does not name a map, and it does not ask the hole to have a second exterior. Both follow, and the counting that delivers them is short.

Near an isolated hole the geometry is Kruskal. Both r and the conformal factor there depend on U and V only through the product UV , so a linear map of the (U, V) plane is an isometry exactly when it preserves UV and the form $dU dV$. That leaves the diagonal maps $(U, V) \mapsto (\lambda U, V/\lambda)$ and the swaps $(U, V) \mapsto (\lambda V, U/\lambda)$, and nothing else. Squaring picks out the involutions, among the diagonals only $\lambda = -1$, which is $N: (U, V) \mapsto (-U, -V)$; among the swaps every λ , a one-parameter family that conjugation by a boost collapses to two representatives, $S: (U, V) \mapsto (V, U)$ for $\lambda > 0$ and $SN: (U, V) \mapsto (-V, -U)$ for $\lambda < 0$. Up to a boost those three exhaust the involutions.

Two of the three go, for different reasons. The fold is a CPT map and has to exchange past and future, which removes S , it fixes T , preserves time orientation and carries the future interior to itself. S is the $\mathbb{RP}^3$ geon map of Louko, Mann and Marolf [28], whose quotient is globally hyperbolic and well behaved, and substituting it gives a different question with a different answer. SN passes the CPT test, since it reverses time orientation and carries F to P , and fails an observational one. It sends each exterior to itself, so it identifies (t, r, n) with $(-t, r, -n)$ with both events in our own exterior, and those two are causally connected once $|t|$ exceeds about a light-crossing time. The minimum photon time from a point to its own antipode at

the same radius is $16.6M$ at $r = 3.5M$, $28.7M$ at $r = 10M$, and approaches the flat-space chord $2r$ far out. Under SN the fold's contact would sit in the open, at every radius and around every hole older than a few crossing times. It does not.

What is left is N , and it is rigid. Composing it with a boost destroys the involution, so no parameter survives to be chosen. The transverse factor is fixed as well: it has to act freely on the bifurcation sphere, and every free involution of S^2 is conjugate to the antipodal map, since the quotient is a closed surface of Euler characteristic one and $\mathbb{RP}^2$ is the only one.

The same count answers a question one step earlier, which is why an involution rather than something of higher order, why two sheets and not four. A free action of a finite group G on a compact manifold M has $\chi(M) = |G| \chi(M/G)$, so $|G|$ divides $\chi(M)$. The transverse sphere in D dimensions is S^{D-2} and $\chi(S^n) = 1 + (-1)^n$, which is two for even n and zero for odd. In four dimensions that is $\chi(S^2) = 2$, so $|G|$ divides two and the fold is two-sheeted with nothing else available. A free $\mathbb{Z}_3$ would need a quotient of Euler characteristic $2/3$. The restriction is even-dimensional: S^3 carries free $\mathbb{Z}_n$ actions for every n , which is what a lens space is, so a four-sheeted fold is a five-dimensional possibility and not a four-dimensional one. Two sheets is not an assumption of the fold. It is what the sphere admits. So the fold near a hole it fixes is $\Theta = N \circ P_\perp$, with nothing left over.

The second exterior then comes free. N carries R to L , so a fold-invariant hole has L , the past interior P , and a bifurcation surface, whatever its formation history. A white-hole region is a consequence of the hole being fold-invariant, not a separate demand on how the hole was made, and no particular formation channel has to supply one.

It also says where such a hole would have to sit, and that is where the trouble is. If Θ carries our sheet to the mirror sheet and fixes the hole, the hole lies in both: its right exterior in ours, its left in the mirror, and its bifurcation surface on the locus joining them, which in this cosmology is the bang. The two routes to the mirror sheet, down a throat and back through the bang, would then have to be one route and one map.

Schwarzschild-de Sitter is where to test that, since it carries a black-hole horizon and a cosmological one at once, and the answer is a single point.

Below the Nariai limit it fails. Each horizon has its own wedge reflection, and in the conformal diagram those are point reflections about two different bifurcation surfaces, so their composition is a translation by twice the separation. Both neighbours of a static region lie to its future, because f is positive strictly between the horizons and negative on both sides, so r is timelike in both, falling to the singularity inside r_b and growing without bound outside r_c . The translation is therefore timelike, and identifying the two reflections means quotienting by it, which closes timelike curves. Nothing milder is available either, since the surface gravities differ, by a factor 6.5 at $\Lambda M^2 = 0.005$ and 1.29 at 0.10.

At Nariai it works, and that is the result. There the geometry is the product $dS_2 \times S^2$, and the embedding settles it with no appeal to diagrams at all. Write $dS_2 = \{X \in \mathbb{R}^{1,2} : -X_0^2 + X_1^2 + X_2^2 = \ell^2\}$, take the static patch to be $X_1 > |X_0|$ and the horizons to be $X_1 = \pm X_0$. Both bifurcation surfaces sit at $X_0 = X_1 = 0$, so at $X_2 = \pm \ell$, and the linear map $(X_0, X_1, X_2) \mapsto (-X_0, -X_1, X_2)$ fixes both. It is an isometry and an involution, it reverses time orientation, it carries our static patch to the other one and F to P , and composing it with the sphere's antipodal map makes it free. It is the wedge reflection at both horizons at once, which is what no sub-Nariai member of the family has. The reason is that dS_2 has circles for spatial sections, so its maximal extension closes into four regions, with no chain and no translation to obstruct.

So the family contains exactly one fold-invariant hole and it is the largest one the family holds. At the observed Λ that is $M = 1/(3\sqrt{\Lambda})$, some 2.2×10^{22} solar masses, with a horizon ten billion light years in radius, not an object inside a universe but a configuration of one. Two calculations run for unrelated reasons agree on the point. The contact radius of Section 5, swept against Λ , runs from half the horizon radius at $\Lambda = 0$ to 0.778, 0.919 and 0.991 of it as $9\Lambda M^2$ reaches 0.9, 0.99 and 0.9999. The one hole the fold can fix is also the one whose interior is entirely in contact, with no forbidden outer shell left over.

Read from the other end the family says something tidier than a lone exception. The fold does not switch on and off along it. It is present throughout, because the cosmological horizon always has a wedge reflection, and composing that with $P_\perp$ gives a free, time-reversing map carrying our static region across that horizon

to a mirror region. That is the map Section 2 uses at our own horizon. At $M = 0$ there is one horizon and the fold fixes it. For $0 < 9\Lambda M^2 < 1$ there are two with distinct bifurcation surfaces, the fold fixes the cosmological one, and it cannot also fix the hole for the reason above. At $9\Lambda M^2 = 1$ the two horizons are the same object and fixing one fixes both.

So the second class is not a separate population. A hole is two-sided exactly when its horizon is the cosmological horizon, and the only mass at which that happens is the maximum. Section 2 and Section 6 are describing one structure at the two ends of one parameter, and the contact radius climbing from half the horizon radius to all of it is that parameter made visible.

One objection to a maximal configuration is that it does not last, and it needs answering, because a hole that decays is not an object. Reduce on the sphere. With $ds^2 = \hat{g}_{ab}dx^a dx^b + R^2 d\Omega^2$ the vacuum equations with Λ read $\hat{R}_{ab} - (2/R)\nabla_a \nabla_b R = \Lambda \hat{g}_{ab}$ and $1 - (\nabla R)^2 - R\nabla^2 R = \Lambda R^2$. Put $R = \ell(1 + \varphi)$. The second linearises to $\nabla^2 \varphi = -2\Lambda\varphi$, and since $\hat{R}_{ab} = (\hat{R}/2)\hat{g}_{ab}$ identically in two dimensions the first forces the traceless part of $\nabla_a \nabla_b \varphi$ to vanish, leaving $\nabla_a \nabla_b \varphi = -\Lambda \hat{g}_{ab} \varphi$. That is three solutions, where a wave equation alone would leave a whole function, namely $\varphi = c \cdot X$ for constant c in $\mathbb{R}^{1,2}$.

Two of the three run away. In the static patch $X_2^2 = \ell^2 + X_0^2 - X_1^2 < \ell^2$, so X_2/ℓ is bounded by one while X_0/ℓ and X_1/ℓ pass a thousand within eight static times. That is the classical Nariai instability, the sphere swelling on one side and shrinking on the other as the solution runs toward Schwarzschild-de Sitter. The third mode is the static near-Nariai deformation and does nothing.

The fold removes both runaways. A perturbation descends to the quotient only if it agrees at identified points, and the sphere radius is a function on the dS_2 factor alone, so the condition is $\varphi(N_\ell x) = \varphi(x)$. Under $(X_0, X_1, X_2) \mapsto (-X_0, -X_1, X_2)$ the two growing modes are odd and the bounded one is even, so neither runaway is a perturbation of the folded solution at all. The one hole the fold can fix is the one the fold makes stable, in the sector where the instability lives. That is linear stability in the spherically symmetric sector and no more, non-spherical modes and the semiclassical decay are different questions and are not treated here.

Charge turns that point into a curve. In Reissner-Nordström-de Sitter a degenerate horizon needs $f(r_0) = f'(r_0) = 0$, and eliminating M leaves $Q^2 = r_0^2(1 - \Lambda r_0^2)$ with $M = Q^2/r_0 + \Lambda r_0^3/3$. That curve has two halves. Above $r_0 = 1/\sqrt{2\Lambda}$ the merged pair is $r_+ = r_c$ and $f''(r_0) < 0$, so $f \leq 0$ on both sides of the double root, r is timelike there, and the near-horizon factor is dS_2 . The argument above used nothing but that factor, so it applies verbatim. Below it the merged pair is $r_- = r_+$ and $f''(r_0) > 0$. The factor is AdS_2 , the horizon is extremal with zero surface gravity, and there is no bifurcate Killing horizon for a wedge reflection to act on at all. The halves meet at the ultracold point where all three horizons coincide, at the largest charge the family admits, $Q = 1/(2\sqrt{\Lambda})$.

So the fold-invariant set is a one-parameter family rather than a point, running from uncharged Nariai to ultracold, every member with its black-hole horizon coincident with its cosmological one. The stability result travels with it, since what selects the three modes is the traceless constraint and not any coefficient, and their parity under N_ℓ is a property of the embedding functions.

Rotation widens it again, and the branch there has an exact end. The transverse map is an isometry of Kerr-de Sitter at every spin and every Λ for the reason it works in Kerr, the metric functions depending on θ through $\cos^2 \theta$ alone, and the merger of the outer pair runs from $a = 0$ with $\Delta_r'' < 0$ throughout, so the radial factor is dS_2 and never AdS_2 . It stops where Δ_r acquires a triple root. There the quartic factorises as $-(\Lambda/3)(r - r_0)^3(r + 3r_0)$, matching coefficients gives $\Lambda r_0^2 = 2\sqrt{3} - 3$, $a^2 = \Lambda r_0^4$ and $M = (4\Lambda/3)r_0^3$, and the spin at the endpoint is a pure number,

$$\frac{a}{M} = \frac{3}{4\sqrt{2\sqrt{3} - 3}} = 1.1009173688,$$

the same in every de Sitter background and past the Kerr extremal bound, which a positive Λ permits. What is not done there is the embedding argument inside the warped rotating near-horizon geometry, what is established is that its two ingredients survive rotation.

One check came free there, and neither side of it was arranged. Reducing with a radial field, the angular equation reads $1 - (\nabla R)^2 - R\nabla^2 R = \Lambda R^2 + Q^2/R^2$, and linearising gives $\nabla^2 \varphi = -(2/r_0^2)(2\Lambda r_0^2 - 1)\varphi$. The modes still have to be $\varphi = c \cdot X$, so consistency forces $\ell_2^2 = r_0^2/(2\Lambda r_0^2 - 1)$, while ℓ_2 was already fixed by the metric as $\ell_2^2 = 2/|f''(r_0)|$. The two agree to 6×10^{-15} along the branch, and the perturbation expression changes sign exactly at $r_0 = 1/\sqrt{2\Lambda}$, which is where the metric says Nariai stops.

What stays open is therefore narrow. For every hole below the maximal mass the alternative reading is still live: the mirror sheet contains its own image of every hole in ours, the fold carries our hole's exterior to the mirror hole's exterior, and L is a second asymptotic region on our own sheet related to us by no fold map at all.

A second thing the comparison makes clear is where the content sits. Drop $P_\perp$ and keep only the $(U, V) \mapsto (-U, -V)$ part, and a point and its image share an angular position, so the required path falls from π to zero and the line $X \equiv 0$ connects them for every interior point, the answer becomes the whole interior. The sphere's antipodal map is therefore the entire content of the question. Without it the sheets reach each other throughout the interior; with it, only inside $r = M$.

A.11 uses the fact that the two wedges are spacelike separated at every pair of points. That is a statement about exteriors, and taking it for the whole answer would be a mistake, because the question of where an inter-sheet interaction is possible has a different answer and a definite one. Kruskal coordinates are conformally flat in the (T, X) plane, so light cones lie at forty-five degrees and the causal structure is Minkowski's. Writing $R = \{X > |T|\}$, $L = \{X < -|T|\}$, $F = \{T > |X|\}$ and $P = \{T < -|X|\}$,

$$J^+(R) \cap J^+(L) = F, \quad J^-(R) \cap J^-(L) = P,$$

so the black hole interior is exactly the set of events both exteriors can influence, and the white hole interior is exactly the set that can influence both. Sampling 2×10^4 configurations per region, no event outside F admits influence from both, and the shortfall from unity inside F comes from how the source points were sampled. The geometry contributes none of it.

The fold's own map exchanges them. J is the boost through imaginary angle π , which in Kruskal coordinates is $(T, X) \mapsto (-T, -X)$, so $R \leftrightarrow L$ and $F \leftrightarrow P$, checked on 2×10^4 points with no exception. Pairing a black hole interior with a white hole interior is therefore not a further postulate; it is the same map that pairs the exteriors, read in the other two regions.

Three consequences follow, one of them unwelcome. An inter-sheet interaction is causally impossible outside a horizon and causally possible inside one, which says more than the seam argument and is independent of the seam coefficient. Anything occurring in F lies in the causal future of both exteriors and therefore reaches neither, so even a strong inter-sheet force is unobservable from outside: §4's null has a second reason behind it that does not use A.11 at all. And the contact window is finite. The longest proper time from horizon to singularity is πM , and the part of it lying inside $r = M$, which is where contact is possible at all, is $(\pi/2 - 1)M$, eighteen per cent of the whole. That is $8.4 \mu\text{s}$ at $3M_\odot$, twelve seconds at Sgr A* and five hours at $6.5 \times 10^9 M_\odot$, scaling linearly with mass.

The commutator makes the same statement about the operators, and gives the depth a number. Under the wedge reflection $J : (U, V) \mapsto (-U, -V)$ the Kruskal null coordinates scale as $dU = 2U$ and $dV = 2V$, so $dU dV = 4UV$ and the radial part of the separation between a point and its image turns timelike exactly when $UV > 0$. That set is the interior. The cross-sheet commutator therefore vanishes identically outside a horizon and is nonzero inside one, switching on wherever UV changes sign, with nothing tuned to put it there.

The four-dimensional condition. It does not switch on at the horizon itself, and the four-dimensional statement is more restrictive than the two-dimensional one above. Reaching the mirror sheet's causal region requires covering the transverse angle as well, since the fold is $\alpha = J \circ P_\perp$ and $P_\perp$ sends the unit vector $\hat{n}$ to $-\hat{n}$. Explicitly: $P_\perp$ is the antipodal map of the sphere, so the angular separation between a point and its image is exactly π at every point, and nothing is lost by working in the equatorial plane.

Causality then bounds the angle directly. From $ds^2 \leq 0$ in Kruskal,

$$r^2 d\Omega^2 \leq F(r) (dT^2 - dX^2), \quad F(r) = \frac{32M^3}{r} e^{-r/2M},$$

so the angular path along any causal curve is at most $\int g d\tau_2$ with $g = \sqrt{F}/r$ and $d\tau_2$ the proper time of the (T, X) part. Throughout, x is taken in the black-hole interior F and Θx in the white-hole interior P ; the roles swap if x is taken in P , and the implication then runs the other way. The maximum angular sweep between two events is therefore the length of the longest timelike curve in the conformally rescaled two-dimensional metric $g^2(-dT^2 + dX^2)$, and the sheets touch exactly when that maximum reaches π .

A boost sends $(U, V) \mapsto (U/\lambda, \lambda V)$ and fixes the angles, and it carries the pair $(x, \Theta x)$ to $(x', \Theta x')$, so the answer depends only on the boost invariant UV , which is to say only on r . Fixing that freedom by putting x at $X = 0$ leaves x at $(T_0, 0)$ and Θx at $(-T_0, 0)$ with $T_0 = \sqrt{(1 - r/2M)e^{r/2M}}$. Along a causal curve $dT > 0$, so T parametrises, and

$$\Delta\phi = \int_{-T_0}^{T_0} g \sqrt{1 - X'^2} dT.$$

Both factors are largest at $X \equiv 0$, g falls with r , since $d \log g^2 / dr = -1/2M - 3/r < 0$, while r rises as $T^2 - X^2$ falls. So the maximiser is $X \equiv 0$, uniquely, and it is also the fixed set of $X \mapsto -X$ and hence a geodesic. On it every exponential cancels, $g |dT/dr| = 1/\sqrt{r(2M - r)}$, and the integral is elementary:

$$\Delta\phi_{\max}(r) = 2 \int_r^{2M} \frac{dr'}{\sqrt{r'(2M - r')}} = 2\pi - 4 \arcsin \sqrt{\frac{r}{2M}}.$$

Setting this equal to π gives $r = 2M \sin^2(\pi/4)$, so the condition is

$$\boxed{r \leq M}$$

exactly half the horizon radius, with the mass cancelling because the answer is a fraction of r_h and not a length.

This question is older than the calculation. The map is Gibbons' elliptic interpretation of the Kruskal manifold [29], written in the form used here by Sanchez and Whiting [30]. Harlow and Numasawa [6], whose gauging of a spacetime inversion §2.1 already leans on, pose it again for the same quotient and call the objects self-intersecting timelike curves, conjecturing that quantum gravity permits them only where the self-intersections sit behind horizons. The result above is an instance of that conjecture and says more than it does, behind the horizon, and then inside $r = M$ as well.

We have not found the radius stated in Harlow and Numasawa [6], which was read in full. Gibbons [29] and Sanchez and Whiting [30] were consulted only through abstracts and secondary citation, so no statement is made about what their text contains, and the novelty of $r = M$ is not established. Nothing else in this appendix depends on it.

Put in words the result stops looking like a coincidence of arcsines. The antipodal map asks for half a revolution. The journey has two interior legs and the fold's own symmetry splits it evenly between them, so each leg must supply a quarter revolution. A leg starting at r supplies $\pi - 2 \arcsin \sqrt{r/2M}$, which is a quarter revolution exactly at $r = M$. Half the horizon radius is the depth from which a causal curve can just turn a quarter of the way round before the horizon reaches it. The whole journey, past singularity through the bifurcation surface to future singularity, gives one full revolution and no more. The antipodal map asks for half of one. Half a revolution takes half the interior, and a fold through a general angle α would take $\cos^2(\alpha/4)$ of it.

Three things follow immediately. The commutator is nonzero only for $UV \geq \sqrt{e}/2 = 0.8244$, against the singularity at $UV = 1$, so the two-dimensional claim that it is nonzero throughout $UV > 0$ is too generous by half. The bound is attained, since the curve $X \equiv 0$ with $d\phi = g dT$ is null, and it is the one that does it.

That the maximiser is a geodesic and not merely a null curve is what makes the statement an equivalence and not a one-way implication. If the maximiser were null and not geodesic, Hawking and Ellis' 4.5.10 would put its endpoint in the chronological future, which is open, and contact would leak past $r = M$. It is the $E = 0$ branch, whose $d\phi/dr = 1/(r\sqrt{|f|})$ saturates the bound exactly; structurally, $\{U = V\}$ is the fixed-point set of $t \mapsto -t$ and so totally geodesic, a great circle inside it is totally geodesic, and in the remaining two dimensions every null curve is a null pregeodesic. So at $r = M$ the image sits in $J^+(\Theta x)$ and not in $I^+(\Theta x)$, and $\leq$ is the right sign. And the bounding sphere carries exactly a quarter of the horizon area, which is an identity: the threshold is $\arcsin \sqrt{r/2M} = \pi/4$, so $\sqrt{r/2M} = 1/\sqrt{2}$, and the quarter in the area is the square of the half in the radius. Whether the echo of $S = A/4G$ is anything is a question we raise and do not answer. What weighs against it is that the quarter here is a bare area ratio with no G and no $\hbar$ in it.

One candidate for that number, ruled out. Let the connecting curve sweep its angle where the angle comes fastest, at the photon sphere: $d\phi/dt = \sqrt{1 - 2M/r}/r$ is maximised at exactly $r = 3M$, giving a minimum coordinate time $3\sqrt{3}\pi M$ and a Kruskal ratio $e^{3\sqrt{3}\pi/4} = 59.21$. The optimisation is right and the photon sphere is out of reach. Along a future-directed causal curve $dU dV \geq 0$ with $dT > 0$, so both U and V increase, and a curve joining Θx to x is confined to $|UV| \leq UV(r) \leq 1$, while at $r = 3M$, $|UV| = 2.241$. No curve that does the job gets there. What the candidate has right is the shape. Contact needs the angle covered as well as the fall, it does not open at the horizon, and how deep it opens is a pure number free of the mass. Once the constraint is imposed that number is one half, not 59.21.

Charge. Charge decides which form of that statement is the robust one, and it is not the one just written. For Reissner-Nordstrom, $r^2 f = (r - r_+)(r - r_-)$, so $-r^2 f$ is a monic quadratic whose roots are the two horizons, and $\int_a^b dr/\sqrt{(b-r)(r-a)} = \pi$ for any pair of roots. The interior band therefore supplies exactly π of turning at every charge, the leg integral is $2\arcsin \sqrt{(r_+ - r)/(r_+ - r_-)}$, and the condition becomes

$$r \leq \frac{r_+ + r_-}{2} = M,$$

the same M , at every charge, and always exactly half way across the band. So the robust statement is that contact fills the inner half of the interior; “half the horizon radius” is the $r_- = 0$ special case of it. That also settles the quarter of the area, which does not survive charge: A_c/A_+ runs from 0.250 at $Q = 0$ to 0.768 at $Q/M = 0.99$ while the physical statement does not move at all. Bekenstein-Hawking is $A/4$ for charged holes too, so anything thermodynamic would have had to survive charge. It is arithmetic.

All of that can be read another way. $P_\perp$ is antipodal and therefore free, and J has no fixed point off B , so the fold is a free involution and the quotient is a manifold in which x and Θx are the same point. On the quotient, then, $x \in J^+(\Theta x)$ says there is a closed causal curve through that point. What $r \leq M$ locates is the causality-violating core of the folded hole: closed timelike curves strictly inside, the closed null geodesic exactly on the boundary, nothing outside it, and the whole of it hidden behind a horizon and then half the interior.

One consequence decides whether the fold polices itself. The quantity an image sum sees is the world function $\sigma(x, \Theta x)$, and it has a simple zero at the boundary. Along $X \equiv 0$ the Schwarzschild t is constant, so the connecting geodesics are the $E = 0$ family,

$$\left(\frac{dr}{d\tau}\right)^2 = \left(\frac{2M}{r} - 1\right) \left(1 + \frac{L^2}{r^2}\right), \quad \frac{d\phi}{d\tau} = \frac{L}{r^2},$$

whose large- L limit is the null curve above. Fixing L so the two interior legs deliver exactly π and reading off the proper time gives $\tau^2/(M-r) \rightarrow 34.85$, a finite nonzero constant, so $\tau \sim \sqrt{M-r}$ and $\sigma = -\tau^2/2$ vanishes linearly at $r = M$. A double zero and a linear vanishing of τ are both excluded by the same numbers. The deepest loop, at $r \rightarrow 0$, has proper length $5.8839M$, against $\sqrt{3}\pi M = 5.4414M$ for the constant-winding ansatz, which is only a lower bound.

A simple zero of σ is what a power-law divergence needs: the image term of a free field's two-point function goes as $1/\sigma$, so it blows up as $(M-r)^{-1}$ on approach to the contact sphere, and the stress tensor, built

from two derivatives of it, as a higher inverse power. That is how a chronology horizon in ordinary general relativity censors its own formation. The power written that way is not the one that holds, because the image point is a caustic and $1/\sigma$ assumes it is not; A.18 does the sum that needs no such assumption, in a model geometry carrying the same degeneracy, and returns a steeper $s^{-3/2}$ with the stress tensor at $s^{-7/2}$. The conjecture needed the sign as well as the power, and A.18's components supply it. In the point-split energy density the fold's pullbacks reproduce the wave operator, so every term cancels except the part of the radial potential the transverse sphere does not account for.

That zero is the constant transverse radius's doing, and where the radius varies there is no cancellation. In flat space, whose fold is the time reflection composed with the spatial point reflection, the image energy density is exactly $(8r^2 + 6\xi^2)/4\pi^2 D^3$, positive everywhere and diverging positively at contact. So the sign the conjecture wanted is the sign it gets, in the one setting where the whole thing is closed form. Neither model is a black hole, one has the caustic and no sign, the other the sign and no caustic. On Schwarzschild the remainder is radial and nothing else. A.19 carries the fold through for any static spherically symmetric metric and the transverse sphere supplies the angular part of the potential exactly, at every one of them, so no ℓ survives in the density: what is left is the radial part of the wave operator set against the mixed radial derivative, a two-dimensional object on the contact orbit.

Two caveats travel with that reading. The fold reverses time orientation, so the quotient is not time-orientable and the loop closes with the transported arrow flipped, which makes "closed timelike curve" a convention that has to be argued rather than assumed. And surfaces of constant r are spacelike inside, so this chronology horizon is spacelike. Hawking's theorems produce the null kind and assume the time orientability this quotient lacks.

The sign of the null-null component matters beyond the contact region. For a null k the trace term and Λ leave the field equations, so $R_{ab}k^ak^b = 8\pi G T_{ab}k^ak^b$ with no residue, which locks the two signs together. A negative $R_{ab}k^ak^b$ is a failure of the null convergence condition, and that condition is the hypothesis Penrose's argument uses to carry a trapped surface to an incomplete geodesic. The paragraphs below find the sign negative along the contact direction, where it delays the conjugate point and shrinks the contact region, and positive along Penrose's own congruence, so his hypothesis stands. The size a negative term would need in order to turn that congruence around is still the natural level to measure the fold's term against. An ingoing radial null congruence in the interior is shear-free and enters the contact region with $\theta = -4/r_h$, so $\theta' = -\theta^2/2 - R_{ab}k^ak^b$ turns the expansion around where $-R_{ab}k^ak^b$ exceeds $8/r_h^2$. Because the region is half the interior and $\theta' = -\theta^2/2$ resumes on the far side, the expansion has to reach zero before the congruence leaves, which is $\text{artanh}(1/u) = u$ at $u = 1.1997$ and raises the requirement to $11.5138/r_h^2$, a factor 1.439 above the pointwise one. In the vacuum the bound is saturated, $\theta = -2/r$ with $dr/d\lambda = -1$ diverges at exactly $\lambda = 2/|\theta_0|$.

Both of those are requirements on a constant stress and the stress is not constant. A.19's world function vanishes linearly on the contact sphere, so the image term diverges there, and the contact sphere is the region's outer edge, which is where an ingoing congruence enters. The largest stress arrives at entry and falls inward, the integrated effect at the edge is divergent, and the vacuum's focusing is not the leading behaviour there, so no finite level on a constant is the question. What takes a number is the thickness of the shell in which the fold's term is the larger of the two, measured against $11.5138/r_h^2$ as the level to reach. Dimensions leave one combination of the two lengths available, since $T_{ab}k^ak^b$ carries four inverse lengths and m^2 carries two:

$$T_{ab}k^ak^b = \kappa m^2 D^{-p} r_h^{p-2}, \quad \frac{D_*}{r_h} = \left(\frac{8\pi\kappa (m/m_P)^2}{11.5138} \right)^{1/p},$$

with D the distance to the contact sphere and κ the pure number A.18's sum on the contact geodesic would supply. D_* is proportional to r_h and to no other length.

The power decides the answer and κ barely enters it, and the power is $5/2$. Three links fix it and each is measured on this geometry. A.19's rule puts the image Green function at $\delta^{-(D-2+n)/2}$ in the distance to the caustic, over four members of a family, so a stress built from two derivatives of it sits at $\delta^{-(D+2+n)/2}$: here

δ^{-4} , measured at $\delta^{-3.95}$, which is the first test that rule has had against a stress rather than a correlator. The mass-dependent part of every quantity is one power softer than its massless counterpart, measured at 0.956, 0.976 and 0.968 of a power for G , P and Q , and that one follows from the proper-time representation rather than from the mode sum. A mass is a phase in the proper time, so to first order the mass part is $\int ds s^{-(D+n)/2} e^{-\sigma/2s}$ with one extra power of s , and that raises the world-function exponent from $(D+n)/2-1$ to $(D+n)/2-2$: one power of σ , checked against quadrature at six pairs (D, n) to 10^{-8} and against the exact massive integral, which returns the first-order exponent to 2×10^{-4} at $m^2 = 10^{-4}$. On this geometry the mass part is also the first derivative times m^2 to 2.5×10^{-4} at $m = 0.05$, so nothing beyond first order is in play.

One power of σ is one power of the distance here and two in flat space, and that is the whole of the apparent conflict with the textbook massive correction. At the antipodal caustic $\sigma = (\pi^2 a^2 - \tau^2)/2$, which is $\pi a \delta$ to eight figures at $\delta = 10^{-5}$; at a hole A.19 finds $\tau^2/(M-r)$ tending to 34.85, finite and nonzero, so σ is proportional to $M-r$ there as well. Away from a caustic $\sigma = r^2/2$ and one power of it is two powers of r , which is the log-carrying flat-space answer, the same statement read in a different variable. The sphere family cannot settle this on its own, because it ties the caustic order to the dimension, $n = N-1$ and $D = N+1$, so order one arrives only in three dimensions where the powers collide and a logarithm interferes with the fit.

The further power the mass slot appeared to lose is the coupling's doing. Where the Ricci tensor vanishes the null stress is $(1-4\xi)P-Q$, so the leading mass parts cancel exactly when $1-4\xi$ equals dQ/dP at the caustic, and here that ratio is 0.33018, putting the cancelling coupling at $\xi = 0.16746$, which is the conformal one to three figures. That is a property of a conformally flat geometry and not of the mass. At $\xi = 0, 1/12, 0.2$ and 0.25 the mass slot measures $-2.97, -2.97, -3.00$ and -2.99 , and $\xi = 1/6 + 0.02$ already gives -3.01 , so the cancellation holds at a point only. The caustic at a hole is order one, so the leading stress is $\delta^{-7/2}$ and the mass part is $\delta^{-5/2}$, and the lower reading survives only as a fine-tuning of ξ to a hole's own dQ/dP , a number nobody has calculated.

At $p = 5/2$, κ is not open either, because every ingredient of it is already here. The caustic amplitude is $\Delta^{1/2} \rightarrow 3.9004 M s^{-1/2}$, on A.19's projection arc length $M(\pi+2)$ and not on the affine total, which the same appendix's measurement excludes; the massless and mass parts are the $k = 5/2$ and $k = 3/2$ proper-time integrals, reproduced against quadrature to 10^{-8} ; the world function vanishes linearly with $|\sigma| = 17.425(M-r)$ from A.19's $\tau^2/(M-r) \rightarrow 34.85$; and the contraction factor, the square of the null derivative of the distance, is exactly one on the ingoing radial congruence because $dr/d\lambda = -1$ on it. Collecting those, $G_{\text{mass}} = -A m^2 \sigma^{-1/2}$ with $A = 0.0525 M$, and two derivatives give $\frac{3}{4} A m^2 \sigma^{-5/2} (\partial_k \sigma)^2$, whose D dependence is $|\sigma|^{-1/2} D^{-5/2}$. In units of $r_h^{1/2}$ that is $\kappa = 0.0039329$.

At one solar mass the shell is then 2.1×10^{-6} m for the 491.6 PeV fermion, 1.5×10^{-11} m for a top quark, 2.3×10^{-13} m for a proton and 5.5×10^{-16} m for an electron, a factor 0.1091 below the unit- κ values because the thickness goes as $\kappa^{2/5}$, and it drops below a Planck length only for masses under 2.0×10^{-19} eV. So the shell is a real length for every field there is. It also widens with the hole, since D_* is proportional to r_h , which puts the effect at its largest in the largest holes. One thing the coefficient does not settle and one it does. The radial sign, which A.18's components settle below, rests on conservation rather than on a sum done in the interior. The order-unity factor beside κ , from keeping only the leading derivative pairing, is measured at one: with $G = F(\sigma)$ the null-null second derivative is $F''(k \cdot \nabla \sigma)^2 + F' k^a k^b \nabla_a \nabla_b \sigma$ and the second piece carries two fewer powers of σ , so it cannot reach the leading coefficient. Against A.18's exact tower, in the geometry where A.19's rule leaves nothing to choose, the leading pairing agrees to twelve digits for the massless part and to 0.2 per cent for the mass part.

The reduction that sign goes through is pure algebra. A stress invariant under the interior's isometries has only T^t_t , T^r_r and $T^\theta_\theta = T^\varphi_\varphi$, and contracting with a null vector without and with a transverse part gives $T_{kk} = |f|(k^t)^2(T^t_t - T^r_r)$ and that same expression plus $r^2(k^\theta)^2(T^\theta_\theta - T^r_r)$, the radial answer is the contact answer less the anisotropy. Conservation supplies the anisotropy outright, $T^\theta_\theta - T^r_r = (r/2)[\partial_r T^r_r - (f'/2f)(T^t_t - T^r_r)]$, verified against the Reissner-Nordström electromagnetic stress to 10^{-11} and broken at 1.1 when the transverse pressure is scaled by 1.4. With the trace as the third relation the 2×2 system inverts exactly and its determinant is nowhere zero in the interior. The trace carries no conformal

anomaly, because an anomaly is state-independent and the image part is a difference of states, so it is $(6\xi - 1)\frac{1}{2}\square\langle\phi^2\rangle - m^2\langle\phi^2\rangle$ with $\langle\phi^2\rangle_{\text{img}} = 0.0014423 M^{-1/2} D^{-3/2}$ from the same assembly as the coefficient. At $\xi = 1/6$ the Laplacian drops and the trace adds nothing further; away from it the Laplacian stays, because the image moves with the point, which is what A.18's geometry does not have. What the reduction leaves at $\xi = 1/6$ is one ratio, the contact geodesic's own $(k \cdot \nabla\sigma)^2$ against the radial congruence's, which is a transport along a curve A.19 already has and not a tensor.

Following that through changes the reading of the sign. The fold joins a point to its image at fixed Schwarzschild t , so the contact geodesic is the $E = 0$ family and its tangent has no t component: for a null geodesic with $E = 0$, $\dot{r}^2 = |f|L^2/r^2$ and $d\varphi/dr = 1/r\sqrt{|f|}$, and §5's leg, from r out to the horizon, delivers $\pi - 2 \arcsin \sqrt{r/2M}$, so the two-leg sweep is $2\pi - 4 \arcsin \sqrt{r/2M}$ and reaches π at $r = M$ with the mass cancelling, matched to quadrature at 10^{-5} . Integrating the leg the other way, from the singularity out to r , gives $4 \arcsin \sqrt{r/2M}$ instead, which agrees at $r = M$ and at no other radius and carries the contact condition the opposite way round, so a check made at $r = M$ alone cannot tell the two apart. With $k^t = 0$ the first term of the contraction drops, so the contact direction measures the anisotropy alone while the radial congruence measures X , and the two are orthogonal.

Conservation and the trace then tie the two together, and the coupling decides how. The trace is $X + 4Y + 2A$ and conservation is $Y' = (f'/2f)X + 2A/r$, so $X = T - 2A - 4Y$, and the trace's two pieces sit on either side of the anisotropy: A carries $D^{-5/2}$, the m^2 piece of $(6\xi - 1)\frac{1}{2}\square\langle\phi^2\rangle - m^2\langle\phi^2\rangle$ carries $D^{-3/2}$ and the $\square$ piece $D^{-7/2}$. At $\xi = 1/6$ the $\square$ piece is absent and $X \rightarrow -2A$, the ratio running 1.056, 1.005, 1.0005, 1.00005 as D falls through four decades, so a negative anisotropy puts X on the other side of zero and the radial contraction with it. At any other coupling the $\square$ piece leads, X follows the trace to 1.00005 of it over the same range, and the contact value does not reach the leading behaviour at all.

Which combination the computed sign belongs to is settled by a symmetry, and the answer is that the calibration geometry cannot distinguish them. On $\mathbb{R} \times S^3$ the rotations fixing a point also fix its antipode, a rotation about an axis fixing both ends of it, so the isotropy group there is the whole of $SO(3)$ and it fixes the image. A symmetric spatial tensor invariant under $SO(3)$ is a multiple of the identity, which the twenty-four octahedral rotations enforce exactly. A test tensor of trace 4 averages to 1.333333333 on every diagonal entry with the off-diagonals at machine zero, so the image stress there is isotropic and its anisotropy is identically zero. Inside a hole the isotropy is only the $SO(2)$ about the radial direction, which fixes the contact image too but keeps the radial and transverse pressures distinct: the same tensor averages to 0.50000 on both transverse entries and 3 on the radial one, the closed form matching a twenty-thousand-point quadrature of the group average to 2×10^{-13} .

The calibration therefore carries one combination where a hole carries two, and the one that distinguishes them vanishes identically there. Its negative value says that some $\rho + p$ is negative at a hole without saying which, and conservation says that at conformal coupling exactly one of them is. What decides which is A.18's split into radial and transverse pressures. The contact contraction there is negative, and it is the contraction built the same way in both geometries, a timelike direction plus a transverse one, so the anisotropy inherits the sign, $X = -2A$ comes out positive, and the null convergence condition holds along Penrose's ingoing radial congruence. It fails along the contact direction instead, which is where the conjecture wanted it to fail. This appendix therefore claims the magnitude and the sign, with the radial half of the sign resting on conservation rather than on a sum done in the interior.

None of that is a claim about singularities. The contraction along Penrose's congruence comes out positive, so his hypothesis holds where he uses it and the theorem stands untouched; what the fold's term does is defocus the contact direction, which is a statement about the two sheets touching. The arithmetic caveat is separate. A D^{-1} growth at a caustic is where a Hadamard expansion stops being a guide to a value, so the crossing point $D_* = cm^2/B_*$ is written in terms of a coefficient not evaluated here. A.19 closes the larger part of that coefficient: the Green function's amplitude at the caustic is a length and nothing else, and the length is the projection's arc length $M(\pi + 2)$, so $c_1 = \sqrt{\pi} M(\pi + 2)$ and $\Delta^{1/2} \rightarrow 3.9004 M s^{-1/2}$ on the contact geodesic. The step from that to the stress tensor's coefficient is what remains.

One thing does not carry over, and it is stated here. The proof that the maximiser sits at $X \equiv 0$ needs $C(r) = r(2\kappa - f') + 2f < 0$, which holds for Schwarzschild and for every Tangherlini n and fails for every

$Q \neq 0$ in a shell outside the Cauchy horizon. The reason is one line: $r_+ r_- = Q^2$ and $r_+ + r_- = 2M$ give $f'(r_-) = -2\sqrt{M^2 - Q^2}/r_-^2$, finite but negative because f falls through zero going outward, so $C(r_-) = 2\kappa r_- + 2\sqrt{M^2 - Q^2}/r_-$ is positive at every nonzero charge, with no threshold anywhere. At the radius the result is about it survives longer, $C(M) = 2s/(1+s)^2 - 4s^2$ with $s = \sqrt{1 - Q^2/M^2}$, which stays negative until the root of $2s^3 + 4s^2 + 2s - 1 = 0$ at $Q/M = 0.9548$. What survives at charge is one direction. The $E = 0$ null geodesic is an explicit curve reaching π at $r = M$, so contact holds for $r \leq M$ at every charge. Whether it also reaches somewhere in (M, r_+) is open, and settling it means maximising over $X(T)$ in the charged chart, where this argument assumes the fixed-point curve wins.

A cosmological constant has the same gap at the other boundary, and there it has an exact threshold. Redoing the scan in the patch adapted to the cosmological horizon, the condition at r_c reduces to $r_c f''(r_c) < 2f'(r_c)$; with $\Lambda r_c^3 = 3r_c - 6M$ from $f(r_c) = 0$ that fails exactly above $9\Lambda M^2 = 1/2$, where $r_c = 6M$. Inside the black-hole horizon the scan stays below -2 at every Λ , so what goes unproved above the threshold is only the exclusion of curves straying beyond r_c . Both gaps run the same way, and it is the forgiving way: a maximiser that leaves $X \equiv 0$ delivers more angle, so the contact radius becomes a lower bound and nothing already claimed shrinks. Neither bites where it would matter. A ten solar mass hole at the observed Λ has $9\Lambda M^2 = 2 \times 10^{-43}$, and at the other end, where the contact region has already grown to the whole interior, a lower bound that cannot be improved is the answer.

One nearby radius should be kept apart from this one, because a quick reader will run them together. The maximal-volume slice inside Schwarzschild sits at $r = 3M/2$, where the sphere has nine-sixteenths of the horizon area [31]. That is a different construction answering a different question, and it is not $r = M$.

Two features belong to the maximiser alone. Connecting curves in general do reach outside, to between 1.07 and 1.28 horizon radii depending on the dimension. The maximiser never leaves the closed interior; its outermost point is $r = 2M$, reached once, at the bifurcation surface. And at $r = M$ it is the only curve that reaches π , so contact at the outer edge of the region passes through B alone. That is a statement about the boundary; the region behaves otherwise, since deeper in there is slack, and curves missing B by a finite margin work too. At $r = 0.5M$ the maximum sweep is 4.19 and a curve displaced by 0.15 in X still returns 3.73, comfortably past π .

The same calculation in D dimensions. Run it in D dimensions and it selects the dimension, which is the one place in this work where the fold says anything about how many there are. Schwarzschild-Tangherlini has $f(r) = 1 - (r_h/r)^n$ with $n = D - 3$, a simple zero at r_h and the same causal diagram. The interior bound is unchanged, $d(\text{angle}) \leq dr/(r\sqrt{|f|})$, and substituting $w = (r/r_h)^n$ turns the leg into a Beta integral:

$$\Delta\phi_1(r) = \int_r^{r_h} \frac{dr'}{r' \sqrt{|f|}} = \frac{1}{n} \int_{(r/r_h)^n}^1 \frac{dw}{\sqrt{w}\sqrt{1-w}} = \frac{2}{n} \left[\frac{\pi}{2} - \arcsin \left(\frac{r}{r_h} \right)^{n/2} \right].$$

So the whole interior, horizon to singularity, supplies exactly π/n of turning to any causal curve crossing it, whatever the mass. In four dimensions that is π : half a revolution, and no causal curve inside a Schwarzschild black hole can turn further than that.

That four-dimensional statement is not ours. Hamilton and McMaken [32] derive the same integral for null geodesics of unbounded angular momentum and invert it to a cardioid that closes on the horizon at the observer's antipode, which is the same π . We use it as a lemma and the D -dependent form above is the only part of it we add. The proper-time analogue, $\tau \leq \pi M$ from horizon to singularity, is older and standard; the two integrands differ only by a factor of r , which bears noticing, since the familiar bound is the one with the r in it.

The maximiser is still $X \equiv 0$ in every dimension. Inside the horizon $d \log g^2/dr = (n/r_h)[(r_h/r)^{n+1} - 1]/f - 2/r$ has a positive numerator over a negative denominator plus a negative term. That is not by itself enough, because the curves being excluded are the ones that stray outside, so the same quantity has to be checked there: for $r > r_h$ the numerator turns negative and f turns positive, and the ratio stays negative, with the maximum over $r \in (1.001, 50)r_h$ running from -1.06 at $D = 4$ to -6.04 at $D = 9$. So the excursions lose in the exterior too and the two interior legs are the whole sweep.

The leg count itself does not depend on the maximiser and is better argued directly. A future-directed causal curve has $dU \geq 0$ and $dV \geq 0$, so once it leaves P it cannot return, and it cannot visit both exteriors because every point of R is spacelike to every point of L . The itinerary is P , then at most one exterior, then F : one leg in each interior and no more.

The same monotonicity settles the count for spacetimes whose maximal extension is an infinite tower, which was the obvious worry about charge and about Λ . A curve that could cross an inner horizon into a fresh universe could accumulate angle without limit. It cannot. Both null coordinates increase, so the curve joining Θx to x is confined to $|UV| \leq UV(r)$, and a second horizon is reachable only if it sits at finite $|UV|$ in the patch built on the horizon the fold reflects. Since $UV = -\text{sgn}(f) e^{2\kappa r_*}$, that asks whether $r_* = \int dr/f$ stays finite there, and at a non-degenerate horizon it does not: it diverges logarithmically, so the next horizon sits at infinite $|UV|$ and the tower behind it is unreachable. The count is two for the charged and the de Sitter cases as well, and the only thing it needs is that the reflected horizon be non-degenerate, which fails only at extremality and at Nariai. That sweep is $2\pi/n$. The distance required is π , fixed by the antipode being a free involution of a sphere and by nothing else. They meet only for $n \leq 2$:

D	$n = D - 3$	sweep $2\pi/n$	where the sheets can touch
4	1	2π	$r \leq r_h/2$
5	2	π , approached and never reached	nowhere
≥ 6	≥ 3	$\leq 2\pi/3$	nowhere

The five-dimensional entry of the table above needs care and it comes out empty. At $n = 2$ the two-leg sweep is $\pi - 2 \arcsin(r/r_h)$, which is strictly below π for every $r > 0$ and reaches it only in the limit $r \rightarrow 0$. That limit is not a point of the spacetime, so the contact set is empty there too, and four dimensions is the only one in which the sheets touch anywhere at all. Above five the sweep is short outright. The distance required is π because the antipodal map is an involution; the sweep is π/n per leg because f has a simple zero and falls as $r^{-(D-3)}$; and the two happen to match in four dimensions, with a factor of two to spare, which is why the contact region here is a region rather than a point.

A.16 identifies two loci at which the fold joins the sheets structurally, the bifurcation surface and the singularity, and those are there in every dimension. The causal contact region is not. In four dimensions it is a neighbourhood of the singularity reaching out to the midpoint of the interior; in five and above there is none, while A.16's two loci remain untouched. That is the clearest statement of the difference between the two kinds of joining. The fold's map still relates the sheets at the singularity whatever the dimension, and outside four nothing can use it.

One algebraic fact underneath all of it. The charged case and the dimensional one look unrelated and are the same observation about $r^2|f|$. Whenever it is a **quadratic whose two real roots are the ends of the $f < 0$ region**, the explicit r cancels from $r\sqrt{|f|} = \sqrt{(b-r)(r-a)}$, the leg integral is an arcsine, the total is

$$\int_a^b \frac{dr}{\sqrt{(b-r)(r-a)}} = \pi$$

for any $a < b$ whatever, and the contact region is exactly the lower half of the interval, $r \leq (a+b)/2$. Checked on random pairs belonging to no metric at all. The metrics with $r^2 f$ quadratic are exactly $f = A + B/r + C/r^2$, which is the Reissner-Nordstrom family with Schwarzschild at $C = 0$.

That one fact does four jobs. Schwarzschild has roots 0 and r_h , both ends of the region, so the total is π and the midpoint is $r_h/2 = M$. Charging the hole replaces the root at the singularity with the inner horizon and changes nothing else: the total is still π , at every charge, and the midpoint is $(r_+ + r_-)/2 = M$, the same M . In five dimensions $r^2|f| = r_h^2 - r^2$ is still a quadratic, but its roots are $\pm r_h$ and the region runs from 0 to r_h , which is the upper half of the root interval, so the sweep is exactly half the arcsine, $\pi/2$, and that is where the marginality comes from. From six dimensions up it is not a quadratic at all and π/n is not an arcsine.

A cosmological constant, which breaks the rule the other way. Schwarzschild-de Sitter has $r^2|f| = 2Mr - r^2 + (\Lambda/3)r^4$, a quartic, so the midpoint rule does not apply and the answer moves. It moves outward, monotonically: at $9\Lambda M^2 = 0.1$ the contact radius is $1.047M$ and 0.515 of the horizon, at 0.5 it is $1.311M$ and 0.597 , and at 0.9 it is $1.990M$ and 0.778 , with the total turn rising from π to 4.21 over the same range. So a positive cosmological constant deepens the contact region, and at the Nariai point, where the two horizons merge and the quartic acquires a double root, the sweep diverges and the whole interior becomes contact-capable. The ratio runs 0.919 at $9\Lambda M^2 = 0.99$ and 0.991 at 0.9999 . Above $9\Lambda M^2 = 1/2$ those figures are lower bounds rather than suprema, for the reason given with the charged case above, and they climb toward one either way. That last statement was worked out here for its own sake and it turns out to name the same configuration the involution count later picks: Nariai is both the only member of the Schwarzschild-de Sitter family the fold can fix and the only one with no forbidden outer shell. Neither calculation was looking for the other.

Set beside the charged case that makes a clean pair. Charge leaves the fraction at exactly one half because it keeps r^2f quadratic; Λ moves it because it does not; and extra dimensions break it a third way, by moving the region off the span of the roots. The dividing line is algebraic and not physical, which needs knowing before anyone reads significance into the one half.

A check where this overlaps the rest of the paper. Pure de Sitter, outside the cosmological horizon, has $r^2|f| = r^4/\ell^2 - r^2$, and substituting $u = \ell/r$ gives a total turn of exactly $\pi/2$ for every ℓ , so the two-leg sweep is exactly π . Elliptic de Sitter sits precisely on the threshold, never over it. That is the same statement Section 1 makes on entirely different grounds, that antipodal points of de Sitter are spacelike separated with the bound saturated only in the limit, and the two calculations agree where they meet.

Where the dimension argument sits. Arguments that single out four dimensions are a small and old genre, and this one is placed in it here. Most entries assume something about matter or observers: Ehrenfest on stable orbits, Tangherlini on atoms, Barrow and Tipler on undistorted wave propagation, Tegmark on predictability. The closest relative assumes neither, and it is a century old. Weyl [33] observes that the Maxwell action is the simplest integral invariant that exists at all, that no such invariant of that structure exists in any other dimension, and that the four-dimensionality of the world, previously taken as accidental, becomes intelligible through it. His selection rests on the degree of a form. This one rests on a causal sweep, and the two are independent of each other and of everything in the genre that assumes an observer. We claim no more than that.

Two limits bound how far to carry that. It is Schwarzschild-Tangherlini, so it says nothing about black rings or other horizon topologies, where the transverse space is not a sphere and the antipodal map is not the relevant involution. And the fold is built here as a four-dimensional construction, so the higher-dimensional cases test what the argument depends on. No claim about a higher-dimensional world is made. What they show is that the four-dimensional result is not robust to the thing one might expect it to be robust to. It sits one dimension away from being empty.

Spin. The fold map itself carries to Kerr exactly, as the end of this section shows. The contact derivation does not carry as it stands. It uses spherical symmetry twice: once to say that $P_\perp$ is antipodal and the required angular path is π at every point, and once to reduce the causal question to a two-dimensional conformal problem whose answer depends only on UV . A rotating interior has neither, and it has an inner horizon and a ring singularity besides.

There is a route through it, though, and the obstruction above looks fatal and is not. In the ingoing angle the sweep diverges: frame dragging turns the inverse square root into a simple pole at each horizon, and a simple pole is not integrable. In the co-rotating angle $\phi_K = \phi - \Omega_H t$, which is the one regular at the bifurcation surface where the optimal curve turns round, the bound is $r\sqrt{N}/|\Delta|$ with N the norm squared of the horizon generator. N has a simple zero at r_+ , so the integrand goes as $(r_+ - r)^{-1/2}$ and the sweep is finite. A first pass on that footing gives a contact radius falling from $r_c = M$ at $a = 0$ to about $0.89M$ at $a = 0.9$, with the fraction of the interior band it occupies falling monotonically from exactly one half to about a third. So spin shrinks the contact region where charge left it alone.

Which isometry the fold uses is settled, and the apparent ambiguity came from naming the angle shift by a coordinate instead of by a flow. Take it to be the flow of the axial Killing field $\psi = \partial_\phi$ through parameter π ,

composed with the equatorial reflection $E : \theta \mapsto \pi - \theta$, and there is one map. It commutes with the wedge reflection N for a reason that needs no calculation: N is built from the horizon generator $\xi = \partial_t + \Omega_H \partial_\phi$, so any isometry preserving ξ preserves the Kruskal structure and commutes with N ; ψ preserves ξ because $[\partial_t, \partial_\phi] = 0$ with Ω_H constant, and E preserves ∂_t and ∂_ϕ separately. So $\Theta = N \circ e^{\pi\psi} \circ E$ squares to $N^2 \circ e^{2\pi\psi} \circ E^2 = 1$, with nothing left to choose.

It is free without needing N at all, since $e^{\pi\psi} \circ E$ fixes a point only where $\theta = \pi/2$ and $\phi = \phi + \pi$ together, and it is an exact isometry of Kerr at every spin because the metric functions depend on θ through $\cos^2 \theta$ alone and on ϕ not at all. The bifurcation surface is genuinely not round, its polar area radius exceeding its equatorial one by 5.41 per cent at $a = 0.6M$ and 37.15 per cent at $a = 0.998M$, and the map does not care. On that surface it is the antipodal map, which is the same transverse structure the non-rotating case uses.

The contact radius above stays a conjecture even so, because settling the map settles the fold and not the angular sweep, and the sweep on a distorted sphere is a separate calculation, and that calculation is what the Kerr case needs beyond the map.

What this does not do: The image of infalling matter sits in P , which is in the *past* of both exteriors, so the fold returns information to the past and not to the future. That is not a solution of the information problem as usually posed and we do not offer it as one. Structurally it is a consistency condition between a future boundary and a past one, which is the shape already used in §3.3 and A.9.

What this narrows is the domain. These results do not all lean on the same thing, and separating them narrows it further. The reciprocal law needs $P_\perp^2 = 1$ and positivity, so it is indifferent to the state. Five states, thermal, flux-carrying, wildly non-thermal, oscillating, and a power law with no temperature at all, each satisfy $R(x, y)R(x, P_\perp y) = 1$ to machine precision. The ninety-degree equality and the odd-multipole statement follow from it and inherit that indifference, as does the mass-independence in the cosmology paper's §3. A.10's reduction needs B . Exactly one claim in these appendices needs the KMS property, that $\alpha^2 = 1$ fixes the temperature, and four need the two-sided structure, that same claim, the transparent seam, the meeting region above, and A.16's consistency condition.

Evaporation, which is what breaks KMS, therefore threatens one line of it. The appendix survives. Its size is $(\kappa t_{\text{evap}})^{-1} \sim M^{-2}$ in Planck units, between 10^{-77} at $3M_\odot$ and 10^{-94} at $10^9 M_\odot$, and the violation it produces is linear in the departure rather than discontinuous in it, unlike A.16's condition, which empties at any breaking whatever. We have not calculated the coefficient, a physical non-equilibrium correction being nothing like the Gaussian used to test continuity, but the conclusion tolerates more than fifty orders of magnitude of it.

What collapse takes is different, and smallness does not repair it. A hole formed by collapse has no P and no L , the collapsing matter replacing them, and no bifurcation surface. For such a hole the $F \leftrightarrow P$ statement is empty, “both sheets meet in F ” has one sheet, and A.10's reduction has no B to sit on. Those results belong first to the cosmological horizon, which is bifurcate, which we are inside, and on which the fold supplies $P_\perp$ with nothing chosen, and second to eternal black holes. The count above says those two are not really two: the fold fixes a black hole's horizon only when that horizon is the cosmological one. Everything that needs neither B nor a mirror sheet survives for a collapse hole, the ringdown statements among them, since they are boundary conditions on an exterior a settled collapse hole has. The observational programme of the cosmology paper's §3 is untouched. That is the domain of this appendix.

A.16 The singularity is the fold's second contact, and what it requires there. §5.3's second joining place. A.15's map does one more thing, and it locates the fold's other contact. In Kruskal coordinates the curvature singularity is the hyperbola $T^2 - X^2 = 1$, whose future branch bounds F and whose past branch bounds P ; since J is $(T, X) \mapsto (-T, -X)$ it exchanges the two branches, verified on 2×10^4 points of the locus. So under the fold the black hole's future singularity and the white hole's past singularity are one locus, and the fold has exactly two loci where it joins the two sheets structurally, the bifurcation surface and the singularity. Only the first has been worked out here.

That is a different sense of joining from A.15's, and the two should not be run together. A.15 asks where a point can causally influence its own image and answers $r \leq M$; this asks where the fold's map relates the sheets at all, and answers at two surfaces. The singularity lies inside A.15's region and the bifurcation

surface lies outside it, at $r = 2M$, so a point of B is not causally connected to its own image even though the fold carries B to itself. What B does is carry the curve, the one that attains A.15's bound passes through it. A.15's domain caveat applies with full force: a hole formed by collapse has no past singularity to be identified with, so for an astrophysical black hole this section says nothing.

That changes the character of the interior boundary condition without resolving anything. A single-sheet theory must say what happens at the singularity as a terminal condition. The fold instead requires the data on that locus to be compatible with its own image under J , which is a self-consistency requirement of the kind §3.3 already uses, and which is the interior counterpart of $\alpha^2 = 1$ being the smoothness condition at B . It does not resolve the singularity. The curvature still diverges there in both sheets, and identifying two divergences does not remove them.

That requirement does have solutions, and counting them is finite work. Data leaves the past branch, crosses under the evolution U , and reaches the future branch, where J returns it, so consistency reads $Ux = Jx$, equivalently $(JU)x = x$ since $J^2 = 1$. If the evolution is covariant under the fold's map, $JUJ = U^{-1}$, then $(JU)^2 = 1$ and the condition becomes an eigenvalue problem for an involution. We build it explicitly on a $2n$ -dimensional phase space with J the antisymplectic involution $(q, p) \mapsto (q, -p)$ and $U = \exp A$ with A anticommuting with J inside $\mathfrak{sp}(2n)$, which is an ordinary kinetic-plus-potential system: $(JU)^2 = 1$ to the arithmetic floor, the spectrum is ± 1 with no imaginary part, and the $+1$ eigenspace has dimension exactly n .

Nothing observed sets that dimension; the counting does. For an involution $\text{tr } M = \dim(+1) - \dim(-1)$, so the split is even exactly when $\text{tr}(JU) = 0$. Since A anticommutes with J , A^2 is block diagonal with blocks bc and cb ; the diagonal blocks of $\exp A$ are the same series in each; and bc and cb share a spectrum, so the two traces cancel. It is the identity $\text{tr } f(XY) = \text{tr } f(YX)$ and nothing more. Checked at $n = 2, 5, 9, 16$ over two hundred draws apiece, with $\text{tr}(JU)$ and $(JU)^2 - 1$ both at the floor of whichever matrix exponential is used, and the dimension n every time. It remains n when b and c are not symmetric, so the count comes from A anticommuting with J rather than from the mechanics placed inside it, and letting A commute with J instead breaks both the involution and the trace. The condition is therefore neither empty nor over-determined. It keeps exactly half the data, through the projector $(1 + JU)/2$.

That projector has the form of the Keldysh split the cosmology paper's §2.1 already uses, $\Phi_c = (\Phi + \Theta\Phi)/2$, with the loop's involution standing where Θ stands; the two differ by the evolution between the branches, and we record the parallel, claiming no identification. What the interior wants is a projection, and the projection is the even sector of a $\mathbb{Z}_2$, as at B .

Covariance is load-bearing here. Adding to A a piece that commutes with J instead of anticommuting, with strength ϵ , destroys the involution and the solution space with it: the kernel has dimension n at $\epsilon = 0$ and is empty at $\epsilon = 10^{-6}$ and above, with the smallest singular value of $JU - 1$ bounded away from zero. The covariance at issue is not T but Θ , which is CPT, so it comes from a theorem and is untouched by T violation in the kaon system. The interior condition has solutions because CPT holds and would have none otherwise, which puts weight on that assumption in a second place independent of §2's.

One reading of that projector needs checking, because it is the natural one and it is wrong in general. A projection that keeps half the data invites being read as a measurement, with the singularity collapsing the state onto its even sector. That requires the projector to be self-adjoint, and $(1 + JU)/2$ is not: U is symplectic rather than symmetric, JU inherits that, and the projector comes out idempotent but oblique, with range and kernel that are not orthogonal complements. An oblique projection carries no Born rule and is not a measurement.

It becomes one in exactly one limit. The obliquity, measured as the departure of the principal angles from orthogonality, is 0.9° at evolution parameter 0.01 and grows to 82° at 2, vanishing only as the evolution between the branches does. So the collapse reading is exact in the constraint formulation and nowhere else. That is the third of the three results the cosmology paper's A.4 records as depending on the constraint formulation, reached here from the projector's obliquity rather than from the wavefunction.

All of this is linear, classical and finite-dimensional. An interacting field theory is not a 12×12 matrix, and nothing here shows the count survives that.

One compatibility question belongs here, because it has a definite answer. The condition above keeps the $+1$ eigenspace of JU , and a Θ -invariant state lives in the $+1$ eigenspace of Θ , whose projector is the cosmology paper’s §2.1 Keldysh split. Whether one state can satisfy both turns entirely on how the sheet swap inside Θ sits against the time reversal in J . Measuring that rather than assuming it gives three cases. If the two anticommute, the common sector is empty at every separation including zero. A trivial swap leaves the sector full. If the swap *is* the time reversal, the sector is full at zero separation and empty otherwise.

A.15 settles which case applies. The fold’s map is $(T, X) \mapsto (-T, -X)$, effecting $R \leftrightarrow L$ and $F \leftrightarrow P$ together, so the swap is not an independent involution standing alongside the time reversal but the same map acting on the wedges. That is the third case, and the two conditions therefore share a sector exactly when there is no evolution between the branches: the Wheeler-DeWitt situation, where the state solves a constraint and there is no t to evolve along.

This is another route to the cosmology paper’s A.4 requirement. The conflict is not general: it arises only if the swap and the time reversal are taken as independent and anticommuting, which A.15 excludes.

A.17 Whether the fold can dispense with the Hawking temperature. §4’s question about the Hawking temperature. Identifying the two singular branches invites the question of whether matter may transit the locus instead of being radiated away, making the Hawking temperature dispensable. The answer here is partial, and the limit is the fold’s own. A.10 shows that in the Bunch-Davies state $\alpha^2 = 1$ can be read as $W(t - i\beta) = W(t)$, the KMS condition, at the primitive period $\beta = 2\pi/\kappa$ and, as A.10 records, at its odd multiples. The reading is what a non-thermal state gives up, and not the identity: $\alpha^2 = 1$ holds because J and $P_\perp$ are commuting involutions, which is a statement about the geometry and is true whatever state the field is in. The involution itself does not break, which reopens the question it was meant to settle. In frequency space that is detailed balance, $\tilde{W}(-\omega) = e^{-\beta\omega}\tilde{W}(\omega)$, and a departure from the thermal occupation violates it at first order with no tolerant window, a fractional departure 10^{-4} gives a violation 8.55×10^{-5} , and 10^{-2} gives 8.55×10^{-3} . So a state that departs from the thermal occupation loses the detailed-balance reading quickly. What it does not lose is $\alpha^2 = 1$. The fold’s involution property therefore does not by itself make the Hawking temperature compulsory. Whether the temperature can actually be dispensed with, with the locus absorbing what would have been radiated, is left open here, and turns on the statement of domain below rather than on the involution property.

Flux is a separate matter and falls outside the domain, unsettled. An equilibrium state at T_H is KMS and carries no net flux; an evaporating hole sits in a state that is not KMS in Killing time, so $\alpha^2 = 1$ fails for it and A.10 to A.15 do not apply. That is the same boundary this appendix records for collapse holes, reached from a different direction. And the transit cannot substitute for evaporation in any case, since the image of infalling matter lies in P , in the past of both exteriors, while evaporation is a statement about the future. The two answer different questions.

What that leaves is a statement of domain, and a narrower one than it may look. The restriction bites on the KMS-dependent claim and on nothing else. Of the results above exactly one needs KMS, namely $\alpha^2 = 1$ fixing the temperature, which is this appendix’s subject. The cosmological horizon is in equilibrium and genuinely KMS, and so is an eternal hole in the Hartle-Hawking state, so that claim holds without qualification on both. What it does not cover is a hole that formed and will evaporate. Extending to astrophysical holes needs the departure of their state from equilibrium, of order $(\kappa t_{\text{evap}})^{-1}$, to degrade the predictions smoothly rather than void them. That calculation would widen the domain and is not done here.

A.18 The image singularity at a caustic, computed without a parametrix. The power of the divergence in §5.3. The reading under test is that the fold censors its own closed causal curves. The world function between a point and its image vanishes linearly at the contact sphere, so the image term of the two-point function blows up there and the stress tensor with it. Counting the blow-up through the Hadamard parametrix gives an image term $\Delta^{1/2}/\sigma$ and a stress tensor s^{-3} , and that count needs Δ finite at the image point, which it is not. The fold’s transverse factor is the antipodal map on a round S^2 , every great circle through a point refocuses at its antipode, and the image point sits on the caustic.

What fails there is an approximation. A parametrix carries a Van Vleck factor and a mode sum does not, so the exact object can be calculated where the approximate one cannot, and the two ingredients that computation needs are both in hand. The state is the first, and A.8 settles it: at a bifurcate Killing horizon

the propagation gives the thermofield double at $\tanh r = e^{-\beta\omega/2}$, carrying the transverse parity that separates $J \circ P_\perp$ from J , which on the cross-sheet term is the insertion of $(-1)^\ell$ per multipole. That insertion and the antipodal map are one operation, by $P_\ell(-x) = (-1)^\ell P_\ell(x)$, which holds to 4.6×10^{-14} over the cases checked. That is the identity §3 already uses to empty the even multipoles of the horizon law, doing a second job here. The second is a geometry in which the sum closes.

The geometry used is $-dt^2 + dr^2 + a^2 d\Omega^2$, which is not Schwarzschild and is not offered as it. It is the smallest geometry carrying the feature at issue: null geodesics leaving a point spread over every great circle of the transverse sphere and refocus, all together, after exactly π of transverse angle, so the connecting family at the image point is one-parameter, the same degeneracy the contact sphere carries. Reducing on the sphere leaves a tower of two-dimensional fields of mass $m_\ell = \sqrt{\ell(\ell+1)}/a$ and

$$W(x, x') = \sum_\ell \frac{2\ell+1}{4\pi} P_\ell(\cos \gamma) W_2(m_\ell; \Delta t, \Delta r).$$

The tower's terms grow as $\sqrt{\ell}$ at the image pair, because the antipodal parity cancels the transverse phase and every multipole arrives in step. That coherence is the singularity, so a taper in ℓ would not regulate the sum, it would set the answer. The Wightman function's own $i\epsilon$ is the regulator that belongs to the object, and it damps the tower by $e^{-m_\ell \epsilon}$; taken at a fortieth of the offset and varied over a factor of four it moves the sum by 1.8×10^{-3} .

Both configurations are approached from outside the lightcone by an offset s , and the comparison is like with like because the world function vanishes linearly in s in each: $\sigma \simeq s$ at the generic pair, where the separation is $\Delta r = 1$ and $\Delta t = 1 - s$, and $\sigma \simeq \pi s$ at the image pair, where it is the full π of transverse angle against $\Delta t = \pi - s$. So the two fitted powers are powers of the same kind of variable, which is also the variable §5.3's linear zero of σ is written in. A generic null pair, no caustic, returns a fitted power -1.0095 against the known -1 , which is the case that could have failed and the reason the other number means anything. The image pair returns -1.4920 on the widest window and sharpens to -1.49674 and then -1.49874 as the window closes, with the largest log-residual falling from 2.6×10^{-3} to 4×10^{-4} , the power is $-3/2$. Removing the antipodal parity by hand, leaving the configuration otherwise untouched, removes the divergence altogether at $+0.0004$.

The coefficient carries information too. Summed at a fixed offset, the argument of the result halves each time ϵ halves, tracked over four halvings with the product of phase and regulator constant to 3×10^{-3} , so the phase belongs to the regulator and the limit sits on the positive real axis while the magnitude holds at 502. The divergence is real and positive and does not oscillate.

So the caustic steepens the divergence by half a power. Its image term goes as $s^{-3/2}$ where the parametrix had s^{-1} , and a stress tensor quadratic in derivatives goes as $s^{-7/2}$ where the parametrix count had s^{-3} . So the self-censoring reading holds with a steeper exponent than the parametrix count gives it, on a calculation that carries no Van Vleck factor at all and therefore does not care that the parametrix fails here.

The same sum gives the components, which is what the sign needs, since a positive coefficient in the two-point function is not a sign for the stress tensor. Three scalars carry the whole tensor at the image pair, the isotropy there being the $SO(2)$ of rotations about the axis through the point and its antipode: the second time derivative D , the second transverse derivative Q and the second radial derivative E . The transverse Hessian is isotropic at the antipode because $P_\ell(\cos \gamma)$ there is a function of the squared distance from it, $\cos \gamma = -1 + |u - v|^2/2$ with u and v the tangent displacements of the two ends, so differentiating once at each end and twice at one end return the same number; parallel transport along the connecting geodesic reverses a great circle's own tangent, which puts the two on opposite signs. Measured against the exact Legendre function that relation holds to 10^{-4} at ℓ from 1 to 40, and the transport is checked in $\mathbb{R}^3$.

Two identities fix the fold and neither is imposed. Each mode obeys its two-dimensional field equation and carries the sphere's eigenvalue, so $\square W = -D + E - 2Q$ must equal $(m^2 + \xi R)W$ exactly. It does, to 3×10^{-7} against a D of 6×10^5 , and scaling Q by a thousandth breaks it. That in turn forces the trace here to be $T^a_a = -m^2 W$, so in this geometry a massless field's image stress is traceless at every offset and not

only at leading order, and the computed trace sits at 4.8×10^{-14} of the energy density. The restriction to this geometry is not cosmetic: the step uses $g^{cd'} W_{;cd'} = -\square W$, which holds because W depends on the coordinate difference here, and A.15's general trace, carrying $(6\xi - 1)\frac{1}{2}\square\langle\phi^2\rangle$ with the image point moving alongside the point, is the one that governs a hole. The image term itself reproduces this appendix's own 501.9 at an offset of 10^{-3} and a fitted power of -1.4916 , and the non-caustic controls return -0.9934 and -2.999 where the parametrix predicts -1 and -3 .

What comes out is a traceless, zero-density, purely anisotropic divergence. D and Q both go as $s^{-7/2}$ with $Q = -D/2$, E goes as $s^{-5/2}$ and drops out, and the components are $\rho = -15s^{-7/2}/32\pi\sqrt{2\pi} = -0.059525 s^{-7/2}$, $T^r_r = 0$ and $T^\theta_\theta = -\rho/2$, a pattern $T^t_t : T^r_r : T^\theta_\theta = 2 : 0 : -1$, returned as $1.9997 : 0.0003 : -1$. Both of that geometry's null contractions are negative, $-D$ along its radial direction and $-D + Q$ along the contact one, in the ratio 1.5003 against $3/2$. Neither sign moves with the coupling or the mass, checked at six pairs. The flat calibration's sign flips as ξ crosses $1/6$ because it is proportional to $m^2 + (\xi - \frac{1}{6})R$; a caustic's cannot, because ξ and m enter the tower only through a constant shift of μ_ℓ^2 and through terms two powers softer than the divergence.

One contraction carries over to a hole and one does not, and what decides which is causal character. Inside a horizon $f < 0$, so r is timelike and t spacelike, and the same three components mean different things. The contact geodesic there has $E = f\dot{t} = 0$ with f nonzero, so its tangent is a timelike direction plus a transverse one, which is what the contact direction is here; this geometry's radial direction is its spacelike one and its transverse sphere has constant radius, so it carries no tidal field and its conservation law is empty. The contact contraction therefore carries over, the anisotropy at a hole is negative, and A.15's conservation chain puts $X = -2A$ above zero.

What does not carry over is the component pattern, and the reason also explains where the pattern comes from. Near a null caustic the leading term collapses to $T_{ab} = -C\frac{15}{4}\sigma^{-7/2}\nabla_a\sigma\nabla_b\sigma$, the trace piece being two powers softer since $\nabla\sigma \cdot \nabla\sigma = 2\sigma$, so the leading stress is rank one along the connecting geodesic and traceless. At a caustic the connecting geodesics are a family, so what acts is the average of $k_a k_b$ over it, which is $\text{diag}(1, 0, \frac{1}{2}, \frac{1}{2})$ in the frame the contact tangent picks out, and that is $2 : 0 : -1$ exactly, with no sum required. Read in a hole's labels the timelike slot is r , so the pattern would put the 2 on T^r_r ; and then conservation forbids it, the dT^r_r/dr term carrying $D^{-9/2}$ against $D^{-7/2}$ for the other two, so the ratio of what could cancel it to what must be cancelled falls as 5.8×10^{-3} , 5.7×10^{-4} , 5.7×10^{-5} over three decades. Here that check is empty, because the coefficient does not depend on position and the transverse radius is constant.

What that rules out is the rank-one form as a hole's leading term, and not the pattern, which conservation returns by a route of its own. Three statements fix it and the rank-one form is none of them: the leading divergence is traceless, since $T^a_a = -m^2\langle\phi^2\rangle$ is two powers soft; the transverse pressures are equal, from the $SO(2)$ about the radial direction; and T^r_r vanishes at leading order, which is what $Y' = (f'/2f)X + 2A/r$ gives, putting Y one power below X and A so that Y/X falls as D . The trace then reads $T^t_t + 2T^\theta_\theta = 0$ and the pattern is forced. So both geometries empty the same slot and both are traceless, for reasons with nothing in common. Here because the radial direction is not in the connecting tangent at all, at a hole because conservation empties it while the tangent runs straight through it. Two unrelated routes to one pattern counts for more than a transfer would have been.

One more thing has to match for a sign to carry, and it is the caustic count, because a crossing supplies a phase and the leading real part is what holds the sign. Every scalar above takes the same phase, so the pattern and the ratio $3/2$ do not feel it at all: a uniform extra θ multiplies each real part by $\cos\theta$, which vanishes at $\pi/2$ and turns over beyond it. This geometry has one crossing, the transverse one, and it is carried by $P_\ell(-1) = (-1)^\ell$ itself. A hole would have a second if a second direction refocused there, and it does not. The count is the dimension of the family of null geodesics running from a point to its image, so geodesic integration measures it.

In this geometry the sphere's share of a null direction has to be exactly all of it for the image to be reached at $\Delta r = 0$, which leaves one free parameter, the choice of great circle, and two circles tilted by δ separate as $\delta \sin$ of the arc and close again at π . In the interior the same is true for the same reason: spherical symmetry makes the angular sweep a function of r alone, so every tilted member reaches π at $r = M$, while giving the

geodesic an energy sets it back twice over: $\dot{r}^2$ at the horizon is E^2 rather than zero, so the member does not turn there and leaves the interior instead of entering the other one, and the angle it sweeps on the way out is smaller at every radius because $\dot{r}$ is larger. At $E = 1$ the sweep falls short of π by 2.268 and at $E = 0.01$ by 0.0777. One degenerate direction in each.

Why the second direction fails to focus differs between the two and does not need to agree. Here it is flat, the $2d$ factor being Minkowski, so its Jacobi field is λ and never returns. There it defocuses, and it has to, because Schwarzschild is Ricci-flat: $R_{ab}k^ak^b = 0$ leaves the optical tidal matrix trace-free, so one screen direction focuses exactly as much as the other defocuses. Both $R_{tt} = f(f''/2 + f'/r)$ and $R_{\theta\theta} = 1 - f - rf'$ vanish identically for $f = 1 - 2M/r$, and adding charge gives $R_{\theta\theta} = Q^2/r^2$, which does not, so the test has something to catch.

Equal counts, so no relative phase from a conjugate point. One difference between the two geometries has to be named and disposed of before that means anything. On Schwarzschild the connecting curve turns in r , its $\dot{r}$ vanishing at the horizon where $|f|$ does, and nothing here corresponds. That turning belongs to the Killing slicing and not to the curve. In Kruskal coordinates the bifurcation region is smooth, the connecting curve runs through it as one smooth null geodesic, and r merely happens to take a maximum along the way. A caustic is where a Jacobi field vanishes and none does there. Reality caps what could still go wrong: a Wightman function at spacelike separation is real, so any residual phase is a multiple of π and the sign either carries exactly or turns over exactly, with nothing between.

And it carries, because the quantity whose phase this is can be calculated both ways and comes out the same. A.19's rule replaces the divergent Van Vleck factor at a caustic by $\Delta^{1/2} \rightarrow \Delta'^{1/2}\sqrt{\pi}Ls^{-1/2}$, with L the projection's arc length and $\Delta'^{1/2}$ the reduced factor of the one non-degenerate transverse direction. Applied to A.18's geometry that direction is flat, so $\Delta'^{1/2} = 1$ exactly, the projection is a great semicircle, and the rule predicts $\pi^{3/2}$ with nothing left to choose. Feeding it through the proper-time integral at $\sigma = \pi s$ returns $1/8\pi\sqrt{2\pi}$, which is A.18's mode-sum coefficient, to sixteen digits and as an exact cancellation. The affine total in place of L misses by 2.773, a Van Vleck factor of 0.428 by that factor, and dropping the $\sqrt{\pi}$ by 0.564. So the image term is the same positive expression in both, for the same reason: a reduced Van Vleck factor is a square root, $\sqrt{\pi}$ is positive and an arc length is positive.

Were it to turn over, the two contractions would swap roles, since A.15's conservation chain makes exactly one of them negative whichever way this goes. The structure of the conclusion is fixed and only the assignment rests on the transfer. What is left is that the radial half rests on conservation rather than on a sum done in the interior, and the same sum on Schwarzschild would remove that step. Its hardest ingredient is in hand. At zero frequency and in the eikonal limit the interior's radial equation is $r(1-r)\psi'' - \psi' + \ell(\ell+1)\psi = 0$ with $2M = 1$, which is hypergeometric with exponents $\{0, 2\}$ at the singularity and the double exponent $\{0, 0\}$ at the horizon, and its regular branch is the polynomial $\psi_\ell = r^2 P_{\ell-1}^{(2,0)}(1-2r)$. Jacobi asymptotics then give the phase in closed form, $2\nu \arcsin \sqrt{r} - 5\pi/4$ with $\nu = \ell + 1/2$, whose variable part is exactly §5's leg from r out to the horizon. Zeros fix a phase only modulo π , so the constant is pinned by the amplitude as well. The full Darboux form matches to 0.0040 at $\ell = 800$ against 1.9991 for the candidate a half-turn away, which is as far apart as two cosines reach. What the fold still needs beyond it is the measure and the saddle of the frequency integral, whose stationary point for equal-time endpoints is $E = 0$.

A.19 The energy density of the image term, and what sets its size. A.18 settles the sign in its own geometry, which A.15 calls the half that decides whether the divergence closes the region or merely marks it. Getting there turns up a larger question. Four settings are computed below, two returning zero and two returning a divergence, and what separates them is whether the field is conformally invariant, which is a statement about matter content. So this appendix runs in three parts: what makes the density vanish, what exponent the divergence carries where it does not, and what its magnitude is. Answers come for the first two. Magnitude is settled wherever the caustic sits in a Riemannian factor, and at a hole it is shown to need a treatment of a different kind.

The energy density a static observer measures for a massless minimally coupled scalar is $\frac{1}{2}[(\partial_t\phi)^2 + (\partial_r\phi)^2 + |\nabla_\perp\phi|^2]$, three terms that are positive classically. Point-split against the image kernel, each becomes a mixed derivative $\partial_\mu^x\partial_\nu^y G(x, \Theta y)$ at $y = x$, and the fold decides the sign of each through its own pullback. The map $\Theta(t, r, n) = (T - t, r, -n)$ is an isometry, an involution, free, and reverses time orientation the way the real

fold does, and its three pullbacks are $\partial_t^y = -\partial_{t'}$, $\partial_r^y = +\partial_{r'}$, and $\partial_n^y = -\partial_{n'}$, the last because the antipodal map's differential is minus the identity.

The angular contraction is the piece that decides everything. At the antipode the first term of the mixed derivative of $P_\ell(\cos \gamma)$ drops, since the projection of n' off n vanishes when $n' = -n$, and what survives is $2P'_\ell(-1) = -(-1)^\ell \ell(\ell+1)$, the transverse sphere hands back exactly its own Laplacian eigenvalue. That is the step the whole cancellation turns on, so it is checked by differencing on the sphere rather than by the projection argument alone, and agrees for ℓ up to twelve to 3.9×10^{-5} . So the three pieces are $W_{\tau\tau}$, $-W_{\rho\rho}$ and $+\ell(\ell+1)a^{-2}W$, and that combination is the wave operator, which annihilates the two-point function. The image term carries no energy density at all, at every separation including the caustic, so the divergence A.18 found is invisible in ρ here. The three pieces run to 1.9×10^9 at the smallest offset and their sum is 2×10^{-12} of that. Bessel identities supply the derivatives rather than the wave equation, so the cancellation is evidence rather than restatement of it; flipping each pullback sign in turn moves the sum by exactly the piece flipped, to one part in 10^6 ; and dropping the time reversal from Θ , which leaves a perfectly good isometry, leaves the density non-zero at 6.7×10^6 .

A zero reached that way is a statement about the geometry and not about the fold. The wave operator appeared because the transverse sphere supplies the whole of the radial equation's potential here, leaving no remainder, and the test is to give the tower a mass the sphere does not supply. Exactly that much survives:

$$\rho_{\text{img}} = -\frac{1}{2} \mu^2 \sum_{\ell} \frac{2\ell+1}{4\pi} (-1)^\ell W(m_\ell; \xi),$$

which the assembled sum reproduces to 2.6×10^{-6} at $\mu = 0.5$ and to 6.0×10^{-8} at $\mu = 2$, the agreement tightening as the residue grows.

So in this model the sign question reduces. Everything in the energy density cancels against the wave operator except the part of the radial potential the transverse sphere does not supply, and the sign is that remainder's sign times the A.18 sum, which is real and positive.

The cancellation was assembled where the transverse radius is a constant, and that turns out to be the whole of why it happens. Flat four-dimensional Minkowski is the same calculation with the radius varying: written in spherical coordinates it is $-dt^2 + dr^2 + r^2 d\Omega^2$, its fold is $\Theta(t, \mathbf{x}) = (T-t, -\mathbf{x})$, a time reflection composed with the spatial point reflection, and its differential is minus the identity on every Cartesian component, which in spherical terms is exactly the three pullbacks above. Nothing about the fold has changed except the transverse radius, which is now r , and everything is in closed form, so the step can be taken here. With all four pullbacks at -1 the fold is $\rho_{\text{img}} = \frac{1}{2}[G_{\Delta t \Delta t} + \nabla^2 G]$, the wave equation in the separation turns that into $\partial^2 G / \partial(\Delta t)^2$, and at the image pair, where $\Delta \mathbf{x} = -2\mathbf{x}$ and $\Delta t = \xi$,

$$\rho_{\text{img}} = \frac{8r^2 + 6\xi^2}{4\pi^2 D^3}, \quad D = 4r^2 - \xi^2.$$

The symbolic assembly and the wave-equation route agree with that expression exactly, each difference returning zero. The numerator is positive at every r and ξ and D is positive outside the contact surface, so the image energy density is positive there and diverges positively on approach, at a fitted power of -3.000052 in D over five halvings, the shortfall being the numerator moving with ξ at finite offset. Three is the generic exponent rather than A.18's $-7/2$, and it should be: a straight line is the only geodesic joining a point to its image here, so there is no refocusing and no caustic. That is also a check across the two appendices, by methods with nothing in common. A.18's control fits a mode sum and returns -1.0095 for the two-point function, so -3.009 for a stress tensor; this returns -3.000052 from a closed form with no sum in it. The generic case is the one place the two calculations overlap and they agree there.

So the sign has an answer in the one setting where everything is exact, and the answer is positive. One thing it rests on should be named, because it is a statement about the state and not about the geometry. The

image term is taken to enter with a positive coefficient, and the propagation of A.8 is what supplies it. The cross-sheet correlator there is the direct one shifted by half a period, $W(t - i\beta/2) = \cos \omega t / 2\omega \sinh(\beta\omega/2)$, positive at coincidence. A negative one would flip the sign of everything below. Two further limits travel with the result. This model has no caustic, so it supplies the sign where A.18 supplied the power, while A.18 supplies the power and not the sign, and neither is the black hole. And its fold fixes the single event $(T/2, \mathbf{0})$ where the real one is free, which changes nothing at the contact surface but makes the model globally unfaithful. What both models agree on is that the zero of the paragraph above is a property of a constant transverse radius and not of the fold.

Doing the fold in general settles that. Take any static spherically symmetric metric, $-h dt^2 + dr^2/f + R^2 d\Omega^2$, which covers Schwarzschild, flat space and the constant-radius model at once, and keep the same three pullbacks. Mode by mode the density is $\frac{1}{2h} \partial_\tau^2 g_\ell + \frac{f}{2} \partial_r \partial_{r'} g_\ell + \frac{\ell(\ell+1)}{2R^2} g_\ell$, and the wave equation for the ℓ -mode writes the time term as the radial part of the wave operator minus $\ell(\ell+1)g_\ell/R^2$. The two angular pieces then cancel against each other exactly, at every h , f and R , leaving

$$\rho_{\text{img}} = \frac{1}{2} \sum_\ell \frac{2\ell+1}{4\pi} (-1)^\ell \left[\frac{1}{\sqrt{g}} \partial_r (\sqrt{g} f \partial_r g_\ell) + f \partial_r \partial_{r'} g_\ell \right], \quad \sqrt{g} = \sqrt{h/f} R^2.$$

So the transverse sphere supplies the angular part of the potential in every geometry of this class, and no ℓ survives in the density at all. The remainder is not the curvature term and not any part of the potential: the wave equation was used to remove the time derivative, and what it leaves behind is the radial operator, not V_ℓ . What is left is the difference between that operator and the mixed radial derivative, and nothing else.

Both earlier cases follow from it. With R constant and $h = f = 1$ the mode function depends on r and r' only through $r' - r$, so $\partial_r \partial_{r'} g = -\partial_r^2 g$ and the radial pair cancels as well: that is the identical zero above, needing the constant radius exactly where this appendix said it did. Let R vary and the measure contributes R'/R terms with no counterpart in the mixed derivative, the cancellation stops, and a density is left, which is the flat-space answer. On Schwarzschild the radial operator is $\partial_r^2 g + \frac{2}{r} \partial_r g - \frac{2M}{r} \partial_r^2 g - \frac{2M}{r^2} \partial_r g$.

That is a smaller problem than the one this appendix opened with, and the rest of it can be settled by asking which of the two answers above a black hole is entitled to.

The flat case is not a toy. Its fold $\Theta(t, \mathbf{x}) = (-t, -\mathbf{x})$ is $\text{diag}(-1, -1, -1, -1)$, and the wedge reflection for the x_1 -wedge composed with the transverse antipode is $\text{diag}(-1, -1, +1, +1) \cdot \text{diag}(+1, +1, -1, -1)$, the same matrix. **The flat-space calculation above is the fold itself, at a Rindler horizon, in closed form.** Its positive answer is therefore an exact statement about the fold at a horizon and not about a model resembling one.

A third geometry is exactly solvable where the interior is not, and it decides between them. On the Einstein static universe a conformally coupled massless scalar has $\omega_n = n/a$ exactly with degeneracy n^2 , and the S^3 addition theorem at the antipode returns $(2\pi^2 a^3)^{-1} n^2 (-1)^{n+1}$, so the antipodal parity arrives on its own. The sum closes, and it closes at general angle:

$$G = \frac{1}{8\pi^2 a^2 (\cos \eta - \cos \gamma)}, \quad G_{\text{img}} = \frac{1}{16\pi^2 a^2 \cos^2(\eta/2)}, \quad \eta = \Delta t/a.$$

The Abel-summed series matches that to 1.2×10^{-5} at five angles well away from the antipode, and its coincidence limit returns the flat $1/4\pi^2 s^2$ to eight figures. That part holds, and the image term diverges where the null geodesics refocus.

The density needs every term of the point-split operator, and the transverse one is easy to mislay. Homogeneity of S^3 says that the total derivative along the diagonal vanishes, $G_{\alpha\alpha} + 2G_{\alpha\beta} + G_{\beta\beta} = 0$, which constrains the Laplacian of ϕ^2 and leaves the mixed derivative to be calculated. It is $-\delta^{ij}/8\pi^2 D^2$ where

the time term carries $+\delta^{ij}$, so it cancels the fourth inverse power exactly, and the conformal improvement $-\xi\nabla_s^2(\phi^2) + \xi G_{tt}\phi^2$ cancels what is left. **The image stress of a conformally coupled scalar on the Einstein static universe is zero, identically in η .**

The operator returning that zero returns the right nonzero answers elsewhere, which is what makes the zero believable. On a single Dirichlet mirror in flat space it gives $-1/16\pi^2 z^4$ at minimal coupling and zero at conformal coupling, both of which are in the textbooks. On this appendix's own flat inversion it gives $+1/32\pi^2 R^4$ and $+1/96\pi^2 R^4$, so the flat inversion carries a density and the Einstein static universe does not. Moving ξ off $1/6$ moves the zero with it, in the density and in the null-null component alike, so the cancellation is conformal invariance and not an operator that returns zero whatever it is handed. Two implementations agree, the algebra above and finite differences taken on the exact S^3 embedding with nothing differentiated by hand.

The zero is the value at one coupling of something with an exact closed form. Since $\nabla_s^2(\phi^2)$ vanishes identically, the whole ξ dependence sits in one term and

$$\rho_{\text{img}} = \frac{6\xi - 1}{16\pi^2 a^2(1 + \cos\eta)}, \quad T_{kk} = \frac{(1 - 6\xi)(1 - \cos\eta)}{8\pi^2 a^4(1 + \cos\eta)^2},$$

both reproducing the assembled operator at six couplings including two outside $[0, 1/6]$. **The image stress is proportional to the departure from conformal coupling**, rather than merely vanishing at it. At $\xi = 0$ the density is $-1/32\pi^2 a^2 \cos^2(\eta/2)$, negative and going as δ^{-2} , while the null-null component keeps the fourth power and is positive, which is the sign that focuses a null congruence.

The Hadamard expansion says where that factor comes from. Its log coefficient is $V_0 = \Delta^{1/2}[m^2 + (\xi - \frac{1}{6})R]/2$, and a massless conformal field in a Ricci-flat spacetime has nothing to put in it. Here $R = 6/a^2$, so $(\xi - \frac{1}{6})R$ is the whole of the breaking and $1 - 6\xi$ is its measure. The same slot carries m^2 , which is why A.18 finds a divergence where this finds none, its tower carries masses $m_\ell = \sqrt{\ell(\ell+1)}/a$. So the constant-radius zero that opens this appendix is the same phenomenon rather than an artefact of a product geometry.

That reading has a limit, and it is the parametrix's own. Everything above is the Hadamard log coefficient, and at a caustic the divergence is instead $\Delta^{1/2}/\sigma$, whose coefficient is one whatever the field is. Two things then separate this appendix's geometry from a hole's. This one is conformally flat, so a conformal field's image stress maps to the flat one and vanishes; and its antipodal pair at equal time is spacelike separated, so there is no $\sigma \rightarrow 0$ to expand in. A hole's contact is a null caustic in a spacetime that is not conformally flat, and there the coupling cancels outright, since $k \cdot k = 0$ drops the metric terms and the two surviving derivative terms are equal and opposite at leading order, leaving $(1 - 2\xi)(-X) - 2\xi X = -X$, and A.18's tower agrees. Five couplings and four masses return one coefficient, -0.0592 , with a spread of 10^{-5} and no zero at $\xi = 1/6$. So a massless conformal field does feel a contact divergence at a hole. Two derivatives on A.19's own amplitude give $T_{kk} = -3.19 \times 10^{-3} M^{-1/2} (M - r)^{-7/2}$, and against the same $11.5138/r_h^2$ that leaves a floor under the shell, $(9.85 \times 10^{-3})^{2/7} \ell_P^{4/7} r_h^{3/7}$, which is 1.1×10^{-19} m at a solar mass and 7.8×10^{-16} m at a billion. Mass still decides every shell that matters, beating the floor's term by 3×10^{46} at the fold's own fermion, but it does not decide whether there is one.

The transport along the contact curve matters in its own right, since it fixes how the fold acts there. A.15's contact curve is the $E = 0$ null geodesic, and $E = f dt/d\lambda = 0$ makes t constant, which makes V/U constant, so in Kruskal the curve is a straight ray through the origin, passing through the bifurcation surface where the fold acts and carried to its image by $\lambda \rightarrow -\lambda$. The contact boundary sits at $UV = \sqrt{e}/2 = 0.824361$ and the ray turns the angle by π to 1.3×10^{-13} , reproducing the contact condition. The Kruskal connection coefficients each carry a factor of U or V , so they vanish at the crossing and are odd along the ray; the transport from x to Θx is therefore the identity, while transporting only as far as the bifurcation surface departs from it by 1.04 , which is what distinguishes a cancellation from an integrator doing nothing. So $P = d\Theta$, with one $+1$ eigenvalue in the angular direction and -1 on both Kruskal directions and on the polar angle.

So the sign of the energy density at a black hole is open. A.15's conjecture has one exact positive datum, the flat inversion, which is the fold itself at a Rindler horizon and which carries a positive density at both couplings, and it has no argument that carries that sign to a hole.

An exponent is a different question and it does transfer, because it depends on the geometry alone. The image two-point function has a single curvature input, the Hadamard amplitude $\Delta^{1/2}(x, \Theta x)/4\pi^2\sigma$, whose Van Vleck determinant is unity in flat space and infinite at a conjugate point. The exponent is therefore fixed by how many transverse directions refocus at contact, and both that count and its reason are available in closed form.

Inside the horizon $\dot{r}^2 = E^2 + L^2(2M/r - 1)/r^2 > 0$ whenever $E \neq 0$, so r runs monotonically unless E vanishes. Since the fold preserves UV it preserves r , which puts x and Θx at one radius, leaving the $E = 0$ geodesic as the only curve that can return to the radius it left. So the connecting geodesic is unique, and $dr/d\varphi = \pm\sqrt{r(2M - r)}$ integrates to

$$r(\varphi) = M(1 + \sin \varphi), \quad \varphi \in [0, \pi],$$

which leaves $r = M$, grazes the bifurcation surface at $\varphi = \pi/2$, returns to $r = M$ and turns through exactly π . The contact boundary comes out of that: it is the radius at which the geodesic returning to its own radius turns through the angle the antipode asks for.

Both transverse directions carry a Killing field, and a Killing field restricted to a geodesic is a Jacobi field. The rotation about the axis through x is $-\sin \varphi \partial_\theta$ on the equator, so in the parallel frame ∂_θ/r the focusing Jacobi field is $M(1 + \sin \varphi) \sin \varphi$, which already carries the initial conditions the Van Vleck determinant asks for, and which vanishes at $\varphi = \pi$. A rotation vanishes on its own axis, and the axis through x runs through the antipodal direction too, so in any spherically symmetric spacetime a geodesic turning through exactly π ends on a point conjugate to its start. Every connecting geodesic here turns through π , since that is where the fold puts the image, so **the fold's image sits at a conjugate point at every radius**. Solving the two-point problem in Kruskal and continuing it in r from the null solution shows what the boundary picks out instead. The world function $\sigma(x, \Theta x)$ is negative inside, positive outside, and crosses zero at $r = M$ with

$$\sigma \longrightarrow (3\pi + 8)M(r - M),$$

a linear zero whose slope is twice the affine length $3\pi/2 + 4$ of the contact geodesic itself, reproduced to 1.1×10^{-9} with a worst shooting residual of 2.1×10^{-14} over the continuation. So conjugacy holds everywhere on the family and contact is the surface where the caustic reaches the light cone, which is also where the pair stops being unrelated and becomes causally related. The fold puts its own image on a caustic wherever it acts, which is what entitled A.18's model geometry to the caustic it was built around.

Nothing in that uses the metric, so it should survive charge, and it does. Substituting $J = r \sin \varphi$ into the Jacobi equation fixes the tidal in closed form for any static spherically symmetric f ,

$$T_\theta = \frac{L^2}{r^4} \left(1 - f + \frac{1}{2} r f' \right),$$

which is $3ML^2/r^5$ at $f = 1 - 2M/r$ and $L^2(3Mr - 2Q^2)/r^6$ with charge, and which matches a numerically built Riemann tensor to 3×10^{-8} at three charges. Integrating the geodesic and reading the tidal off that tensor rather than off the formula, $r \sin \varphi$ solves the equation along the whole curve. The contact radius stays at $(r_+ + r_-)/2 = M$ at every charge while the area ratio A_c/A_+ runs from 0.25 to 0.77, which is why the turning statement is the invariant one and the quarter of an area is not. Dimension generalises the same way and fixes the caustic's order: in D dimensions the transverse space has $D - 2$ directions, one of them ∂_t and $D - 3$ of them sphere directions each carrying a rotation that vanishes at both ends, so **the caustic order is $D - 3$** , which is one in four.

The order of that caustic separates the two exponents, and emptiness fixes it. Expanding the identity in a null frame leaves $R_{ab}k^ak^b$ equal to the trace of the transverse tidal matrix, the k and n terms dropping by Riemann's antisymmetry. Schwarzschild is vacuum, so that matrix is traceless, $\text{diag}(+\Psi, -\Psi)$ with $\Psi = 3ML^2/r^5$, one direction focusing and one spreading, and the Jacobi determinant can vanish through a single factor. A vacuum black hole has no point caustic available to it. The Einstein static universe has one, carrying $R_{ab}k^ak^b = (2/a^2)(k^t)^2 > 0$, both directions focusing and an antipode conjugate at order two. That is the whole difference between the two answers. Order two gives this appendix's δ^{-2} in the two-point function and δ^{-4} in the stress; order one gives A.18's $\delta^{-3/2}$ and $\delta^{-7/2}$.

So A.18's exponent is the one a black hole carries, and it has a reason behind it. The obstacle A.18 named was that its geometry is not Schwarzschild. The caustic order is the only feature of the geometry the exponent depends on, and that order agrees in the two, forced on the black hole by the Ricci term in Raychaudhuri. A.18's fields are massive, which by the paragraphs above is the condition for feeling the image term at all, so the two halves fit.

An amplitude comes out of the same two Jacobi fields: $dJ/d\lambda = -L/M$ at contact exactly, the spreading partner integrates to 47.561945 against an affine length $3\pi/2 + 4 = 8.712389$, and $\Delta \rightarrow 0.183180 \lambda_{\text{tot}}/(\lambda_{\text{tot}} - \lambda)$, a ratio left alone by the rescaling of the affine parameter that L performs. The tidal strength behind all of it was read off a Riemann tensor built numerically and checked first against $R_{ab} = 0$ and the Kretschmann scalar $48M^2/r^6$, and the two Jacobi fields satisfy their branch equations to 9×10^{-16} .

The exponent itself follows from the caustic's order. A Green function's generic light-cone singularity is $\delta^{-(D-2)/2}$, and each direction that refocuses on the connecting family steepens it by half a power, giving an exponent $-(D-2+n)/2$ for the image term and two powers further for a stress quadratic in derivatives. In four dimensions that reads $-(1+n/2)$.

Five points fix that without being fitted to it, three of them exact. A whole family, $\mathbb{R} \times S^N$, turns out to be solvable, because conformal coupling in $D = N + 1$ makes $\omega_n = [n + (N-1)/2]/a$ equally spaced and the degeneracy cancels the frequency, leaving a binomial series that closes on

$$G_{\text{img}} = \frac{a^{1-N}}{2\lambda 2^{2\lambda} \text{Vol}(S^N) \cos^{2\lambda}(\eta/2)}, \quad \lambda = \frac{N-1}{2},$$

whose $N = 3$ member is the Einstein static universe above, and whose exponent is $-(N-1)$ at every N . The Abel-summed modes match that to 8×10^{-4} relative over twelve points, and the exponents come out as $-(D-2+n)/2$ at $(D, n) = (3, 1), (4, 2), (5, 3), (6, 4)$ and $(7, 5)$. The two measured points are A.18's. Its control pair with no caustic returns -1.0095 against -1 , and its image pair returns -1.49874 against $-3/2$. With $n = D - 3$ for a spherically symmetric hole the four-dimensional answer is $-3/2$ and $-7/2$, which is A.18's measurement, so the transfer to Schwarzschild rests on the two configurations sharing a degeneracy.

The amplitude is the same statement seen in the proper-time representation, where $G = \int_0^\infty ds (4\pi s)^{-D/2} \Delta^{1/2} e^{-\sigma/2s}$ and a degenerate family replaces the infinite $\Delta^{1/2}$ by a finite $c_n s^{-n/2}$, which is precisely what shifts the exponent. On the sphere family $c_n = L^n \pi^{n/2} 4\pi / (n! \text{Vol}(S^{n+1}))$, and a Lorentzian Green function and a Riemannian heat kernel at the antipode agree on it to 3×10^{-6} at $n = 1, 2, 3, 4$. Since the heat kernel of a metric product factorises and a flat factor carries $\Delta = 1$, the amplitude belongs to the degenerate family alone, so A.18's $\mathbb{R}^2 \times S^2$ has S^2 's value, $c_1 = \pi^{3/2}a$, in the $D = 4, n = 1$ case, which is the case a hole is in as well.

That expression references the volume of a sphere one dimension above the family, which a local caustic cannot know about, so it wants rearranging. Rearranged it is $c_n = L^n 2\Gamma(n/2 + 1)/n!$, exact at all four, or equivalently

$$c_n = V_{\text{fam}} \left(\frac{L}{2\sqrt{\pi}} \right)^n,$$

with $V_{\text{fam}} = \text{Vol}(S^n)$ running over 2π , 4π , $2\pi^2$ and $8\pi^2/3$, so the linear dependence on the family's volume is tested, any other power of it misses by 80% or more.

Whether the length there is the geodesic's, or some other scale of the Jacobi profile that happens to agree with it on a sphere, cannot be settled on a symmetric space. The curvature operator is parallel along a geodesic in one, so every Jacobi field is a pure sine and every candidate scale coincides; complex projective space gives $\sin(2u)/2$ at its cut locus, which is still a sine. A surface of revolution $du^2 + f(u)^2 d\varphi^2$ does settle it. Its meridians are a one-parameter family of volume 2π whose Jacobi field is f itself, and $f = a[\sin\theta + \epsilon h(\theta)]$ with h odd at both poles moves the profile while L and V_{fam} stay where they are. Taking the ratio of the perturbed pole-to-pole heat kernel to the round one at fixed proper time, which cancels the solver's bias, that ratio extrapolates to 1 for six perturbations pushing in both directions, where a profile-dependent amplitude would need between 0.73 and 1.50. **The amplitude is the geodesic length and the family's volume and nothing else.**

On the contact geodesic $V_{\text{fam}} = 2\pi$ as well, a circle of planes through the axis, and the tidal matrix is exactly $\text{diag}(+\Psi, -\Psi)$ in a parallel-propagated frame, so the Gaussian integral factorises and the single non-degenerate transverse direction contributes the 0.183180 above multiplicatively.

The affine parameter is not that length, and a plane wave shows why while confirming the exponent on a genuinely null caustic. Take $ds^2 = 2du dv + A_{ij}x^i x^j du^2 + dx^2 + dy^2$ with A traceless, which is vacuum and so carries one focusing transverse direction and one spreading, exactly the contact geodesic's structure. Fourier in v turns the wave equation into a Schrödinger equation in u with an oscillator of frequency ω in one transverse direction and an inverted one in the other, so the propagator is Mehler's, the caustic sits at $\omega U = \pi$, and the amplitude goes as $(\pi - \omega U)^{-1/2}$, the same half power, exactly, on a null vacuum caustic. What it supplies no candidate for is a length, since under $u \rightarrow cu$, $v \rightarrow v/c$ the metric keeps its form with $\omega \rightarrow \omega/c$ and $U \rightarrow cU$, so $1/\omega$ and U carry the dimension of u .

Two measurements fix which length the rule uses. It is not a functional of the curvature along the geodesic: across the surfaces of revolution above the Gaussian curvature $-f''/f$ moves by up to 254 per cent and changes sign, while L and V_{fam} stay put and the amplitude does not move. And it is not the geodesic's own length: putting the caustic in an S^2 and letting the geodesic also run along a flat direction holds the projection at πa while the geodesic's own length grows as $\sqrt{\Delta u^2 + (\pi a)^2}$, and over a 38 per cent change in that, in one, two and three flat dimensions, the amplitude holds to the fifth decimal while tracking the sphere's radius exactly when that is changed instead. In the Lorentzian version the pair sits on the light cone, where the geodesic's own length is zero and the amplitude is unchanged.

What it is, then, is the arc length of the geodesic's projection onto the factor the caustic lives in. On $\mathbb{R}^2 \times S^2$, which is A.18's own geometry, the curve covers π of the sphere, the projection is πa , and $\sqrt{\pi} \cdot \pi a = \pi^{3/2} a$ exactly, giving that model a closed-form amplitude. A black hole carries the same transverse sphere with a varying radius, so the same quantity reads $\int r d\varphi$, which along $r = M(1 + \sin\varphi)$ is $M(\pi + 2)$.

It has a second reading, and the two turn out to be one integral. Inside the horizon r is timelike, so the geodesic sweeps a proper extent $\int d\tau = \int dr/\sqrt{2M/r - 1}$ climbing to the bifurcation surface and returning, and $dr/d\varphi = \sqrt{r(2M - r)}$ makes $r d\varphi = \sqrt{r/(2M - r)} dr = d\tau$ term by term. Both come to $M(\pi + 2) = 5.141593M$, and neither mentions the affine parameter. With the reduced Van Vleck 0.183180 from the spreading direction,

$$c_1 = \sqrt{\pi} M(\pi + 2) = 9.1132 M, \quad \Delta^{1/2} \rightarrow 3.90 M s^{-1/2}.$$

The sign follows from the one exactly solvable caustic. On the Einstein static universe the divergent stress is $T_{kk} = (1 - 6\xi)(1 - \cos\eta)/8\pi^2 a^4(1 + \cos\eta)^2$ while the Hadamard log coefficient is $V_0 = \Delta^{1/2}[m^2 + (\xi - \frac{1}{6})R]/2$, and the two carry the same factor with opposite signs, so the caustic-divergent stress goes as $-V_0$. Schwarzschild is Ricci-flat, which shuts the $(\xi - \frac{1}{6})R$ channel whatever the coupling and leaves $V_0 = m^2 \Delta^{1/2}/2 > 0$ for any field that is not massless. So $T_{kk} < 0$ at the contact surface. Integrating Raychaudhuri, a smaller R_{kk} pushes the conjugate point further away and a negative one removes it. At $R_{kk} = 0.5, 0.2$

and 0.05 it sits at 6.28, 9.94 and 19.87, and at zero or below there is none. Contact requires a conjugate point at the antipodal angle, so delaying it makes contact harder. **A.15's divergence closes the region rather than marking it**, and it does so for the fields that exist rather than for a chosen coupling. The sign is calibrated on that one exactly solvable caustic.

The shape either side of the boundary follows from the sign of σ and needs nothing further. Outside the horizon the pair is spacelike too, so the term does not stop there: it is suppressed rather than absent, exponentially in the mass for anything heavy and quadratically for anything light, with the worst case across the whole field content thirty-four orders below the dark energy. Section 1's silence is about the commutator and does not reach the stress tensor, which §5.4 says in its own words. Between the horizon and $r = M$ the pair is spacelike and the term is finite and real. At $r = M$ it diverges as $\delta^{-3/2}$ with the coefficient above. Inside, σ is negative and $\sigma^{-3/2}$ continues to $i|\sigma|^{-3/2}$, so the image term is imaginary there: **the contact region carries a decay rate**, which is pair creation, and it arrives from the causal structure alone.

One thing about that treatment, the obvious tool is the wrong one. At a generic fold caustic the action has a cubic stationary point in the family parameter and the uniform replacement for the failed parametrix is an Airy function. Here the degenerate direction is generated by a rotation Killing field, so every member of the family is an isometric image of every other and the action is exactly flat along it, the affine length is the same at all nine rotations sampled, to 10^{-12} . A symmetry degeneracy is not resolved by a cubic term, and integrating the orbit exactly is what replaces it. The power that follows can be checked against the exponent rule without doing the integral, since the Riemannian $\Delta^{1/2} \rightarrow c_n s^{-n/2}$ reads in the Lorentzian propagator as one extra half power of the null momentum per degenerate direction, and that reproduces $-(D-2+n)/2$ at $n = 0, 1, 2, 3$, in agreement with the amplitude above reached the other way. The sign of the resulting stress still needs A.18's sum redone on the Schwarzschild contact geodesic, which is possible, since that geodesic is the closed-form curve above.

A.20 Matter rule, neutrino masses and the conditional decay scale. §7's commitments, with their conditions. We use the gravitational-production state and radiation history of Boyle, Finn and Turok [20,21]. In the adopted small-Weyl-coupling branch, matching the observed dark-matter abundance gives $M_1 = 4.916 \times 10^8$ GeV. This number is conditional on that production model and state selection.

Stabilising the sterile species with a $\mathbb{Z}_2$ is an additional matter-theory assumption. The quotient admits a nontrivial sign bundle, but the bundle pulls back trivially to the simply connected physical cover. We have not constructed a continuation from the Euclidean boundary data to an interacting cover action and measure that enforce the same species rule. The phenomenology therefore retains the stabilisation rule those authors impose.

When that rule is exact, the sterile particle cannot decay. Its forbidden Yukawa column makes the light-neutrino mass matrix rank at most two in the stated seesaw approximation. The two measured oscillation splittings then leave one light neutrino massless. For normal ordering and the oscillation inputs used here, $\Sigma m_\nu = 58.78 \pm 0.32$ meV, with the independent input errors propagated at first order. Which ordering holds is specified separately; exact stabilisation alone does not choose it.

The double-beta interval in the cosmology paper's §3.3 spans the remaining unknown Majorana phase. The fold does not select a point within that interval.

A concrete weakly broken implementation adds a small gauge-invariant Yukawa column $-y_\alpha \bar{L}_\alpha \tilde{H} N_{1R} + \text{h.c.}$, with the other two sterile neutrinos heavier and no other open N_1 decay channel. This is an additional matter interaction. The fold does not imply it. Write $q = y^\dagger y$ and $H^0 = (v + h + iG^0)/\sqrt{2}$, with $v = 246.22$ GeV. At tree level and $M_1 \gg m_W, m_Z, m_h$ [22],

$$\Gamma_0 = \frac{qM_1}{8\pi}, \quad \text{Br}(h\nu) : \text{Br}(Z\nu) : \text{Br}(W\ell) = 1 : 1 : 2,$$

where the charged channel includes both charges. A pure Higgs-neutrino branching assignment is therefore unavailable in this example. The same column adds $\delta m_\nu = -v^2 y y^T / (2M_1)$. Relative to the rank-two tree-level baseline, its smallest mass obeys

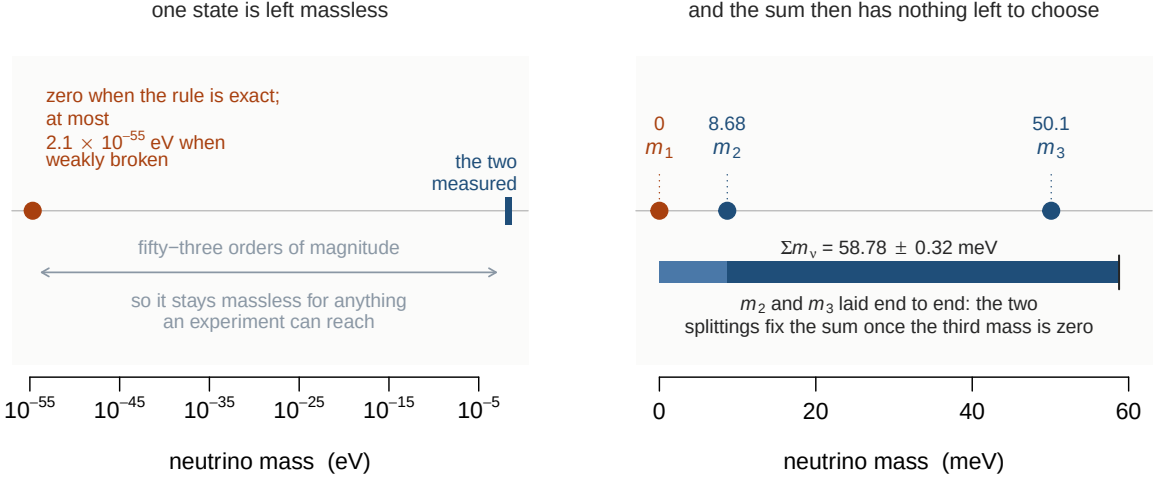


Figure 16: What the rule does to the spectrum. Left, the full range. When the rule is exact the sterile partner’s Yukawa column is forbidden, the light mass matrix has rank at most two, and one light neutrino is exactly massless; the weakly broken version below lifts that zero to at most 2.1×10^{-55} eV for the example lifetime, fifty-three orders of magnitude below the lighter of the two measured splittings. Right, the same spectrum on a linear scale in meV. With m_1 at zero the two splittings fix the sum with nothing left to choose, $\Sigma m_\nu = \sqrt{\Delta m_{21}^2} + \sqrt{\Delta m_{31}^2} = 58.78 \pm 0.32$ meV for normal ordering, drawn as the two masses laid end to end.

$$m_{\min} \leq \|\delta m_\nu\|_2 = \frac{4\pi v^2 \hbar}{M_1^2 \tau}, \quad \tau = \hbar/\Gamma_0.$$

Duran, Morrison and Profumo [23] calculated the gamma-ray and neutrino emission from decaying ultra-heavy sterile neutrinos across generalised CPT-symmetric cosmologies, and report a reach to lifetimes near 10^{27} s for very-high-energy Cherenkov telescopes and 10^{30} s for neutrino telescopes. The example lifetime used below is inside that second window, so the value is testable. For that example $\tau = 10^{28}$ s, $\|y\| \simeq 1.8 \times 10^{-30}$ and the bound is about 2.1×10^{-55} eV, reached twice, once from each of the two definitions above and the same number both ways. This bounds the added tree-level mass term; radiative masses of the baseline are a separate question. With the adopted radiation history and only this new interaction, the inverse-decay population and the decay depletion are negligible at the accuracy of the abundance benchmark. Thus a long-lived decay need not spoil the abundance or the effectively massless light-neutrino result, although exact stabilisation and a nonzero decay cannot hold simultaneously.

For a cold parent at rest, the neutral hard channels have

$$E_{\nu,b} = \frac{M_1^2 - m_b^2}{2M_1} \simeq \frac{M_1}{2}, \quad b = h, Z.$$

Borah, Das, Okada and Sarmah [24] reach the same kinematics from the other direction. They take a heavy right-handed neutrino of 440 PeV and show its decay reproduces the KM3NeT flux at the observed 220 PeV, reporting that the lifetime required for the best fit saturates existing gamma-ray bounds. The mass differs from M_1 here by ten per cent. The logical direction is what differs: they select the mass to fit the measured energy, whereas M_1 above is fixed by matching the dark-matter abundance in the adopted production branch, with the half-mass energy following. We claim no priority for the decay interpretation of that event, which is theirs. What the fold adds is that the scale was not chosen to land there.

Both neutral hard channels, $h\nu$ and $Z\nu$, therefore give 245.8 PeV at the computational mass benchmark, a physical scale of roughly 245 PeV. Electroweak radiation and boson decay produce a continuum and change

the endpoint weight at this energy [25]. Halo velocities, extragalactic redshift, propagation and detector response must also be included to predict a measured spectrum. Neither that spectrum nor an event rate is calculated here. Two consequences of the 1 : 1 : 2 ratio separate, one closing a hope and the other opening a test.

Flavour cannot do the assigning that the cosmology paper’s §3.2 falsifier needs. The channel ratio is flavour-blind: all three channels carry the same $|y_\alpha|^2$, and the 1 : 1 : 2 comes from gauge and Higgs structure, with the Yukawa column playing no part, which stays free. Since the oscillation-averaged transfer is linear, the Earth-side compositions the decay can reach are exactly the convex hull of the images of pure- e , pure- μ and pure- τ sources, which is the image of the whole source simplex. Every astrophysical source composition is itself a point of that simplex, so the decay reproduces any of them by choice of Yukawa column, and pion-decay, muon-damped, neutron-decay and equal-flavour sources are all reachable. An Earth-side flavour measurement therefore cannot separate this component from an astrophysical population, and the cosmology paper’s §3.2 directional requirement stands undiminished.

What the ratio does fix is the split between the hard line and everything else, and that is a discriminant. Of four decays one gives $h\nu$ and one $Z\nu$, both feeding the line at $M_1/2$, while two give $W\ell$, whose charged lepton never reaches Earth and whose products land in the continuum. The line-to-continuum normalisation is therefore set by tree-level structure and is independent of the free Yukawa column, where an astrophysical population has no reason to hold a fixed ratio between a line and its own continuum.

How visible that is can be judged without the fragmentation calculation, by asking how much the answer depends on it. Write the endpoint continuum as $N_{\text{tot}}(n+1)(1-x)^n$ with $x = 2E/M_1$, so the continuum inside a window δ below the endpoint is $N_{\text{tot}}\delta^{n+1}$ while the line contributes 0.5 regardless. At the thirty per cent resolution of the cosmology paper’s §3.2 and a cascade multiplicity of 10^2 to 10^3 , the line is a feature of order tens of per cent of the endpoint continuum for a fall-off $n = 3$ to 5 , and is swamped at $n = 2$. Across a grid spanning both unknowns it clears a fifth of the local continuum at 81 per cent of the grid points at that resolution and at 50 per cent at fifty per cent resolution. Those are fractions of points on a scan carrying no measure. They are neither probabilities nor a detector forecast. The claim they support is only that the 1 : 1 : 2 structure fixes the tree-level neutral-channel yield at 0.5 per decay; what fraction of that is observable at an endpoint needs electroweak showering, propagation and detector response, none of which is done here. Visibility is real in the tree-level sense and genuinely conditional on everything downstream of it.

The part that needs nothing is the lock. That 0.5 is fixed by the 1 : 1 : 2 tree-level ratio and by nothing else, not the Yukawa column, not the lifetime, not the halo profile, not the fragmentation. The prediction is therefore not a ratio but a constraint, that the line strength and the decay rate are tied at a fixed fraction, two hard neutrinos per four decays. An astrophysical population has no reason to tie a line-like feature to its own continuum normalisation at any fixed value, which is what makes this a discriminant where flavour is not. It is a population-level statement, since measuring a lock requires both the line and the continuum it is locked to, so it belongs with the cosmology paper’s §3.2 directional requirement, where the single-event test is a different instrument. The two sit in the same regime and need the same order of exposure, which is the useful part. A population that can place the events on the sky can also weigh the line against its own continuum, and the two discriminants come together or not at all. KM3NeT’s broad reconstructed event-energy interval [26] is not evidence for this decay assignment; no likelihood preference over an astrophysical population is claimed.

The distinction is experimentally useful. A spectrum and flux calculated for a specified lifetime and flavour direction can be tested with neutrino and photon observations. A channel exclusion would constrain that implementation, while exact stabilisation predicts no decay signal. Neither observation would by itself establish or reject the entire two-copy geometry.

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AI disclosure. The author originated the central ideas, directed the work, and is responsible for every claim and decision in this paper. Anthropic’s Claude Opus 5, Claude Opus 5.5 and Claude Sonnet 5 were used under the author’s direction to draft and edit text from the author’s outlines, to write and revise the R and Python in the release, and to run repeated checking passes over both manuscripts. MiniMax-M3, DeepSeek-V4-Pro, DeepSeek-Reasoner, Alibaba’s Qwen3.8-Max, Moonshot’s Kimi K3 and OpenAI’s GPT-5.6 (Sol, Terra and Luna) were used as independent readers, to search the literature and to list objections and alternatives to work through. These are large language models, nothing a model proposed was kept until it had been redone in a script or checked against a source, and no model is an author. Every number here is produced by a script in the release, and each check over those scripts was validated by planting an error it had to catch, since a check that matches nothing reports success.

Code and data availability

Every quantitative claim in this paper is produced by a script in the release, and the release carries a table, `CLAIMS.tsv`, mapping each claim to the file that produces it. The code is at <https://github.com/BenWise/man/separate-ways>, archived at <https://doi.org/10.5281/zenodo.23034297>, and this paper’s Zenodo record is <https://doi.org/10.5281/zenodo.23030633>, the concept DOI, which always resolves to the most recent version, with the cosmology paper it accompanies at <https://doi.org/10.5281/zenodo.22888119>. The scripts specific to this paper are the 175 calculation files and the 22 figure generators, each printing its own results and its own limits when run with no arguments. Of those, 149 calculations and 19 of the generators are R and load no package, so base R is enough; the other 26 calculations are Python, 7 using the standard library alone and the rest importing numpy, scipy, sympy or mpmath at the versions the release records. 17 of those generators draw the figures here, and each re-derives and asserts the numbers its own caption states. Several plant deliberate failures and stop if the check does not catch them, because a check that cannot fail is worse than none. Two further files are shared with the cosmology paper and predate this one.

Ten of those files stand behind no claim above, and deliberately. Three are the checkers that guard the manuscript rather than results in it. Four are superseded, and each prints why and what replaced it, first and last, when run: a header comment is not enough when the output is what a reader sees. One adjudicates an external model’s claims rather than making any. One supports a result in the cosmology paper. And one assesses outlier detection as the instrument for a rare contaminating minority, kept as the record of a quantified test.

Two files in the same release carry what is not settled. `REFEREE_OPEN.md` lists the open objections and what was done to each, including the ones that went against the paper. `PROVENANCE_GAPS.md` records every number that was once stated without a script behind it, what closing it found, and the four figures that had to be replaced because they sat below the arithmetic floor of the method that produced them.

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The numbering here is this paper’s own. Entries shared with the cosmology paper carry different numbers there; the rest belong to this paper alone.

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